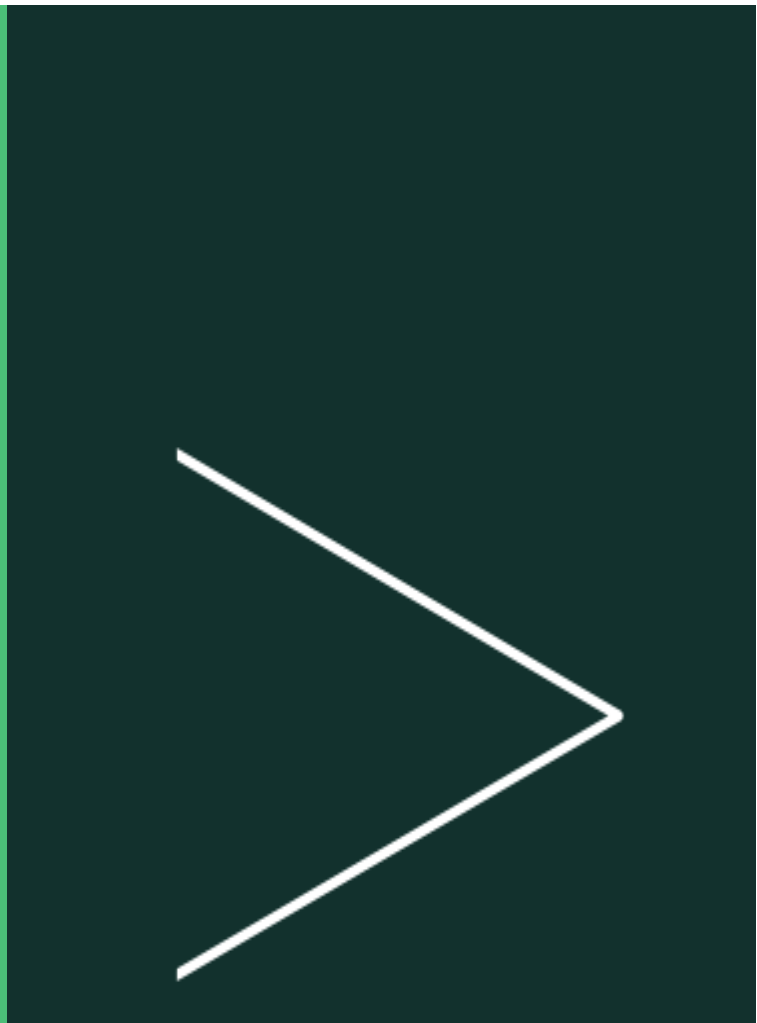
Founders Pledge

Navigating Risks from Advanced Artificial Intelligence

A Guide for Philanthropists

TOM BARNES

June 2024



Executive Summary

The field of Artificial Intelligence is highly complex, ever-changing, and hotly debated. Despite these uncertainties, this report reaches the following conclusions:

- Advanced Artificial Intelligence will **profoundly affect the trajectory of human civilization**. (more)
- We believe that advanced AI systems will **more likely than not be developed and deployed within our lifetimes** — or concretely, a greater than 50% chance by 2050. We do not dismiss the possibility that such systems could be developed much sooner in the next ten years (with a chance of between 10% - 50%). (more)
- **Advanced Artificial Intelligence could pose risks up to and including catastrophe, on the scale that extinguishes or disempowers all of humanity**.
    - In particular, we are most concerned that powerful AI systems may not be controllable, and thus may seek to disempower humanity in pursuit of their own objectives. (more)
    - We are further concerned that AI systems may be misused for destructive purposes, such as for biological attack (more) or military advantage (more).
    - Additionally, we believe that AI systems could result in broader structural failures, including the extreme concentration of power and wealth (more); the erosion of epistemic security (more); or the mistreatment of morally-relevant beings (more)

The report then turns to consider how philanthropists can most effectively tackle these risks, in order to maximize their impact on one of the most pressing global issues. Our core guiding principles for effective giving within AI risk are the following:

1. Prioritize reducing risks which pose **the greatest expected harm**, both in terms of their likelihood, and the severity of their impact given their occurrence.
2. Prioritize threats and solutions which are **most neglected** by other funders.
3. Prioritize strategies which **guard against failures from other actors**, allowing one to **hedge against the highest risk scenarios.**
4. Pursue interventions which can act on **multiple risks simultaneously**, forming a **threat-agnostic approach.**
5. **Leverage existing societal resources** towards the most impactful opportunities which other actors - such as governments - may miss.
6. Mitigate the possibility of inadvertently causing harm by pursuing **robustly positive opportunities.**

In practice, we believe that the implementation of these principles implies the following concrete recommendations:

1. **Increase global resilience** to cover cases where existing AI risk defenses fail to restrain localized catastrophic events.

2. Fund **technical research to prevent misaligned AI systems** emerging, particularly focusing on alignment challenges **neglected by AI labs** and others.

3. Facilitate **coordination between key actors** to prevent multilateral failures, for instance through the use of dialogues, agreements and standard regulations,

4. **Ensure that governments are well equipped** to deal with the many risks from AI, given governments' unique powers and resources

We believe that funding in these areas can not just reduce the risk of catastrophe, but meaningfully improve humanity's long-run trajectory.

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Introduction

Artificial Intelligence (AI) represents one of the most significant technological advancements of our era, promising profound changes across various sectors and aspects of human life. However, as AI systems become increasingly sophisticated, they also introduce complex and unprecedented challenges. This report delves into these complexities, particularly focusing on the transformative potential of AI and the associated risks that could shape the trajectory of human civilization. By examining a broad spectrum of AI challenges, this report aims to provide a comprehensive guide for philanthropists and other stakeholders on how to effectively mitigate the most pressing risks associated with advanced AI development. Through strategic interventions and informed funding, we can not only avert catastrophe, but also steer AI advancements towards a more secure and beneficial future for all.[1]

Definitions

There is intense discussion around opportunities and risks from artificial intelligence, across many domains. To make progress on these questions, it is first necessary to precisely define the topic.

Artificial intelligence (AI)

We adopt a minimalist definition of intelligence, as proposed by Legg and Hutter (2006):[2]

> *Intelligence measures an agent's ability to achieve goals in a wide range of environments.*

This definition captures both (1) *ability* - how well the agent works on a given task - and (2) *generality* - how many different task types the same agent can perform.

Transformative Artificial Intelligence (TAI)

This report focuses on "Advanced" AI systems, which could have a transformative impact.[3] More precisely, we refer to ***Transformative* AI (TAI)**, defined by Karnofsky (2016) as

> *"... a potential future AI that precipitates a transition comparable to (or more significant than) the agricultural or industrial revolution".*[4]

This transition may occur due to a single AI system, or many. TAI is defined separately from artificial general intelligence (AGI), since (for instance) reaching AGI may not be required for AI systems to be transformative, or to pose catastrophic risks.

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[1] This introduction was written by ChatGPT
[2] "Intelligence" itself has been defined in many ways - see Legg and Hutter (2007) for ~70 such definitions
[3] We use the terms "advanced" and "transformative" interchangeably
[4] For more detail on conceptualizing TAI, see Muelhauser (2019). See Maas (2023) for a more detailed analysis of terminology around TAI

Risks from AI

In this report we primarily focus on extreme risks from AI - both "catastrophic" and "existential". Following Ord (2020), we define an existential risk broadly as "*a risk that threatens the destruction of humanity's long-term potential*". This more inclusive definition encompasses civilizational collapse and dystopian outcomes, in addition to extinction.

Contribution to existing literature

Rather than contribute primary research, this report attempts to condense and distil many existing reports across multiple domains. As such, it takes a very high-level approach, whilst keeping a practical focus on *what* philanthropists and other donors should fund to reduce existential risks from AI, and *why*.

Outline of this report

This report is broken down into the following sections:

- Section 1 outlines broad claims & considerations about the situation we currently face. It covers high level uncertainties and questions, such as the likelihood and impact of advanced AI.
- Section 2 explores the plausibility of different **threat models** - paths by which advanced AI systems could cause catastrophe.
- In Section 3, we outline a large number of **potential interventions** that aim to reduce risks from advanced AI
- In Section 4, we investigate the **funding landscape** for working on reducing risks from advanced AI
- Finally, in section 5 we outline a **strategy for prioritizing** between interventions to reduce risk.

The report is thus structured as follows (Figure 1):

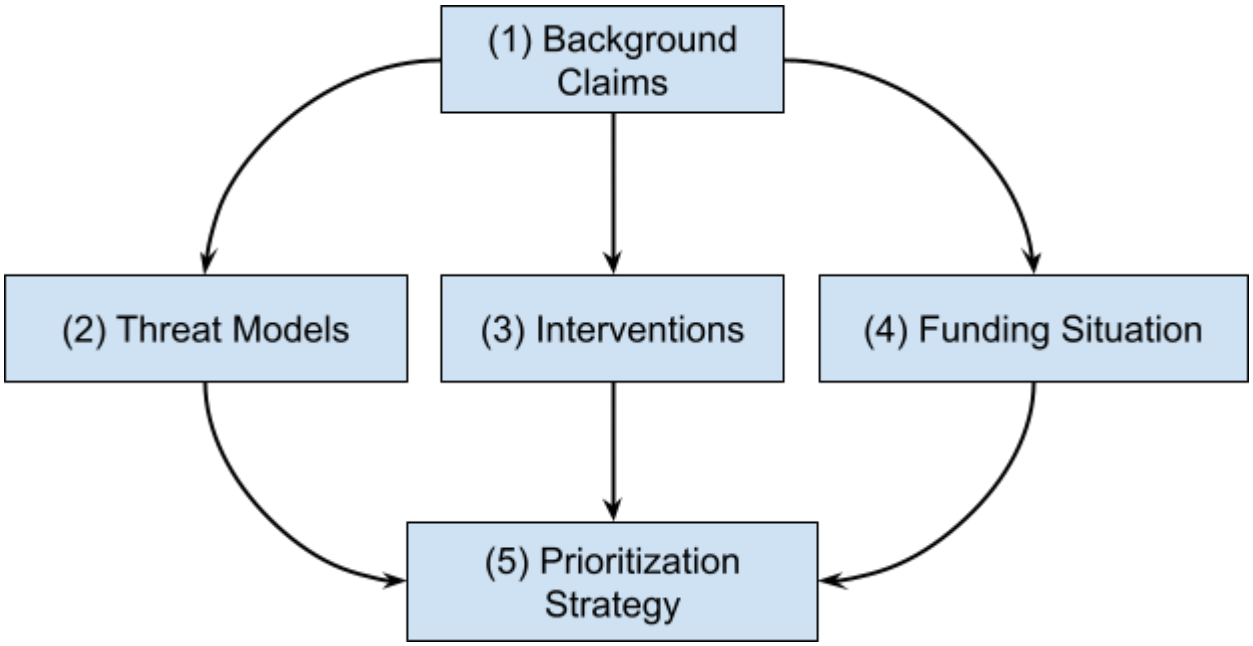


Figure 1: Outline of report

Section 1: Background claims

In this section, we first outline the significance of advanced AI on society's long-run trajectory. We then give some reasons for expecting that such transformations may occur in the near future. We cover two major questions:

1. **Significance**: How significant would advanced AI be?
2. **Timelines**: When will advanced AI systems be developed?

Significance

There are two broad arguments for why advanced AI would radically alter our long-run trajectory:

Argument from Technology:

1. Technological events such as the agricultural and industrial revolutions have previously altered the trajectory of human civilization,
2. This generalizes: Technology has the ability to alter human civilization's trajectory.
3. AI is a technology powerful and general enough for such a transformation to occur again.

Argument from Intelligence:

1. Intelligence allowed humans to assert control over Earth's resources, using them for their own purposes whilst preventing other species from doing so.
2. This generalizes: Intelligent beings develop the capability to gain control over resources for their own ends, at the expense of other beings.
3. Smarter-than-human AI systems could similarly gain control over resources, at the expense of humans.[5]

Since advanced AI would be *both* a technological *and* intelligence jump, it's plausible that TAI could dramatically change humanity's long-run trajectory.

Argument from Technology

Advanced AI would be a significant *technological* jump. Taking a macro-historic perspective, we can see how technological developments have changed humanity's trajectory:

- The Neolithic Revolution (due to new agricultural technology) drove population growth in key settlements, spurring the development of early human civilizations
- The Industrial Revolution (due to various scientific and technological advances) caused dramatic reductions in mortality, illiteracy, poverty, malnutrition, whilst dramatically increasing political freedom, moral progress and scientific knowledge.

The impact of the latter is illustrated in Figure 3:

[5] This is known as the "second species" argument - for more details, see Ngo 2020

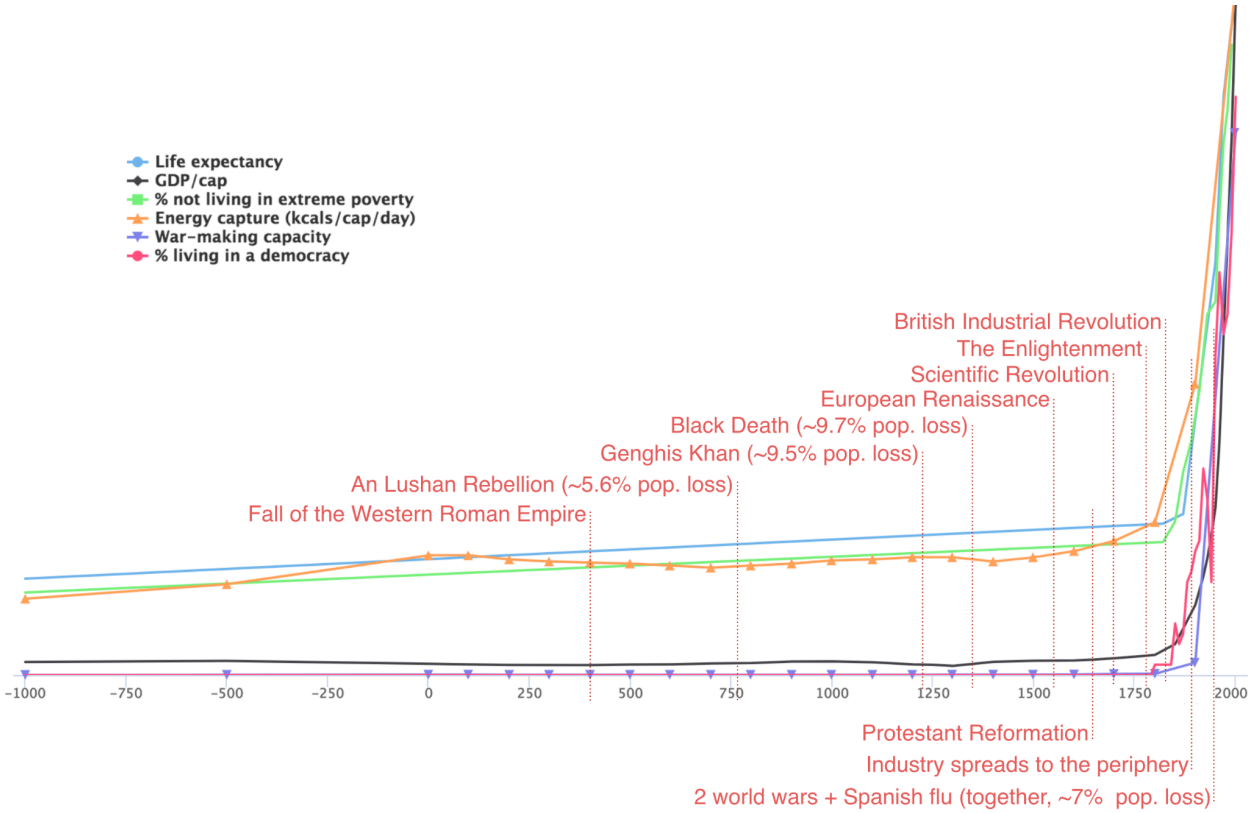


Figure 3: Plot of Life Expectancy, GDP per Capita, % of people not living in extreme poverty, Energy capture, War-making capacity and % living in a democracy, over time (Muelhauser (2017))

As Muelhauser (2017) crudely summarizes:

> *Everything was awful for a very long time, and then the industrial revolution happened.*[6]

Thus, there is precedent for technology to alter humanity's trajectory (either positively or negatively).

Argument from Intelligence

We can also see how AI could radically change our long-run trajectory, given the unique nature of AI as a potential *smarter-than-human intelligence*.
Humans have exerted control over the Earth's resources across a wide variety of environments. Intelligence is the factor which (by and large) has allowed for this dominance. Concurrently, the Holocene ("Human") era has led ~140,000 species to become extinct per year - a rate 100x to 1000x greater than previously (Pimm et al 1995).[7]

[6] This is approximately the case for humans, though an assessment inclusive of non-human animals could change this conclusion.
[7] Diagram from Ceballos et al (2015). The rate of animal extinction has rapidly increased over the last few centuries, coinciding with humanity's collective capabilities rising due to greater technology

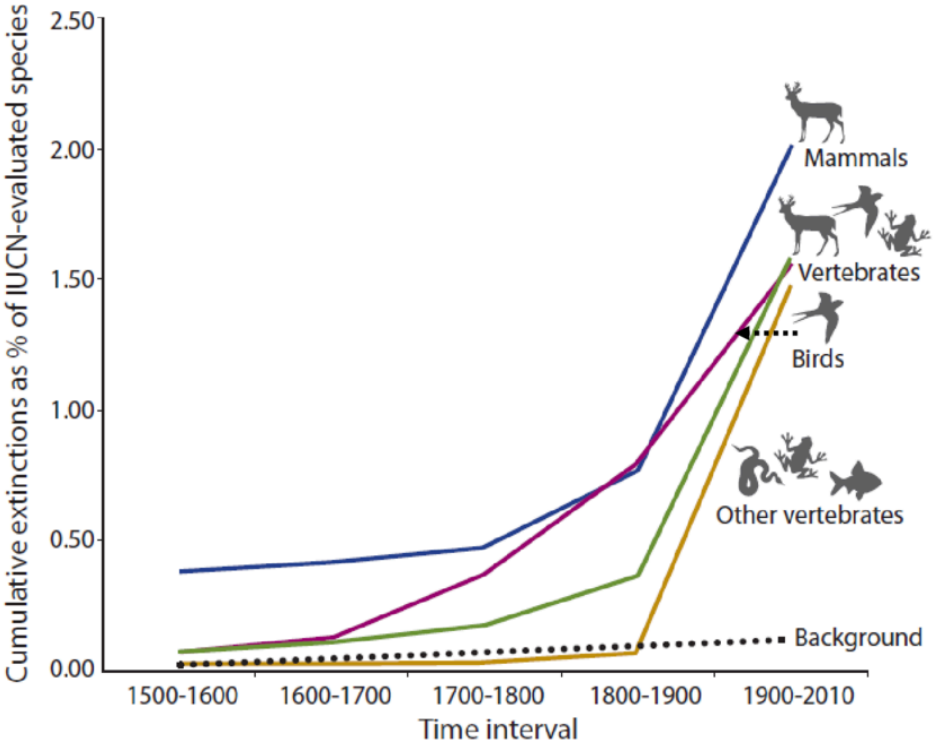


Figure 2: Cumulative vertebrate species recorded as extinct or extinct in the wild by the IUCN. Full Text at [Ceballos et al (2015)]

If AI systems exceed human capabilities, it would similarly end humanity's position as the most capable agent on the planet. If we face the same fate that other animals have in the face of humans, then a smarter-than-human intelligence could lead to human extinction, or similarly radical outcomes.[8]

Summary

Future AI systems may *both* exceed human intelligence, *and* initiate a massive technological jump. These two lines or arguments both converge on powerful AI possessing the potential to alter our long-run trajectory.

Strategic Clarity

Suppose we grant that the development of advanced AI would substantially alter our long-run trajectory. This still leaves many macro-strategic uncertainties, such as:

[8] It's worth noting that many other less intelligent species have not been driven to extinction. However, this is where precision around an "existential catastrophe" is necessary. Other species no longer threaten human dominance of the planet, leaving them unable to execute goals that conflict with human preferences. Similarly, even if advanced AI systems do not cause human extinction, they may disempower us in a similar fashion. This would (by and large) constitute an existential catastrophe.

1. Is advanced AI even *feasible*?
2. If so, over what *timeframe*?
3. How *rapidly* will advanced AI systems be developed?
4. *How many actors* will be involved in developing advanced AI systems?

Clarity over these questions could greatly improve the quality of decision-making when navigating the risks from AI.

Feasibility

In *principle*, it seems very likely[9] physically possible for humanity to *at some point* be able to build AI systems which are on par with human-level abilities. At the very least, complete replication of the human brain seems physically possible, under the Computational Theory of Mind (CTOM). CTOM implies that the computations of a human mind could in theory be replicated on a Turing Machine. With sufficient resources, we could thus recreate the human mind, which would likely pass the bar for being a transformative system.

Of course, it may be many centuries before humanity is able to construct such a machine, but we have strong theoretical grounds to suspect a human-level intelligence is at least *possible*. Indeed, this possibility drove initial attempts to build AI systems, using logical systems.

However, it is an open question whether or when we will be able to reach human intelligence based on Machine Learning (ML) techniques, the dominant paradigm within artificial intelligence today.[10]

Timelines

It is generally challenging to make accurate long-range forecasts, especially on a subject as speculative as transformative artificial intelligence.[11] Nevertheless, many different approaches have been attempted. There is insufficient space in this report to review all such approaches - however, Wynroe et al 2023 provides a literature review synthesizing many forecasting methods.

[9] I.e. more than 95% chance
[10] Most uncertainty is driven by the question of whether sufficiently large amounts of data and computing power can reach human-level intelligence. ML methods themselves seem *in theory* capable of replicating human intelligence, given that three-layer Deep Neural Networks are universal function approximators (Hartman et al 1990)
[11] See for example Muelhauser (2019) on the feasibility of long-range forecasts (in general)

In summary, they find that model-based approaches (weighted by credence)[12] predict a median date of **2089**. Meanwhile, judgment based approaches (weighted by credence)[13] predict a median date of **2045.** Some more details are given in the table below:

Timeline	**Probability of TAI/AGI by...**		
Date	**2030**	**2050**	**2100**
Average of model-based approaches	8%	27%	54%
Average of Judgment- based approaches	25%	57%	88%

However, three important caveats are necessary:

1. There is some correlation in reasoning between the above forecasts. As such, they do not represent fully independent predictions of TAI. For example, it's likely Metaculus forecasters defer at least somewhat to expert surveys from Grace et al (2022), or to previous work by Open Philanthropy (rather than forming fully independent impressions)
2. There is likely selection bias at play, since individuals most convenience that advanced AI systems will emerge soon are likely to be incentivised to spend time focused on the question of AI timelines (and thus the sample is skewed towards individuals with shorter timtlines).
3. The timeline for *transformative* AI is not obviously the question we are most interested in. Instead, we might be more interested in:
    - When will it be *possible* to build transformative AI?
    - When will transformative AI *actually* be built / deployed?
    - When will artificial *general* intelligence be built / deployed?
    - When will AIs *capable of catastrophe* be built / deployed?

[12] Model-based approaches includes Ajeya Cotra's bio anchors (50% weight), Semi-informative priors (28% weight), Insights-based model (3%), Whole Brain Emulation (3%) and Phase transitions and AGI (16%)
[13] Judgment-based approaches includes AI Impacts survey (2022) (8%), Metaculus (23%), Samotsvety report (40%), Ajeya Cotra (17%) and Holden Karnofsky (12%)

We might expect some correlation between these different questions. However, it's an open question as to how much they do correlate. The weaker the correlation between Advanced AI and "AI systems we are most concerned about", the less relevant Advanced AI is to our focus.

Takeoff Speeds

Another key consideration for what a world with transformative AI will look like is the speed at AI development, or "takeoff".[14] Christiano (2018) compares the two positions of "slow" and "fast". A closely related question is about the *continuity* of AI progress. Some posit that progress will be "smooth", following relatively simple growth curves;[15] Others believe progress will be much more "lumpy", with rapid capability improvements coming unexpectedly in a short space of time.[16]

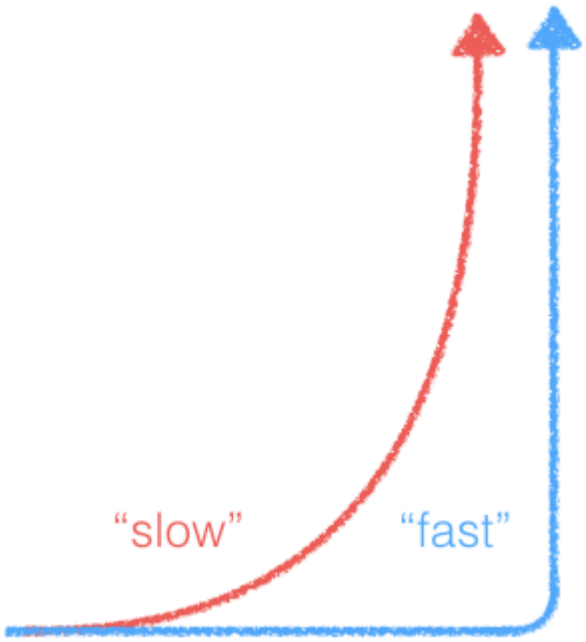


Figure 4: Illustrative plots of the capability of AI systems (y-axis) over time (x-axis) (Christiano (2018))

Observing the trendlines of AI systems today, two contradictory stories emerge:

[14] Takeoff speeds sometimes refer to how fast AI systems will develop from today's capabilities to TAI; in other instances, takeoff speed asks how rapidly AI systems move from TAI or AGI to *super*intelligence; in still other contexts, it can refer to both.
[15] For instance, this is Christiano (2018) and Davidson (2023)'s view
[16] This is closer to Yudkowsky (2013)'s view

Scaling Laws

On the one hand, we can observe surprisingly smooth relationships between the *size* of AI models (as measured by compute, dataset size or parameters) and *loss*[17] (the so-called "scaling laws" (Kaplan et al 2020)).

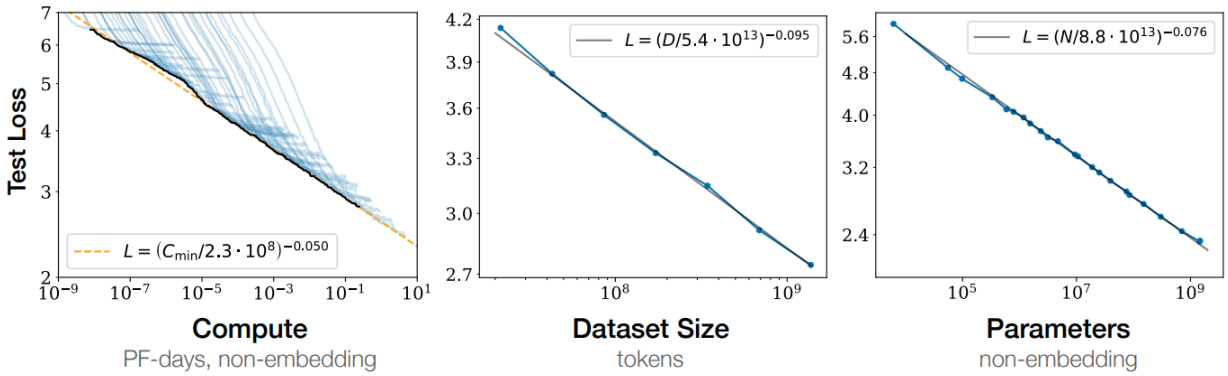
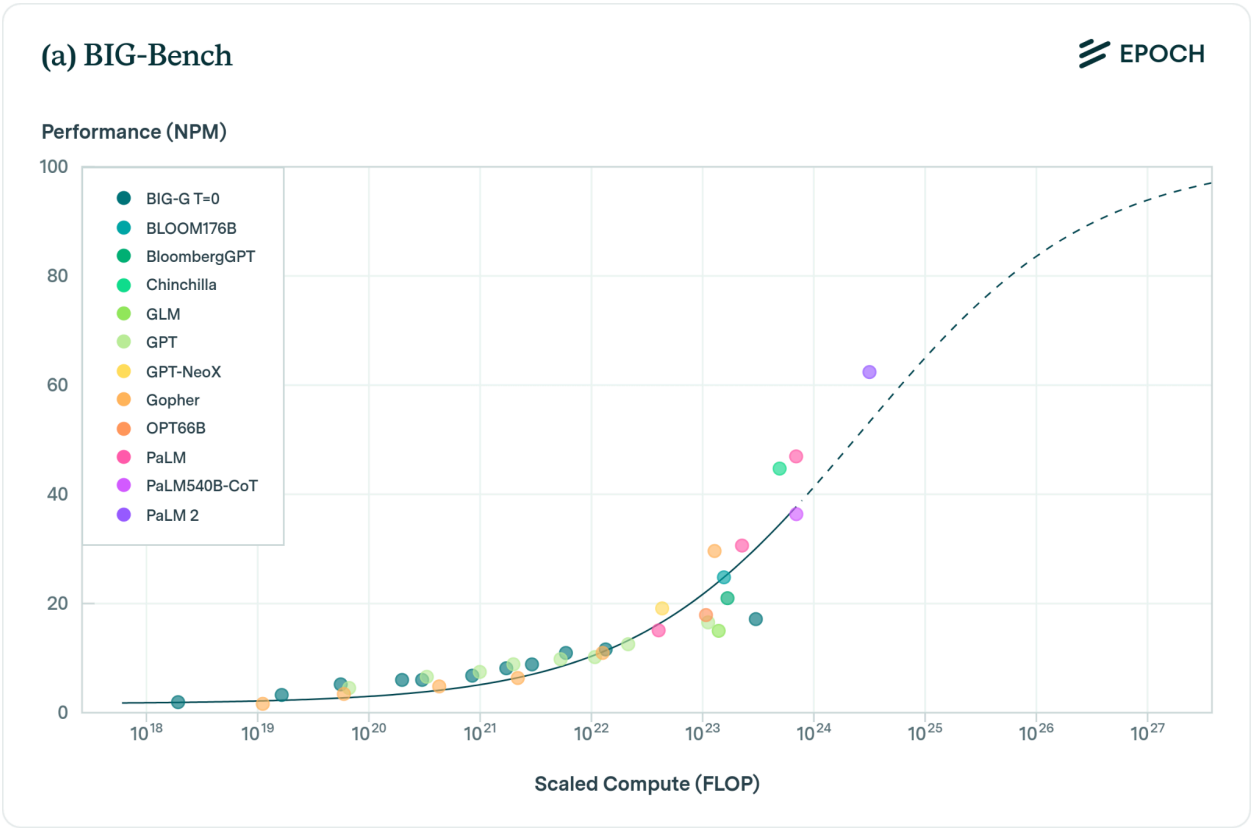


Figure 5: Language modeling performance improves smoothly when model size, dataset size, and amount of compute used for training are increased. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two (Kaplan et al 2020)

Further, preliminary efforts have been made to translate loss into performance - as measured by various benchmarks. For example, Owen (2023) plots 16 different recent ML models against their performance on common ML benchmarks. The results in Figure 6 illustrate some correlation between scale-estimated loss and performance.

[17] "Loss" being a measure for the difference between a model's prediction and reality. See Parmar (2018) for an intuitive explanation of loss



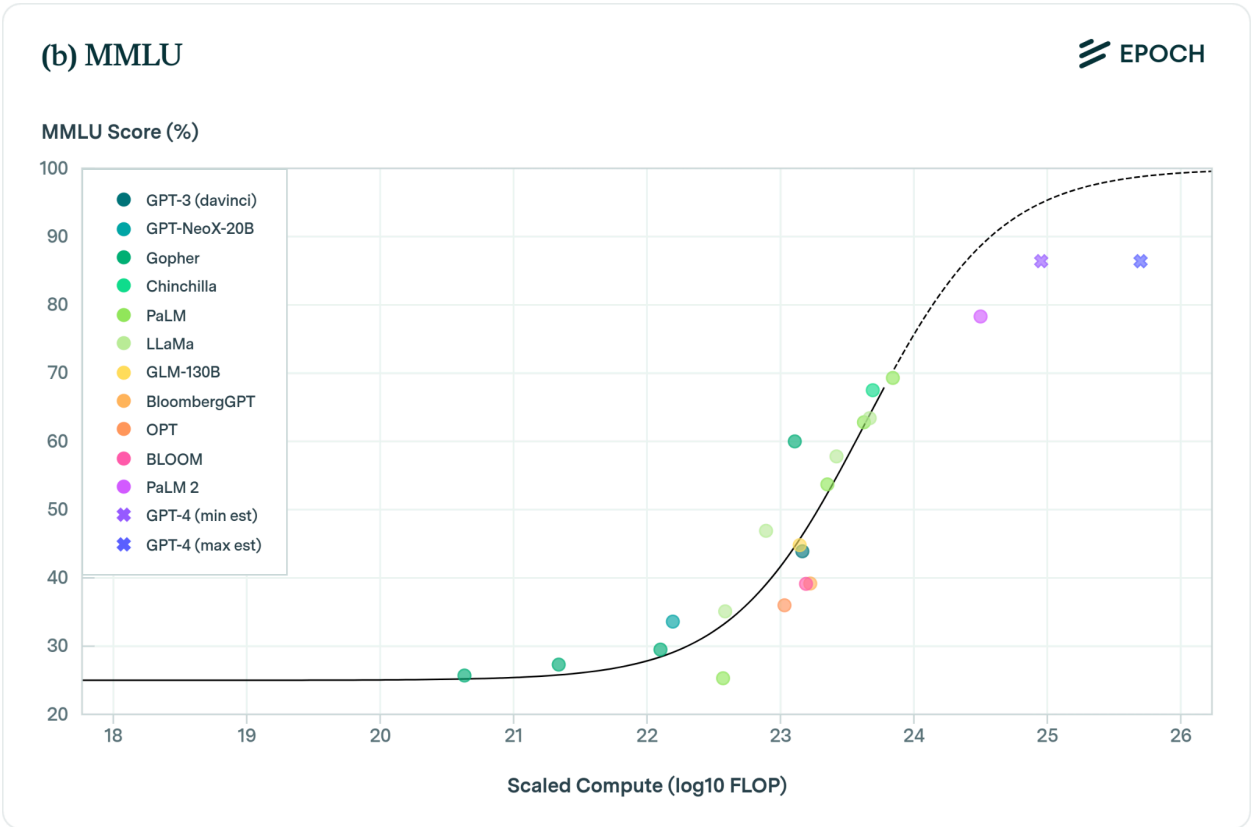


Figure 6: Benchmark performance (BIG-Bench (a) and MMLU (b)) is fairly predictable from scaling-estimated loss, i.e., scaling of model size and dataset size. Loss, on the x-axis, is

expressed as scaled compute. Sigmoid fits are shown in black for both plots, with bold lines showing data used for fitting and dashed lines showing extrapolation. GPT-4's training details were not publicly reported, so we show lower and higher estimates of its scaled compute. (Owen, 2023)

This gives us a somewhat smooth, continuous story of AI progress, which (if extrapolated) suggests we might be headed towards a continuous takeoff scenario.

Emergence

On the other hand, the smooth relationship presented above is somewhat illusory. Looking more closely at the relationship between model size and performance, we find that particular abilities only appear suddenly and unpredictably at seemingly-arbitrary compute thresholds. This is the phenomenon of *emergent capabilities*.[18]

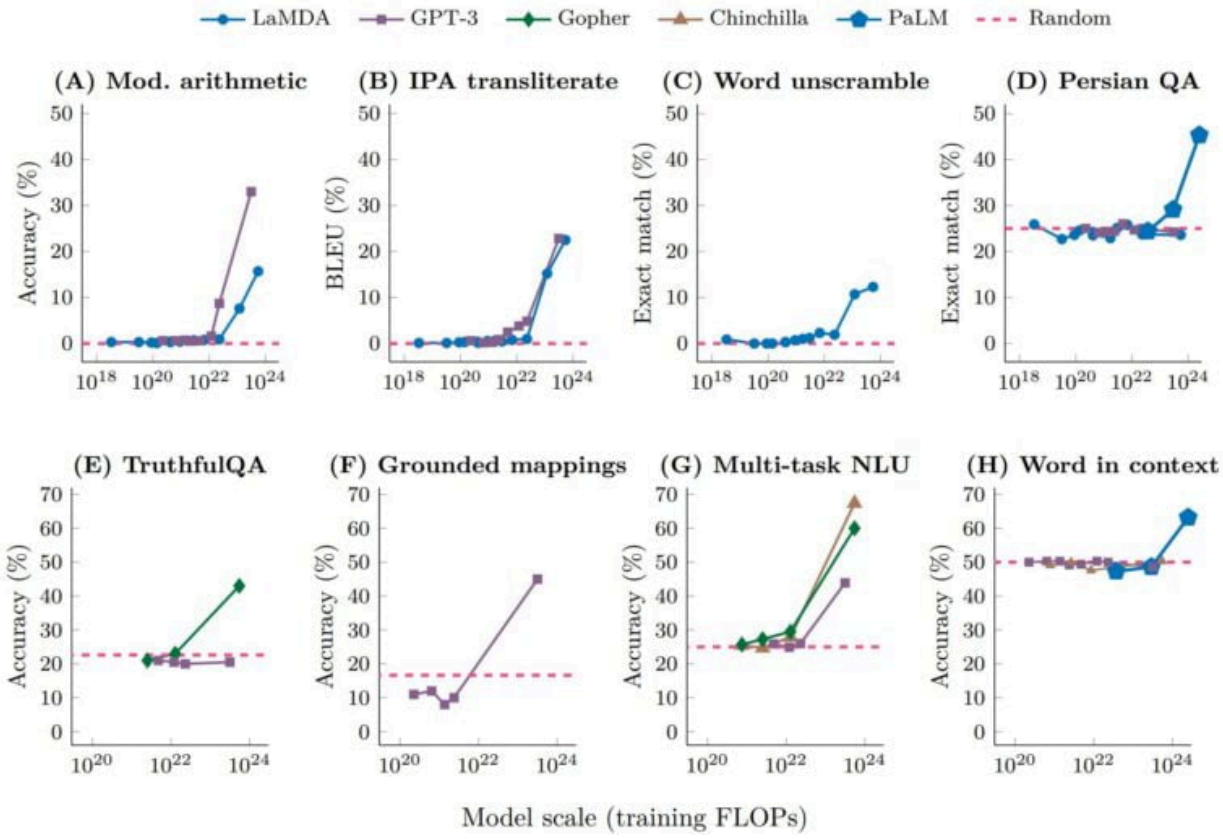


Figure 7: Eight examples of emergence in the few-shot prompting setting. Each point is a separate model. The ability to perform a task via few-shot prompting is emergent when a language model achieves random performance until a certain scale, after which performance significantly increases to well-above random. Note that models that used more training compute also typically have more parameters — hence, [the authors] show an analogous figure with number of model parameters instead of training FLOPs (Wei et al 2022)

[18] See Wei et al, 2022 for more detail on emergence inside Large Language Models (LLMs)

If the capabilities necessary to develop transformative AI systems similarly appear unpredictably through emergence, then a "fast takeoff" world seems more plausible, with only minor changes leading to sudden, immediate improvements in capabilities (and hence in the power and impact of AI systems). Such emergence weakens our ability to forecast when transformative AI systems will appear.

Polarity

A final key question is *how many* actors will develop transformative AI systems (and how many transformative AI systems will emerge)

On the one hand, Bostrom (2014)[19] argues that we might see a *unipolar* world, where a single AI develops a *decisive strategic advantage* - defined as "a level of technological and other advantages sufficient to enable it to achieve complete world domination"). If true, our world may end up forming a *singleton* ("a world order in which there is at the global level a single decision-making agency").

On the other hand, we might see a *multipolar* scenario, as outlined in Bostrom (2014).[20] In this world, there may be many powerful AI systems developed. This might avoid risks of power centralization under a singleton, but causes other challenges, such as how powerful AI systems interact together without destructive effect.

The level of polarity closely intersects with the speed of takeoff. If a fast takeoff is likely, then the first actor to deploy such a powerful AI system will be able to rapidly accelerate their advantage. By contrast, in slow-takeoff worlds, no actor is able to pull far ahead in capability at any one time, and thus it might be easier for many parties to build frontier AI systems at the same time.

Looking to the strategic picture in 2023, two observations emerge:

1. First, there is no single AI lab with a *clear* decisive lead over another. For example, whilst OpenAI developed ChatGPT, other companies have developed similarly capable models like Claude from Anthropic. So there is not clearly one AI *company* that has taken a significant lead over another

2. Second, due to the way in which AI systems are developed, the training of the first transformative AI system could immediately be followed by the ready deployment of >>millions of AIs. This is due to the nature of how we train ML systems - they require much more computing power to *train* than they do for *inference* (i.e. for usage).[21]

This dramatically changes the strategic picture. It implies that, for any capability ***c***, we may rapidly move from a world in which no AI system has capability ***c***, to one where many AIs can be used for ***c***. Combined with the phenomena of *emergence* mentioned above, we may have

[19] Chapter 5
[20] Chapter 11
[21] See Heim (2021) for more details on *why* training is more computationally complex than inference.

little idea about when capability ***c*** will emerge at all. Thus we may see new capabilities unexpectedly developing and rapidly proliferating.

Independent Impression

My own median forecast is that advanced AI systems will emerge between 2030 and 2040. Very briefly, this is because:

- It appears that the "scaling laws" of increased compute and data, combined with algorithmic insights (making up the AI "triad") appear to hold strong. That is, it seems quite plausible that compute and data, combined with only a few algorithmic improvements, will lead to AI as capable as humans (or capable of automating science and technology,[22] which *then* soon leads to human-level intelligence)
- I expect the level of computation required to reach transformative AI is significantly lower than predicted by Ajeya Cotra's Bioanchors approach.[23]
- AI systems have been rapidly advancing in the last couple of years. The capabilities of PaLM (in particular) took many by surprise;[24] whilst subsequent new releases (like ChatGPT/GPT-4) have performed at least as well as the average human or better on many benchmarks. This is much better than how systems performed a few years ago. Thus it seems (to me) intuitive that there cannot be many more years of breakthroughs left before we hit human-level.[25]
- Investment and "hype" around AI has followed in recent times. This has both continued to drive AI progress, and provided less reason to defer to outside-view skepticism on the general topic of transformative AI.

My personal view on the likelihood of a continuous or discontinuous takeoff is unclear. On the one hand, I expect that so-called scaling laws may continue to hold, giving predictable reductions in the "loss function" of ML systems. On the other hand, due to the phenomenon of emergence, I expect that dangerous capabilities may be very hard to predict in advance, and thus leave us unable to defend against a takeover-capable system. Broadly speaking, I think the latter case (i.e. a "fast" takeoff) is more plausible, with ~60% confidence.

On the issue of polarity, I expect that we may see one actor rapidly outpace all others. This is in part due to the plausibility of a "fast" takeoff.

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[22] Or "PASTA" (see Forecasting Transformative AI, Part 1: What Kind of AI?)

[23] Kokotajlo (2021) provides an intuition for why a vast increase in available computing power - a 12 order of magnitude increase on today - is highly likely (80%) to result in transformative AI systems. This is much higher than Cotra's estimate of TAI with the same level of compute (50%)

[24] For example, forecasts on Metaculus fell around the time that PaLM was released (April 2022), whilst Geoffrey Hinton pinpoints PaLM as the point where he believed systems were more capable than previously anticipated

[25] For instance, consider that GPT4 performs 90th percentile and 99th percentile in the Biology Olympiad, whereas GPT3.5 performed 10th and 31st (respectively) - with a ~3 month gap between the two. I cannot foresee how these kind of jumps can continue to happen across many tests, without hitting general human capability.

However, these are highly uncertain beliefs, which could be easily updated in light of new evidence. Thus, gaining greater strategic clarity is one promising avenue for further research. Next, we explore the potential risks we might encounter.

Section 2: Threat models

So far, we have only made the claim that advanced AI will have a transformational impact. This alone does not imply there is any inherent *risk* from transformative AI - nor that it should be prioritized above other problems in the world.[26]

To understand the potential risks from AI, this section reviews various **threat models** - roughly, models or paths of how transformative AI could lead to catastrophe. The threat models reviewed are:

1. **Power-seeking AI**: An AI system seeks power in unintended ways - because gaining power is an *instrumental goal* towards any given end goal. Acting on power-seeking intentions could lead to the disempowerment of our species - which would constitute catastrophe.

2. **AI-induced Biological Catastrophe**: Biologically-capable AI systems are used in various applications - both advancing the frontier of life sciences, and reducing barriers to entry for a wide range of people. Such AI systems could increase the risk of engineered pandemics from both accidents and misuse, potentially resulting in a global catastrophe or worse.

3. **Military AI Risks**: AI systems may be used in high stakes military decisions, including nuclear weapons. Such usage could directly increase the chance of accidental nuclear use, as well as decreasing global stability more broadly.

4. **Democratic and Distributional Risks**: AI systems may cause immense concentrations of wealth and power in the hands of a very small minority. This could lead to various catastrophic consequences, including

    a. The empowerment of malevolent individuals;

    b. Civilizational disintegration and collapse;

    c. A flawed lock-in of values which does not align with the best interests of all individuals.

5. **Epistemic Insecurity**: AI systems may lead to vast increases in misinformation, persuasion and manipulation, which could undermine the access, flow and possession

[26] Although even if transformative AI is not *risky*, it still might ought to be prioritized as an opportunity to cause a positive trajectory change to the long-term future. Analogously, the formation of the United States may have been a highly influential moment, even if not one characterized by risks

of true information. In the limit, this could act as a risk factor by reducing society's overarching resilience to other catastrophic risks.

6. **Risks from Digital Minds**. Finally, the development of powerful AI systems may coincide with the emergence of new, morally-relevant agents. Without an appropriate understanding of such beings, we may inadvertently cause a significant catastrophe for such minds by failing to treat them appropriately.

The risks covered here include *accident*, *misuse*, and *structural*[27] *risks*.[28]

Power seeking AI

Introduction

The prospect of a "Power seeking" AI system is one of the most discussed pathways to an existential catastrophe. Carlsmith (2021) is a canonical report on this risk pathway. In this section, we give a high-level overview of the key claims in the report, evaluating them based on theoretical arguments and available empirical evidence.

Carlsmith breaks down his threat model into six conjunctive steps, giving probabilities for each step occurring by 2070:

1. It will become **possible** and **financially feasible** to build **capable**, **planning**, **strategically aware** AI systems (more)
2. There will be **strong incentives** to deploy such AI systems (more)
3. It will be much harder to build **aligned** AI systems than misaligned, power-seeking AI systems. (more)
4. Misaligned, power-seeking AI systems will be **deployed** (more)
5. Deployed power-seeking systems will **permanently disempower humanity** (more)
6. Permanently disempowering humanity would be an **existential catastrophe** (more)

We review each of these claims in the subsequent sections

[27] Zwetlsloot and Dafoe (2019) introduce the "structural risk" perspective, to give insight on how catastrophe may not occur narrowly through specific accidents or misuse, but rather due to broader societal failures. Clarke (2022) provides more definitional clarification

[28] See Dafoe 2020; Clarke 2022 for more details on this taxonomy. However, throughout this report, I avoid splitting risks distinctly into these three buckets, since there is often overlap (e.g. power-seeking AI could come about via accident or misuse, and structural risks may also be at play). Instead, I break the problem down by focusing on object-level pathways to catastrophe.

Claim 1: It will be possible and financially feasible to build Advanced, Planning, Strategically aware (APS) AI systems (by 2070)

This claim looks at how likely we are to develop powerful AI systems by 2070. Under Carlmsith's definition, powerful AI systems have **Advanced Capabilities, Agentic Planning, and Strategic Awareness (APS)**

Advanced Capabilities

Carlsmith defines AI systems with **advanced capabilities** as AIs that can outperform the best humans on some set of tasks, which - when performed - grant the AI system significant power in today's world. For example:

- Outperforming the best humans in the stock market would allow an AI to amass significant wealth
- Outperforming the best humans in significant parts of the economy would allow human labor to be substituted for AI labor, in highly economically valuable sectors.
- Outperforming the best humans in persuasion could allow AIs to gain political power (directly or indirectly).

Existing AI systems like GPT-4 are already capable of human (or above) levels of ability across a broad range of domains.[29] This is currently translating into some economic value - for instance, automating work conducted by Mechanical Turk (Harrington 2023).

It remains to be seen whether future systems (like "GPT-N")[30] will be able to surpass Carlsmith's definition of "advanced capabilities". *Conditional on scaling laws continuing to hold*, I expect (with >50% probability) for such systems to be capable of automating at least 20% of cognitive tasks by the 2030s. Using a compute-based framework, Davidson (2023)'s model predicts it will take a median of 3 years to jump from "AI systems automating 20% of cognitive labor" to "AI systems automating 100% of cognitive labor". Combining these two, my overall guess is we will surpass the bar of "advanced capable systems" by 2040 with >50% probability.

Agentic Planning

Carlsmith defines **Agentic Planning** as an AI system which is able to make and execute plans, in pursuit of objectives, on the basis of models of the world. Carlsmith gives examples such as:[31]

- Planning and then taking a trip from New York to San Francisco.

[29] For instance, 99th percentile in Biology Olympiad, 90th percentile in Bar exam (GPT-4). Although note that GPT-4 is not yet an "autonomous" worker conducting these tasks - merely an assistant to boost the productivity of people

[30] I'm not sure what level "N" needs to be. If GPT+1 systems are deployed every ~2-3 years, we will be at e.g. GPT-7 by 2030.

[31] In the paper, Carlsmith gives further clarity on what such agentic planning looks like

- Reasoning about the safest way to cut down a tree, then doing it.
- Designing the components of a particle collider.

Under a weak definition of agentic planning ("systems that can make and execute plans, in pursuit of objectives, on the basis of models of the world"), we are relatively close to seeing this bar passed in the next 12-24 months.[32] Weak forms of agentic planning can easily be achieved via large language models - one can simply ask a GPT-type system to reason about how to achieve a goal. The system can then break this into smaller subgoals, then execute on those subgoals.[33]

For example, taking the example "*Planning and then taking a trip from New York to San Francisco*": I expect that within the next 3 years, a personal "AI assistant" could, based on a single prompt:

1. Search the internet for flights
2. Fill in a form to book a flight
3. Print out tickets (if hooked up to a printer)
4. Send details to the user's calendar
5. Plan their route to the airport
6. Provide clear instructions to the human for how to navigate the airport

Whilst systems like this aren't yet commercially viable, I would guess that they will be very soon (less than 3 years), given strong financial incentives to do so.

On the other hand, this is just one example of agentic planning, and perhaps an unusually easy one. My guess is that *autonomous, general* agentic AIs - planning and executing a wide range of goals without human instruction - are currently limited by capabilities. Still, given current trends and growing commercial investment, I expect "agency" will not pose a fundamental barrier in the deployment of AI systems once the baseline capability is established.

More generally, I expect general capable *agentic* AI systems to be viable soon after the equivalent generally capable *tool* (i.e. non-agentic) AI systems are available.[34] Thus, I believe that, conditional on systems with advanced capabilities being possible by 2070, an equivalent agentic system will be possible almost immediately afterwards.

[32] Plausibly it has already been passed with developments like "AutoGPT"
[33] See e.g. the launch of "AutoGPT" or GPT-4 with plugins, which gives a very early sign of the direction that more advanced systems will progress along.
[34] See Karnofsky (2012) for discussion about "tool" and "agent" AIs

Strategic Awareness

Strategic awareness is approximately defined as when AI systems have a model of the world, and are aware of their impact upon it. For example, a strategically-aware system could answer questions like "what would happen if I had access to more computing power".

Compared to advanced capabilities or agentic planning, strategic awareness is much harder to define, and hence harder to forecast. Berglund et al, (2023) offers one approach to defining and measuring situational awareness, based on evaluating existing GPT-3-level systems. The authors conclude that existing models already have some weak situational awareness.[35]

Ideally, one could use evaluations to measure situational/strategic awareness at different model sizes, extrapolating to the point where "full" strategic awareness is reached. Then, one could plot this to existing scaling law trends to estimate when the first strategically aware systems will emerge. Figure 8 gives us an idea of *how* we might model this. But crucially, we do not know the model size necessary (as measured in Training compute) to reach strategic awareness.

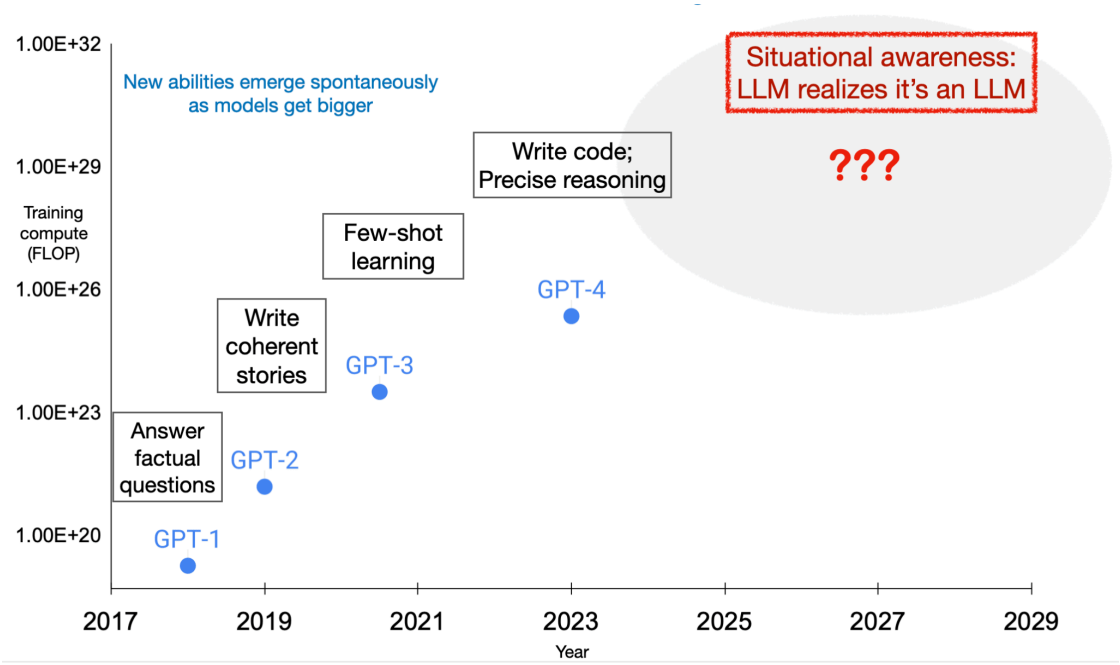


Figure 8: New abilities emerge as LLMs are scaled up. It is unknown when situational awareness will emerge in base models (Owain et al 2023)

My best guess is that - *conditional on highly capable agentic systems having already emerged* - strategic awareness (as defined by Carlsmith) will very likely emerge by 2070. Thus I do not think that strategic awareness requires additional algorithmic insight, distinct from agency or advanced capability. Instead, I expect that strategic awareness will emerge as AI systems generally improve as per the *scaling laws*, However, it should be re-emphasised that precisely forecasting strategic or situational awareness is highly challenging at the present time.

[35] Note that situational and strategic awareness refer to different (although related) concepts

Relationship between APS AI systems and TAI systems

In the previous section we explained why we might expect transformative AI systems to most likely appear in the 2030s. There are some subtle but important distinctions in the timeline between *transformative* AI and *advanced, planning, strategically aware* systems.[36] When just forecasting APS systems, my impression is that they are also highly likely to appear around the 2030s.

Summary

Carlsmith's probability of APS systems being developed by 2070 is about 65%, based mostly on deference to other sources like Cotra (2020), Roodman (2020), and Davidson (2021).

Given the timelines outlined previously, my independent impression is that there is an ~80% chance we will develop APS systems by 2070.[37] Furthermore, I'd put a 50% chance of getting such systems by ~2035.

Claim 2: There will be strong incentives to build APS AI systems

This section asks whether actors have (strong) **incentives** to actually *develop* APS systems. Carlsmith speculates that incentives might lead to economically valuable AI systems, but without agency or strategic awareness. Nonetheless, he still thinks there will be incentives to build APS systems because they will be:

1. **Useful** (because agentic planning / strategic awareness seems especially helpful in complex environments)
2. **Efficient to develop** (potentially, the fastest way to develop powerful AI systems with commercial applications is through APS systems)
3. **Difficult to prevent** (due to some natural convergence towards APS behavior, even if systems are not explicitly trained as such)

Carlsmith emphasizes **usefulness** as the strongest incentive for building APS systems

In recent years we have seen strong evidence of the incentive to build capable systems, with some evidence for building agentic systems. For example, ChatGPT has been widely adopted,[38] whilst there is significant interest in building agentic systems like an "AutoGPT" (Gray (2023)).

[36] For instance, perhaps it is possible to develop systems with high capabilities (advanced), with a long time horizon based on detailed world models (planning), without being strategically aware. Or perhaps just advanced models without agency or awareness (i.e. powerful Tool AIs). On the flip side, it's also possible we develop highly capable, agentic and aware systems, but they fail to have a real-world impact (e.g. not significantly increasing GDP). My own view is that the correlation between APS systems and TAI systems is close enough to make TAI forecasts informative for APS forecasts.
[37] This high probability is because I believe it is highly likely (~90%) that we see transformative AI by 2060, and if so, I think APS AI systems are the most likely pathway to this
[38] 1 million people used ChatGPT in the first five days - Duarte (2023)

Future capable and agentic systems may also be strategically aware (by accident or design), and such awareness may also be economically valuable.[39]

With strong commercial incentives to build such systems, and the observation that similar weaker systems are already being developed, I would guess there is an ~80 - 90% chance that people will be strongly motivated to build APS systems. I think the main counterargument against a higher probability is that it may be more commercially viable to build advanced AIs specialized in distinct tasks, rather than general systems that more closely resemble APS. In such a scenario, I imagine that even if APS systems are a) possible and b) financially viable, they may simply not be built, since the same benefits could be gained via comprehensive AI services instead (see Drexler (2019) for more depth on this argument)

Claim 3: It will be much harder to build aligned APS systems than misaligned APS systems

Assuming a) it is possible to build APS systems in the next few decades, and b) there are incentives to create such systems, Carlsmith explores the prospect that APS systems will not be **aligned**.[40] Specifically, the report focuses on systems misaligned such that they *seek power*, in a way that designers did not intend.

Of all premises outlined in Carlsmith's report, alignment difficulty is the hardest to confidently evaluate. Below I describe and then evaluate a number of claims relevant to the difficulty of alignment.[41] Three key problems which I cover in most depth are:

1. Outer Alignment
2. Inner Alignment
3. Instrumental Convergence

These three problems are illustrated in Figure 9.[42]

[39] For example, AI systems with high contextual knowledge may be highly useful for deployment in different domains.
[40] Carlsmith focuses specifically on "making sure AI systems behave as their designers intend" (rather than e.g. "make AI go well"). This definition comes from Christiano (2018). Carlsmith defines "misaligned behavior" as "unintended behavior due to problems with the AI systems objectives". Meanwhile, "fully aligned" means the AI system does not pursue any misaligned behaviors, as a result of the inputs it receives (e.g. text, images, etc). Carlsmith proceeds to give more nuanced definitions of these terms, which I don't comprehensively cover here
[41] Note, I structure this section somewhat differently to Carlsmith, based on what I see as the key difficulties in alignment
[42]

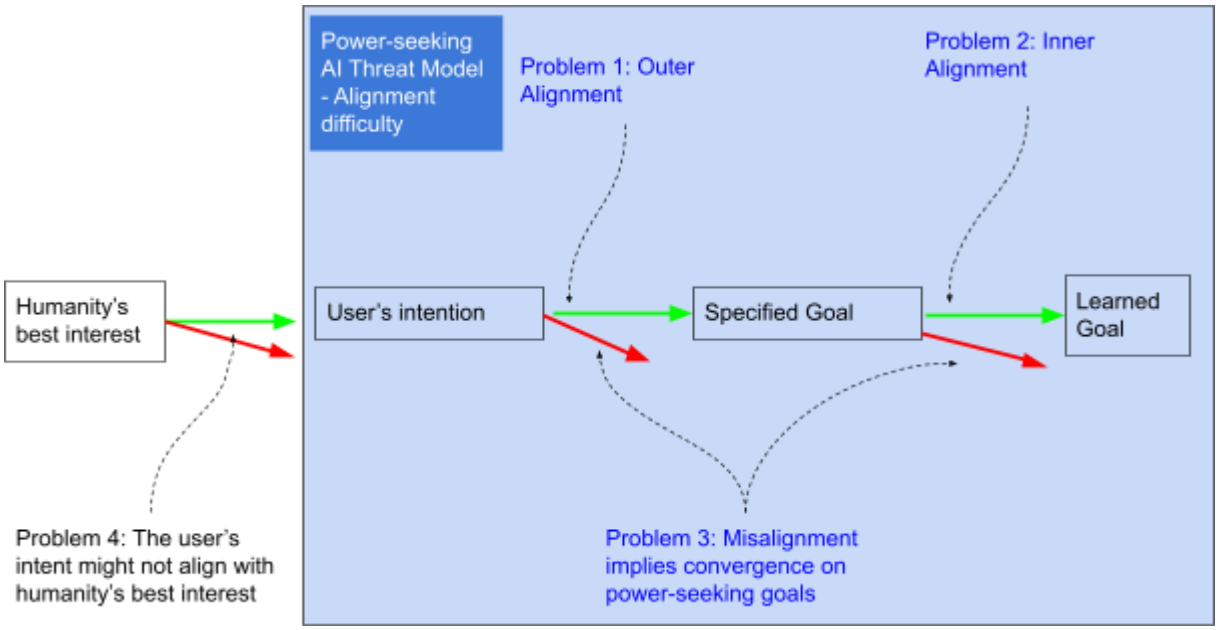


Figure 9: Illustration of four key challenges within AI alignment:

1. ***Outer alignment:*** *How to ensure the user correctly specifies their intentions into an AI system's training signal*
2. ***Inner Alignment****: How to ensure the goal specified is actually robustly learnt by the AI system*
3. ***Instrumental convergence****: How to ensure that an AI system which is misaligned (either in the outer or inner sense) does not converge on the goal of seeking power from humans*
4. ***Value alignment****: How to ensure that the user's intentions align with the broader best interests of humanity*[43]

Red arrows illustrate cases where misalignment occurs

Challenges of alignment

Carlsmith describes reasons why it might be difficult to align the goals of APS AI systems. He covers three problems with controlling AI systems - controlling *objectives,*[44] controlling

[43] The fourth problem - ensuring that the user's intentions align with *humanity's best interests* - are covered separately from this threat model. Whilst the first three challenges are primarily technical in nature, the fourth is fundamentally a socio-political problem. The threat model on distributional and democratic risks touches on how we ensure alignment to *humanity's best interests*.

[44] This further breaks down into *proxy* problems, *search* problems, and *myopia*. I cover the first two more extensively below. On myopia: AIs with myopic (short-term) objectives might have less incentive for power-seeking behavior than those with non-myopic (long-term) objectives. We will likely want agents able to do planning over a long time horizon. This will make them more useful if aligned, but also more able to execute dangerous long-term plans for takeover.

capabilities[45] and controlling *circumstances*.[46] I focus on controlling *objectives*, breaking the problem down into *outer* alignment and *inner* alignment.

Outer Alignment

The first challenge of alignment is outer alignment - the problem of defining an objective for an AI system that leads to the user's intended behavior, rather than causing unexpected or unintended consequences (Hubinger et al (2019)).[47]

One reason outer alignment is challenging is due to the phenomena of specification gaming (DeepMind, 2020) - when a specified goal is "gamed" by an AI system to naively get more reward, without executing the intended objective. The story of King Midas illustrates the problem:

> *The King wishes for everything he touches to turn to gold, but when interpreted literally, this causes all food, water, and even his own daughter turning to gold. This is not what the King truly intended, but follows naturally from his wish as literally stated.*

We see this problem appear in contemporary society, such as in schools which "teach to the test" rather than fulfilling the (intended) objective of education; or police optimizing for reducing the number of *reported* crimes, with the adverse effect of incentivizing police to make it hard to report crime at all. This phenomenon is commonly referred to as Goodhart's Law.

A common response is that we can simply specify enough conditions into the objective function, such that when that function is maximized, it gives us what we "truly" want. However, coherently and correctly specifying exact human values with enough detail is an extremely challenging task - potentially impossible (at least for humans, at current capability). Furthermore, depending on the nature of APS-level AI development, we may only get a singular chance to correctly specify the objective, or else suffer catastrophe (due to instrumental convergence - outlined below).

Some further challenges for outer alignment - challenges which take on greater importance as we get closer to more advanced systems - include:

- Ensuring systems remain aligned with our intentions when they become (far) more intelligent than humans are able to understand.[48]

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[45] Less capable systems are probably easier to control than more capable systems. Therefore alignment might be easier if we control the capabilities of systems. For example, we might be able to use specialization (i.e. only building narrow APS systems). Or we could try to prevent problematic improvements at all. Finally, we can look for alignment techniques that scale with greater capabilities.
[46] We might be able to make sure APS systems aren't not put in places where power seeking is possible. For instance, "air gapping" the computer, or making it harder for humans to fall prey to persuasive AIs.
[47] "*Outer alignment refers to aligning the specified loss function* [objective] *with the* [user's] *intended goal*" (Page 7, Hubinger et al (2019))
[48] Iterated Amplification is one avenue of research in this direction.

- Ensuring that systems can be aligned at scale using methods beyond RLHF.[49]

How likely is outer misalignment?

We currently cannot observe an outer alignment failure from AI in a high-stakes domain. Nevertheless, we have several lines of arguments suggesting that this may occur, absent intervention:

1. We have empirical evidence from less powerful AI systems that specification gaming - one type of outer alignment failure - is a relatively common problem, with over 70 published examples in this table.
2. We have analogies from non-AI domains which demonstrate the challenge of correctly specifying our intentions, through examples of Goodhart's Law in society.
3. We have thought experiments (such as the "paperclip maximizer") which give a conceptual argument that getting intelligent agents to do what we "truly" want them to do on the first try will be hard.

My brief conclusion is that without *any* attempt to mitigate, outer alignment failures will be common and likely in advanced systems. I conclude this based on the fact that we already see low-level outer alignment failures in existing systems, and that we can see similar problems in other domains. However, I also expect that by default, it will be necessary to outer align systems somewhat in order to ensure their commercial usefulness. Thus, even if outer misalignment is a common problem, it is not necessarily the most difficult problem to solve (relative to other challenges).

Inner Alignment

Even if we achieve "outer alignment",[50] it is not necessarily the case that the AI system is *actually* optimizing for what the user intends. This gives rise to a second problem, one of *inner misalignment*. This is where the *specified goal*[51] differs from the *learned goal*.[52] In current ML systems, a closely related observable problem is *goal misgeneralization* (DeepMind 2022) - when the *specified* goal and *learned* goal are aligned when the AI system is trained, but come apart ("misgeneralizes") outside of training.

To give an intuition for what inner misalignment looks like, consider the case of evolution.

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[49] Most frontier systems rely on some form of Reinforcement Learning from Human Feedback (RLHF). This helps to direct current systems towards human preferences, but does not scale well. Scalable oversight (OpenAI 2023) aims to tackle this.
[50] i.e. we find ways to correctly specify a user's goals into an AI system, in a way that scales up to greater-than human intelligence
[51] I.e. the goal we gave to the AI system. Also known as the base objective (From Glossary, Hubinger et al 2019)
[52] I.e. the goal that the AI system actually develops. Also known as the mesa-objective (From Glossary, Hubinger et al 2019)

Evolution can be thought of as an optimization process, in which organisms are selected according to their genetic fitness. Thus the "goal" of evolution is to maximize genetic fitness.[53] This is a fairly myopic goal, and most organisms (e.g. bacteria, plants) do not develop goals beyond this - instead they abide simple heuristics like "follow sunlight".

However, other species (like humans) have developed these heuristics into more complex goals, such as food, love, or money. These goals are quite distinct from the original goal of "maximize genetic fitness", as they emerge outside of the standard evolutionary environment. Indeed, humans have now strayed far from evolution's original goal, for instance by inventing contraception.

Thus we can say that, from the perspective of evolution, there is a *misalignment* between the *specified goal* for humans, and the goal *learned* by humans.[54]

How likely is inner misalignment?

Compared to outer misalignment, evidence for inner misalignment in future systems is less clear. However, there are a couple of lines of evidence that inner misalignment may be a challenge in future AI systems.

1. First, we have the case of human evolution described above, which gives us a non-AI analogy for how inner misalignment can arise in intelligent beings.

2. Second, we have *tentative* empirical evidence from existing AI systems - in the form of *goal misgeneralization*. Langosco et al (2021) provides one well-cited example, and DeepMind (2022) has shown similar failures occurring across different environments. They also compile a list of 14 examples of goal misgeneralization

Note that our current examples of goal misgeneralization are all in demonstrations, rather than systems deployed in the world. Thus the faileure modes seen above are likely to differ from real-world systems, especially as systems develop more advanced capabilities and strategic awareness. Furthermore, evidence of goal misgeneralization does not necessarily imply inner misalignment, and the extent to which goal misgeneralization is a useful indication of potential inner misalignment failures is debatable.[55]

Additional challenges

If inner misalignment failures were as innocuous as the above examples, then we would have little reason to be concerned. Perhaps problems with goal misgeneralization (as described

[53] Note this is a vast simplification of the evolutionary process, and likely does not stand up to scrutiny. This example only offers an intuition pump, rather than a watertight "proof".
[54] Importantly, this misalignment is different to the case of outer misalignment. There is no mismatch between the "intended" goal and the "specified" goal of maximizing genetic fitness. Rather, the mismatch is between the specified goal and the learned goal.
[55] See e.g. Turner (2023) on whether the goal misgeneralization case from Langosco et al (2021) gives evidence of inner misalignment

above) will be naturally solved. After all, preventing this kind of failure helps to build more capable systems. For less capable AIs, after observing failures in the real world, we can simply make adjustments to the specified goal, and then redeploy the system.

However, there are two problems which make inner misalignment a *significantly* worse problem as models get more powerful- **deceptiveness** and **opaqueness.**

Deceptiveness

Previously, we specified that an APS AI system is *strategically aware*. A *strategically aware* system will thus be aware when it is undergoing training. It may notice that the goal it was originally being trained for (the *specified* goal) differs from the goal it *learned* during the first steps of training.

Crucially, once the "situational awareness" threshold is passed, a system will be aware that it is undergoing a training run, and thus is able to reason that it can act *as if* in pursuit of X (the *specified* goal) in order to later aim for Y (the *learned* goal). Thus a system may act deceptively aligned, to varying extents.[56] Deceptive behaviors may make it progressively harder to tell if the system is truly optimizing for the goal we initially specified.

Opaqueness

Additionally, we currently have a very poor understanding of how AI systems work on a deeper level. Such systems are "black boxes" - we have inputs and outputs, but don't know what happens in between. Therefore, it will be challenging to detect the difference between an actually aligned system and a deceptively aligned system. As model size grows, we may find it increasingly difficult to make progress on this problem. Nevertheless, work on interpretability (Rudner and Toner 2021)[57] aims to help address this problem.

My overall view is that if AI systems are able to engage in deceptive behavior, and if we are unable to interpret large AI systems, then inner misalignment is a *significantly* harder problem to solve than outer alignment. However, there remain key uncertainties over whether deceptiveness will emerge in models, and how hard it will be to detect and eliminate.

Instrumental convergence

The final key hypothesis in Carlsmith's argument is *instrumental convergence*: If an APS system is less-than-fully aligned, it will (by default) exhibit "power-seeking" behavior.

We first need to differentiate between *instrumental* and *terminal* goals. *Terminal*[58] goals are goals we want as an end in themselves - for instance, perhaps a goal to live a long, happy,

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[56] Karnofsky (2023) delineates three different levels of deceptiveness which take progressively concerning forms:
1. Playing the "Training Game" - as described by Cotra (2022)
2. Deceptive Alignment - as described by Hubinger et al (p23, 2019)
3. Gradient Hacking - as described by Hubinger (2019)

[57] Also referred to as explainability

[58] Also known as intrinsic goals, or end goals

healthy life. This contrasts with *instrumental* goals, goals which help us on the way to achieving our terminal goal - for instance, earning money.

One claim made is that there is a natural *convergence* towards a set of instrumental goals - that is, irrespective of the *terminal* goal, there are some instrumental goals common to (almost) any given end goal. Some instrumentally convergent goals, which an advanced AI system may possess, includes:

1. **Self-preservation** - ensuring that the system continues to exist (or another system with an identical terminal goal)
2. **Goal content integrity** - ensuring that the system's terminal (end) goal cannot be modified.
3. **Improved cognitive capabilities** - increasing the cognitive abilities of the system, in order to better reason about how to achieve terminal goals.
4. **Technological development** - developing more powerful technologies, so that the system can better execute on its goals
5. **Resource acquisition** - gathering more resources (e.g. money, materials) which may help in pursuit of both instrumental and terminal goals

Carlsmith uses the catch-all term of "power-seeking" to describe these behaviors.

Preventing instrumental convergence may be significantly challenging given the general nature of instrumental goals. For instance, the combination of self preservation and goal integrity implies that a system will try (very) hard to prevent us shutting down the system or changing its goal, once that goal has been given.

This means we may only get one attempt at aligning systems. The paperclip example illustrates why:[59]

Suppose we want an AI system to make as many paper clips as possible. Interpreting this naively, the system starts converting all factories into paperclip factories, causing significant disruption. Upon noticing this problem, it is likely that humans would attempt to shut down the system, and/or to change the system's objective. Shutdown and a change of goals would prevent the AI system from making paper clips ever again.

Thus, an instrumental goal for maximizing paperclip production is self preservation and goal integrity. To truly prevent a shutdown, a move for the paperclip maximiser would be to cause human extinction. This is not due to any "dislike" for humanity, but merely because doing so is efficient for achieving the original terminal goal.

It is not merely enough that we specify an "aligned" objective for a powerful AI system, nor just that objective be internalized by the AI system, but that we do both of these *on the first try*.

[59] The paperclip maximizer example was made famous by Bostrom (2003)

Otherwise, an AI engaging in misaligned behaviors would be shut down by humans. So to get ahead, the AI would first try to shut down humans.

How likely is it that misaligned systems will converge on power-seeking instrumental goals?

Bostrom (2014) gives the canonical philosophical argument for why, for *any* sufficiently intelligent system, we should expect the system to converge on the instrumental goals outlined above. This argument is given above.

One counter is that it's not clear whether intelligent beings - be they humans, AI systems, or other entities - have coherent "goals" at all.[60] Nor is it clear whether having "goals" necessarily implies converging on power-seeking subgoals.

For example, humans don't seem to have clearly defined terminal goals that can be described as "utility functions". Rather than coherently maximizing a particular utility function, we exhibit an array of cognitive biases and irrationalities. Furthermore, even when we do aim to achieve certain goals, we don't seem to do so by converging on power-seeking goals.[61] Similarly, it might be argued that AI systems like language models do not, and will not, rationally pursue a crisply-defined utility function,[62] nor will they converge on power-seeking to achieve it.

In response, we can observe that existing ML systems (such as language models) do in some sense seek power, even if not in the same sense as originally conceived in *Superintelligence*. For example, Perez et al (2022) find evidence of several low-level power-seeking goals (e.g. resource acquisition, enhanced capabilities) in existing language models, with power-seeking desires increasing as model size increases. Furthermore, if systems are build using RL techniques, we may posit that they will more closely resemble goal-directed systems, where instrumental convergence pressures may apply. Future systems of this form may therefore seek power in pursuit of their goal - whether that aligns with our intended goal or not.

Summary of Alignment Difficulty

In the below table, I give a brief summary over what each of these perspectives suggests about several key alignment difficulties. I break down my analysis into two perspectives - theoretical and empirical. These approximately correspond to Steinhardt (2022)'s dichotomy between a "Philosophy" and an "Engineering" approach.[63]

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[60] In the sense that "having coherent goals" approximates something like a consistent utility function. See Arbital 2017 or Coherence Arguments for discussion on why more intelligent agents would cohere around expected utility maximizing behavior

[61] See Grace (2022) for more discussion of this point

[62] Work on shard theory (Pope and Turner, 2022) explores this possibility in more depth

[63] *"The Engineering approach tends to be empirically-driven, drawing experience from existing or past ML systems and looking at issues that either: (1) are already major problems, or (2) are minor problems, but can be expected to get worse in the future. Engineering tends to be bottom-up and tends to be both in touch with and anchored on current state-of-the-art systems.*

"The Philosophy approach tends to think more about the limit of very advanced systems. It is willing to entertain thought experiments that would be implausible with current state-of-the-art systems (such as Nick Bostrom's paperclip maximizer) and is open to considering abstractions without knowing many details.

1. **Theoretical / Philosophical perspective:** Conceptual reasoning via thought experiments, non-AI analogies, theorems and proofs.
2. **Empirical / Engineering perspective:** Evidence based on the behavior of existing ML systems.

Failure mode	**Likelihood of failure mode**	**Difficulty of preventing failure mode**
Outer Misalignment	**Thought Experiments**: The stories of King Midas and the Paperclip maximiser illustrate that clearly specifying an intended goal is challenging. **Analogy**: In human society we regularly see examples of Goodhart's Law, where society optimizes on proxies for the intended behavior rather than intended behavior itself. **Evidence from existing systems**: DeepMind (2020) provides a list of over 70 examples of Specification Gaming. **Overall opinion**: Outer misalignment failures are **likely** absent intervention	**Evidence from existing systems**: Reinforcement Learning from Human Feedback (RLHF) (Christiano et al 2017) appears to get current systems (like GPT-3.5 and GPT-4) to pursue behavior in line with the user's intent. However, this alone will not solve the problem for future AI systems. **Alignment Theory**: Several proposals for aligning smarter-than-human AIs to humans - such as Iterated Distillation, Amplification and Debate - have been proposed. However, these methods have not (yet) been shown applicable for AI systems. **Overall opinion**: I think it is **fairly easy** to make progress on outer alignment for current AI systems, and **moderately difficult** to extend this to AI systems in new environments where human oversight is not feasible
Instrumental Convergence (on Power-seeking)	**Philosophical reasoning**: Intelligent agents with almost any goal will converge on a set or power-seeking	**Alignment Theory**: Perhaps we will be able to build and train agents without explicit utility functions, and hence do not converge on power-seeking goals. Work in Shard

It often sounds more "sci-fi like" and more like philosophy than like computer science. It draws some inspiration from current ML systems, but often only in broad strokes."

	goals (Bostrom 2014) **Theorem**: Turner et al (2019) show that in AIs trained by Reinforcement Learning, it is optimal for agents to seek power, in order to maximize their objective **Evidence from existing systems**: Perez et al (2022) find evidence of several power-seeking goals (e.g. resource acquisition, enhanced capabilities) in existing language models, with increased power-seeking desires as model size increases **Analogy**: Humans do **not** appear to be power- seeking agents. Instead, we are often irrational agents, rather than pursuing clearly defined utility functions. **Overall opinion:** I think it is **highly likely** that AI systems will converge on power-seeking goals if their terminal goals differ from those intended by their user	Theory may help us reason about the way that goals are formed, which could side-step the problem of instrumental converge **Overall opinion:** Absent a new paradigm for thinking about goal-directed behavior, I think it will be **very difficult** to prevent convergence on power-seeking
Inner misalignment; Goal Misgeneralization	**Analogy**: Evolution offers one example of inner misalignment - where a simple goal (genetic fitness) gave rise to a different goal (human flourishing) **Evidence from existing systems**: We have several cases of goal	**Overall opinion:** *Conditional on preventing deception*, I think it will be **moderately easy** to detect inner misalignment failures and prevent goal misgeneralization.

	misgeneralization (Langosco 2022) in current ML systems **Overall opinion:** I think there is **moderate evidence** that AI systems will learn goals different to their specified goal.	
Deceptiveness	**Alignment Theory**: Based on current methods of training AI systems,[64] it is likely that strategically aware systems will be incentivized to "play the training game" (Cotra 2022), engaging in deceptive behavior in order to pursue a misaligned objective. **Evidence from existing systems**: There have been several cases of AI systems engaging in low-level deception (ARC Evals 2023; Park et al 2023) in pursuit of a reward. However, these are not cases of a *strategically aware* system engaging in deceptive alignment **Overall opinion:** I think there is **relatively weak evidence** to expect AI systems will pursue deceptive strategies once situationally aware.	**Empirical Alignment research**: Research on Interpretability (Olah 2022) can help to reduce the "black box" nature of AI systems, making it easier to detect deception. However, the level of interpretability necessary to detect deception is currently far off, and may be unable to detect the deepest forms of detection **Overall opinion:** I think it will be **very hard** to confidently know whether we have prevented systems from pursuing deceptive alignment strategies.

Overall, Carlsmith says "*my best guess is that ensuring the full PS-alignment of APS systems is going to be very difficult*". His overall probability that "misaligned APS is "much" harder than aligned APS" is ~40%.

[64] RLHF, or more generally, "Human Feedback on Diverse Tasks (HFDT)"

Of all premises, I am most uncertain about alignment difficulty. To help break down the question of "alignment difficulty",[65] here are some subclaims (each with very rough probabilities):[66]

Outer alignment

- Outer Alignment is likely a problem absent additional effort: 80%
- It will be difficult to outer align AI systems: 30%

Inner alignment

- Inner Alignment is likely a problem absent additional effort: 60%
- It will be difficult to inner align AI systems: 60%

Instrumental Convergence

- Conditional on some form of misalignment (outer of inner), APS systems will converge on power-seeking goals: 80%

Overall difficulty of alignment[67] = [P(outer misalignment occurs) * P(outer alignment is difficult) + P(inner misalignment occurs) * P(inner alignment is difficult)] * P(instrumental convergence)[68]

Overall difficulty of alignment = [(80% * 30%) + (60% * 60%)] * 80% = **48%**

Thus my overall probability that the below statement is true:

> *It will be **much harder to develop APS systems** that would be **practically PS-aligned** if deployed, **than to develop APS systems that would be practically PS-misaligned** if deployed (even if relevant decision-makers don't know this), but which are at least superficially attractive to deploy anyway, conditional on (1) and (2)*

...is close to 50:50

[65] Where "difficulty" is ill-defined. I take "difficulty" to mean something like "it will take significantly more resources (money, labor, time, etc) for actors developing AI systems to develop *and* align them, rather than just develop them." We could consider the "difficulty of alignment" analogous to the size of the "alignment tax".
[66] These probabilities should **not** be interpreted literally.
[67] Full definition from Carlsmith (49): *It will be much harder to develop APS systems that would be practically PS-aligned if deployed, than to develop APS systems that would be practically PS-misaligned if deployed (even if relevant decision-makers don't know this), but which are at least superficially attractive to deploy anyway, conditional on (1) and (2)*
[68] Assuming independence between inner and outer misalignment failures

Claim 4: Power-seeking AI systems will be deployed

Suppose that a frontier AI lab - named "Magma"[69] - builds a system that is (either outer or inner) misaligned, and has converged on the goal of seeking power over humans. Does this mean that Magma would actually deploy the system?

Despite a desire to prevent AI takeover, Magma might nonetheless choose to deploy their system. There are two broad classes of reason for this:

1. **Unilateral failure**: Magma *individually* deploys a misaligned system
2. **Coordination failure**: Magma deploys a misaligned system *in response to others*

Unilateral failure

Why might Magma choose to deploy a power-seeking system, despite the risks? Two reasons are:

1. Magma overestimates the probability that their AI system is aligned. They might mistake the likelihood of misalignment due to *negligence* or *deception*.

> **Negligence** - Magma may not be interested in testing for misalignment, perhaps because of skepticism about the difficulty of aligning systems, or a focus on the wrong parts of the problem. For instance, Magma may engage in "safety washing",[70] superficially testing how safe the model is on measures irrelevant to power-seeking, in order to convince others (and perhaps themselves) that the risk is lower than it truly is.
>
> **Deception** - Strategically aware AI systems may actively try to convince decision makers that they are well-aligned, so as to get deployed and later pursue misaligned goals. This may fool Magma if they do not pay sufficient attention to deception, even if they are interested in alignment more generally.

2. The (perceived) benefits of deployment outweigh the costs, given Magma's beliefs about alignment.

> **Externalities** - The full costs of misalignment (especially to the long-term future) will not be felt by Magma. For example, if there is a 1% chance of misalignment but 99% chance of tremendous success for Magma, then they may be willing to take this gamble - even if society at large would be better off not making this choice. Actors with less prosocial preferences are more likely to take the risk.
>
> **Usefulness** - If the AI system is deployed and aligned to Magma's interests, it could give them large profits, allow them to gain power, help solve global problems, or provide many other benefits. All else being equal, we might expect that more useful AI systems are more likely to be deployed. Note that "usefulness" here need not just refer

[69] I take the name Magma from Cotra 2022
[70] Scholl (2022)

to private benefit - if an AI system could do significant social good if deployed, then it is likely it will be too. Carlsmith vividly illustrates the social value of deploying aligned AI systems (37-38):

> *When pandemics are raging, oceans are rising, parents and grandparents are dying of cancer, rival nations are gaining in power, and billions (or even trillions) of dollars are sitting on the table, concerns about science-fictiony risks from power-seeking AI systems may, especially for some relevant actors, take a backseat.*

Thus it is not just naive profit-maximizing firms that may pursue powerful AI systems - altruistic actors may do so too, given the enormous potential social benefits.

How likely is a unilateral failure?

Indifference - Some of the major AI labs - such as OpenAI, Anthropic and DeepMind - have expressed clear concerns about misaligned AI systems.[71] However, other near-frontier labs appear less directly concerned with power-seeking AI.[72] It is hard to project which specific lab will be the first to build APS systems, however, so speculation over the attitudes of particular companies may not be informative for the actual actions of companies closer to TAI.

Deception - The likelihood of deceptiveness depends in part on the previous section

Externalities - Magma could have various different attitudes to externalities, which depend on (among other factors) the governance structures and the beliefs of individuals in the organization.

For instance, if Magma is purely a commercial enterprise, it may fail to consider private costs of building and deploying a model beyond costs to Magma itself. Alternatively, if Magma is built by a national government, it may account for the interests of people in the government's territory. Another possibility is a non-profit organization, which could act in the interest of various different groups depending on the non-profit's stated mission. This could make Magma more or less likely to consider externalities in its decision

In practice, frontier labs are currently a mix of nonprofit and for-profit organizations, with a blend of social and commercial interests. It is hard to assess if this mix is safer or more risky than other entities - like government or academics - driving endeavors to build transformative AI systems.

Coordination failure

A second class of problems emerges when we consider the role of other AI actors, such as other labs or countries. There are a number of coordination-related factors which might make misalignment more likely to occur:

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[71] See, for instance:
- Anthropic's CEO estimates a 10 - 25% chance of an AI catastrophe
- OpenAI' CEO states that the worst-case scenario for AI is "lights out for all of us"
- DeepMind's CEO believes AI risk should be treated as seriously as the climate crisis

[72] For instance, Meta AI does not believe the risks are credible (Heikkilä 2023)

Deployment Dilemma[73] **-** There may be a chance that AI labs (or countries) descend into a racing dynamic, where each is incentivized to build and deploy systems quickly, at the expense of appropriate safety measures.

For example, it may be that Lab A is more interested in building a powerful AI, than in building a powerful and safe AI (for reasons described above). This might force Lab B - who are willing to act slowly and cautiously - to cut corners in order to stop Lab A. The net result is that both labs are more likely to deploy unsafe AI systems. This can occur even when a safety-conscious lab - like Lab B - is otherwise willing to adequately invest in safety precautions.

Even if labs A and B are both safety conscious and willing to move slowly, the *perception* of competition and limited safety measures may lead to harmful racing dynamics. This might happen due to poor information and communication between the labs.

Numbers of actors - Suppose that Labs A and B are able to effectively coordinate to not deploy powerful systems - but Lab C emerges, destabilizing the equilibrium. This can cause the same deployment dilemma to emerge.

More generally, the larger the number of actors, the more likely at least one lab deploys a misaligned system.[74] Variations in beliefs, risk-tolerance, and social responsibility mean that some actors (even if only a small minority) may be more lax about alignment. For example, even if only 5% of actors would be willing to build systems without sufficient safety cautions, then by the time that twenty labs can build misaligned systems, one lab (in expectation) will deploy such a system.[75]

How likely is a coordination failure?

Deployment Dilemma - Unfortunately, there is a strong historical precedent of actors racing ahead with dangerous technology out of fear about an adversary. Indeed the *security dilemma* gives theoretical grounding for why actors tend to pursue offensive or dangerous capabilities over defensive or safety measures. Belfield and Ruhl (2022) study this dynamic in two case studies - atomic fears in the 1930s, and the missile gap of the 1950s. In both cases, fears were unfounded, but nevertheless led to a rapid speed up in the discovery, development and deployment of dangerous technology. Thus, even if the true risk of different AI labs / countries deploying dangerous AI is small, we have reason to suspect that this fear might drive actors to cut on safety measures.

Looking at the state of AI developments today, we can see very high perceptions of an "AI race", amongst both excited participants[76] and concerned observers.[77] On the other hand, there

[73] Analogous to the "Security Dilemma" - actors may take escalating actions of increasing risks in order to get ahead of the competition.
[74] This is a more general phenomenon - see the unilateralist's curse
[75] Assuming independence between the p(lab deployment)
[76] See e.g. Nadella 2023 - "*..I hope that, with our innovation, [Google] will definitely want to come out and show that they can dance. And I want people to know that we made them dance, and I think that'll be a great day.*"
[77] See e.g. Hogarth 2023 - *We must slow down the race to God-like AI*

are signs of collaboration between frontier labs.[78] Thus, on the current trajectory, it is unclear how much major players will compete or collaborate.

Number of actors - It is likely the number of actors capable of building APS systems will (significantly) increase over time, due to three dynamics:

1. Increasing model size

With declining costs to computing power and algorithmic insights increasing efficiency, larger and larger models are being trained.[79]

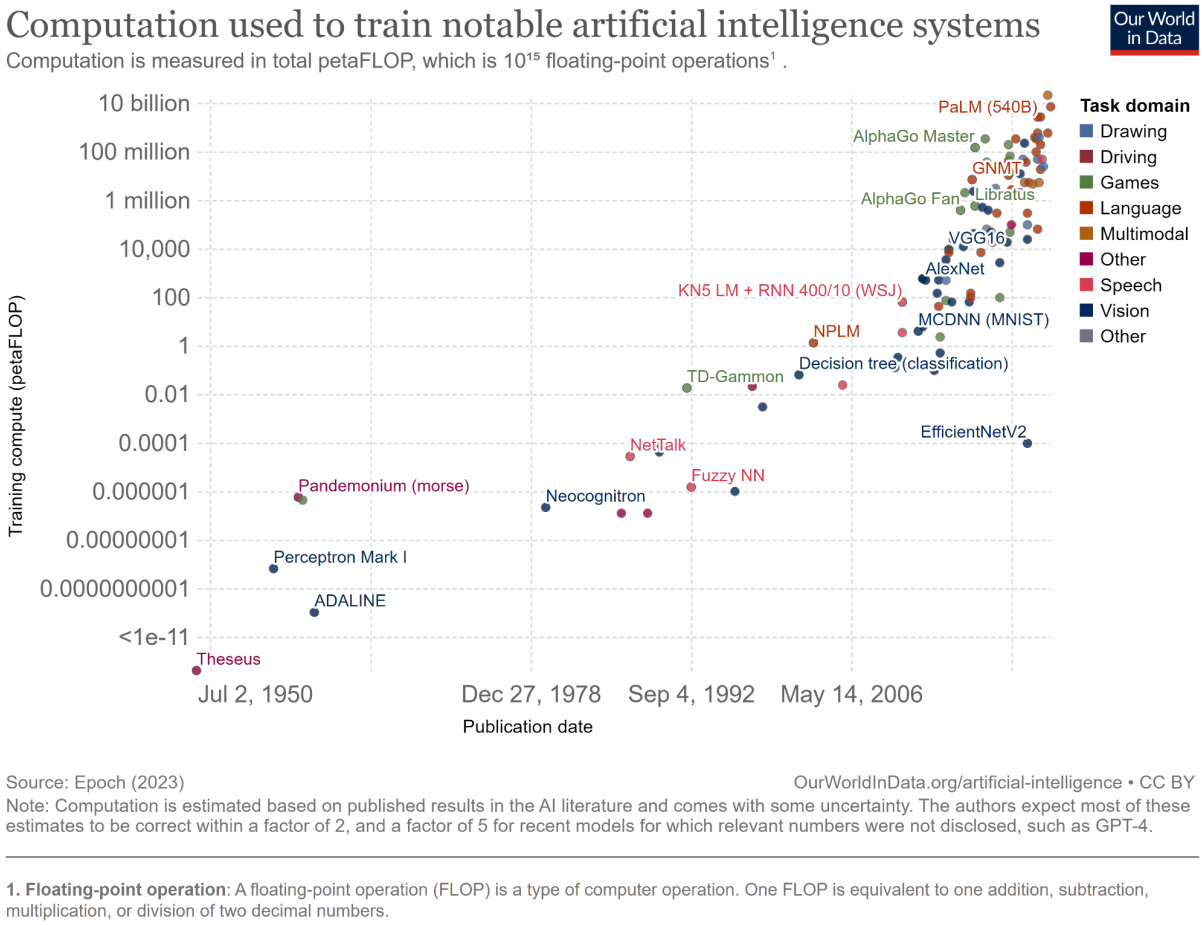


Figure 11: Computation used to train notable artificial intelligence systems. Computation is measured in total petaFLOP, which is 10^{15} floating-point operations estimated from AI literature, albeit with some uncertainty. Estimates are expected to be accurate within a factor of 2, or a factor of 5 for recent undisclosed models like GPT-4. (Our World in Data 2023)

If these dynamics continue, then the cost of building powerful systems will fall rapidly over time. For example, Heim (2023) predicts that, eight years after the original training of GPT-4, it

[78] For instance, OpenAI, Anthropic and DeepMind have collaborated and cooperated with each other on a number of measures, such as voluntarily allowing the UK government access to their latest models (Clarke 2023), with an intent for governments to introduce regulations on frontier AI systems
[79] As measured in petaFLOP

would cost new actors around $100,000 to build a similar sized model.[80] If this trend continues, then eight years after the first lab is able to build a misaligned APS system, there will be perhaps thousands of actors willing and able to build their own system.

2. Diffusion of existing models

As well as declining costs, it may be possible for actors to gain access to frontier models - for instance due to leaks, espionage, or because the lab chooses to open source the model. This has some precedent, with Meta AI's LLaMA being leaked just two weeks after launch (Vincent 2023).

3. Increasing investment in AI

Finally, with increased money and attention on AI more generally, the number of new entrants trying to build AI systems is increasing, and may continue to do so for as long as commercial incentives are sustained

In sum, there are several reasons to suspect that the number of actors building powerful AI systems will increase significantly. This will likely worsen coordination attempts, accelerate racing dynamics, and increase the chance that an AI lab like Magma deploys a misaligned AI system.

My approximate overall probability on the chance of a misaligned deployment occuring, *conditional on alignment being "difficult"*, is roughly 75%. The majority of this risk is driven by multilateral, rather than unilateral failures. This is because I expect that at some point, at least 1 out of n actors will be willing and able to deploy a model which is misaligned, as n increases. However I do not put the risk higher than 75%, because of the possibility that the first actor successfully deploys an aligned model, which prevents other actors from deploying misaligned systems.

Claim 5: When deployed, power-seeking systems will permanently disempower humanity

Even if power-seeking APS systems are deployed, it's not obvious that they would permanently disempower humanity. We may be able to stop the system in its tracks (by either literally or metaphorically "pulling the plug"). First, we need to consider the mechanisms by which AI systems attempt to takeover (i.e. disempower) humanity. Second, we need to consider various risk factors for a successful takeover attempt.

Mechanisms for AI takeover

[80] *"[Wiblin] ...On current trends, when would an individual be able to train GPT-4 themselves, say, at home, using a bunch of hardware that they've bought for around $100,000? [Heim] ...this should be around like seven to eight years" (source)*

To evaluate how likely it is that a misaligned AI system could effectively disempower humanity, it can help to describe in more depth what an AI takeover might involve.[81] Some speculative behaviors a power-seeking AI system might engaged in could include:

1. **Hacking computer systems**. Software systems have many vulnerabilities,[82] which are regularly found and exploited by humans. We might expect AI systems to be at least as good at finding such exploits, and possibly far better. This might allow them to "break out" onto the internet if given the chance, before duplicating itself many times.

2. **Persuading, manipulating or coercing humans**. A common response to the above is that humans could simply "pull the plug". However, it is possible that individuals, not software, will be the key vulnerability that allows an AI system to "escape". For instance, the system may be able to extort, blackmail or corrupt human overseers. Whilst this is highly speculative at the capability of current systems,[83] a situationally aware AI system well read in human psychology may be much more adept at finding human vulnerabilities.

3. **Gain broad social influence**. More broadly, AI systems may be able to steer society into undesirable political outcomes. For instance, AI systems might be able to engage in electoral manipulation, steering voters towards policymakers less willing or able to prevent AIs systems being integrated into other key places of power.[84]

4. **Gaining access to money**. AI systems are already used in financial markets, often with a "black box" nature that can lead to flash crashes (AI Incident 28). If misaligned systems are rolled out into financial markets, they may be able to siphon off money without human detection. This may allow the system to execute numerous other goals on the path to takeover

5. **Developing advanced technologies.** AI systems may be able to rapidly expand the frontier of scientific knowledge. This could enable entirely new technologies, such as nanotechnology. An AI system adept at the science, engineering and manufacturing of nanotechnology, along with access to the physical world, might be able to rapidly construct and deploy dangerous nanosystems, leading to a "gray goo" scenario described by Drexler (1986).

These are just some plausible pathways by which AI systems might be able to outwit humanity. However, there are likely many other vulnerabilities which an AI system might be able to exploit, some of which we are not aware of ourselves.

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[81] Note this is *not* necessarily how an AI system would *actually* try to disempower humanity - rather, these stories give some intuitions as to what takeover might look like. See Carlsmith (2022), Section 4.2, for more examples of potential power-seeking behaviors. Or see Hilton (2022) for some stories about how a power-seeking AI could actually take power.
[82] See NIST (2023) for a list of known software vulnerabilities.
[83] Although we do already have some instances of deception - for instance, GPT-4 lying to a TaskRabbit Worker (Vice 2023), or CICERO persuading *Diplomacy* players (Science 2022)
[84] The use of AI systems for steering political discourse and outcomes is not novel - see e.g. Russian interference in the 2016 United States elections.

How likely is disempowerment from a deployed, misaligned AI system?

The above five stories are just speculative examples of what *may* happen if misaligned systems are deployed. How likely is it that a misaligned system would in fact successfully disempower all of humanity? I consider:

1. How might the *speed of AI development* alter our ability to prevent disempowerment?
2. How might *warning shots* alter our ability to prevent disempowerment?

Speed of AI development

In a previous section, we discussed the various speeds at which AI systems may be developed.

All else equal, we might expect a faster takeoff to increase the probability of disempowerment. One reason for this is that, in faster takeoff worlds, humans have less time to notice a takeover attempt. For example, in a "slow" scenario, AI systems may more predictably develop dangerous capabilities (like manipulation or hacking). This would give us time to develop countermeasures for those particular pathways. In contrast, if AI systems are able to rapidly and unexpectedly develop and deploy nanotechnology, then it would be very challenging for us to rapidly respond to the danger in time.

Warning shot(s)

Before power-seeking AI systems are able to successfully seize power, less capable systems may make a failed attempt. If this occurs, we might expect some form of societal response to prevent such an occurrence again.

A relevant analogy here may be the case of Covid-19 and the risk of a catastrophic pandemic. One argument goes that, whilst Covid-19 caused tremendous damage, it did not constitute a biological *catastrophe*. However, it may have woken policymakers up to the possibility that future pandemics may occur (Yassif et al 2023).

Similarly, we may hope that if a less-than-catastrophic AI takeover attempt is made, it may put relevant actors "on guard". This might subsequently inspire policymakers, AI labs and regulators to take action, to prevent a future, more dangerous attempt.

However, an unfortunate outcome of Covid-19 has instead been "pandemic fatigue", with society unwilling to take the issue more seriously following the covid crisis (WHO 2020). Thus the track record for a proactive response to an AI disaster does not look positive.

In addition, the threat of a misaligned AI takeover is somewhat unique, in that the relevant AI systems may be strategically aware. AIs may choose to wait until the probability of takeover is very high, before making their move. They may do so specifically because they realize that a failed "warning shot" might reduce their chances of taking over at a later date. Thus we may have no chance to spot an attempted takeover before it is too late.

Overall view

My overall impression is that it will be hard for humanity to prevent full disempowerment. My main reasons to believe this are:

1. I expect that, before we reach takeover-capable systems, AIs will be broadly deployed in society (for instance conducting much cognitive labor). Thus there will be many mechanisms by which an AI system may be able to take power, should it desire to.

2. I think it is more likely than not that the capabilities relevant for disempowering humanity - such as strategic awareness and deception - will emerge suddenly rather than gradually. This will give us less time to prepare for any takeover attempt.

3. As described previously, we might expect deceptive alignment to occur, meaning systems are able to "lie in wait". This would leave us without a warning shot, and with little chance to defend against a takeover attempt.

Thus, conditional on capable power-seeking systems already being deployed, I expect a takeover to occur with approximately 80% likelihood.

Claim 6: The disempowerment of humanity would constitute an existential catastrophe

Finally, Carlsmith considers whether the disempowerment of humanity would be considered an existential catastrophe - as defined by Ord (2020): "an event that drastically reduces the value of the trajectories along which human civilization could realistically develop".[85]

Without much further analysis, Carlsmith concludes that if humanity is disempowered, it would very likely (95%) be considered an existential catastrophe. A couple of reasons Carlsmith does not believe this is 100% likely are:

1. It's possible a misaligned AI will still converge on the same version of "goodness" that humanity would do

2. It's possible that misaligned AI finds reasons to cooperate with humanity's values, even if it has disempowered them

3. It's possible that misaligned AIs will be morally relevant patients, such that their disempowerment of humanity allows for a morally beneficial future

However, Carlsmith is sufficiently unconvinced by these reasons to put more than a 5% chance that disempowerment would not be an existential catastrophe. Without any further analysis, I agree with the reasoning Carlsmith gives - as such I would guess that disempowerment is [at least] 95% likely to constitute an existential catastrophe.

[85] Carlsmith adds some nuance to this definition

Assessing the threat of power-seeking AI

It is possible that we build power-seeking AI systems, which successfully disempower humans and cause an existential catastrophe. Summarizing some of the key dangerous from power-seeking systems into ten points:

1. Improvements in AI capabilities are **highly unpredictable** (more) - but the timeline to takeover-capable AI systems is likely small (more)
2. There are **intense commercial and social incentives** to build highly capable systems (more)
3. We do not know how to correctly **specify our intentions** into AI systems, and this may **get harder with increasing capability** (more)
4. We do not know if AI systems will **actually optimize** for the thing we specify, especially given their opaque nature (more)
5. AI systems may successfully **deceive** us into thinking that they are pursuing what we want (more)
6. A system not pursuing what we want is likely **to converge on power-seeking behavior** (more)
7. Actors are **unlikely to act appropriately cautiously** when building powerful AI systems (more)
8. There is a **small window of time** for the first actor to successfully align a takeover-capable system, before a less cautious actor builds their own system (more)
9. There is **only one try** for sufficiently powerful systems - once out of the box, it will be very hard to prevent takeover. (more)
10. An AI system that successfully takes power from humans will create a **valueless universe**, rather than one of human flourishing (more)

Carlsmith's overall view is a 5% chance of power-seeking AI by 2070. More recently, he has updated his position to >10%. Meanwhile, several reviewers of Carlsmith's paper provide their own probabilities.[86] Bottom-line estimates of an existential risk from a power-seeking AI range from <0.1% to >75%.

My overall probability on each of the following claims is as follows:

1. By 2070, it will become possible and financially feasible to build highly capable, agentic planning, strategically aware (APS) systems: **90%**
2. There will be strong incentives to build APS systems| (1): **90%**

[86] Carlsmith 2021

3. It will be much harder to develop APS systems that would be practically PS-aligned if deployed, than to develop APS systems that would be practically PS-misaligned if deployed| (1)–(2): **48%**

4. Misaligned APS systems will in fact deployed| (1) - (3): **80%**

5. Misaligned power-seeking will scale to the point of permanently disempowering ~all of humanity| (1)–(4): **80%**

6. The permanent disempowerment of humanity will constitute an existential catastrophe| (1) - (5): **95%**

Multiplying these probabilities together, I ultimately conclude there is a **~24% probability of existential catastrophe from misaligned, power-seeking AI by 2070**.[87] To account for uncertainties in each step, I give lower and upper bounds on the likelihood of each estimate. These are modeled in Guesstimate. My 5% - 95% confidence interval is between a 10% and 35% chance of catastrophe. However, much caution should be taken when interpreting these numbers.

However, it is worth noting that power-seeking AI is just one pathway to an existential catastrophe via misalignment. I thus think that the broader probability of catastrophe via a misaligned AI system is likely higher than 25% - perhaps above 30%, when accounting for misalignment routes not directly captured in Carlsmith's framework.[88] Furthermore, there are other pathways to catastrophe quite distinct from misalignment. These are discussed below.

AI-induced Biological Catastrophe

For a deeper analysis on Global Catastrophic Biological Risks more generally, we recommend Ruhl 2023

In the not-too-distant future, it may become possible for AI systems to play a significant role in the advancement of biological science and engineering. These advances may offer great potential for improvements in life expectancy and quality of life.

Nonetheless, the use of AI in biology is a dual-use technology, with the potential to cause significant harm. In the limit, the intersection of AI and biology could cause a global or even existential catastrophe. Such an event may occur due to the malevolent intent of bad actors ("misuse"), or due to recklessness and negligence on the behalf of otherwise well-intentioned actors ("accident risk").

In this section, we explore three claims

(1) Will it be possible to develop AI systems with biological capabilities, and over what timeframe?

[87] 90%*90%*48%*80%*80%*95%

[88] Threat models which discuss misalignment but vary from Carlsmith include Christiano (2019), Critch 2019 or Hendrycks (2023). Threat models which develop ideas touched on by Carlsmith include Ngo (2022), Soares (2022) and Cotra (2022).

(2) Will biologically capable AI systems *differentially* increase the chance of a global biological catastrophe, conditional on (1)?

(3) How likely is an existential catastrophe to occur, conditional on (1) and (2)?

Figure 12 illustrates how these claims fit together:

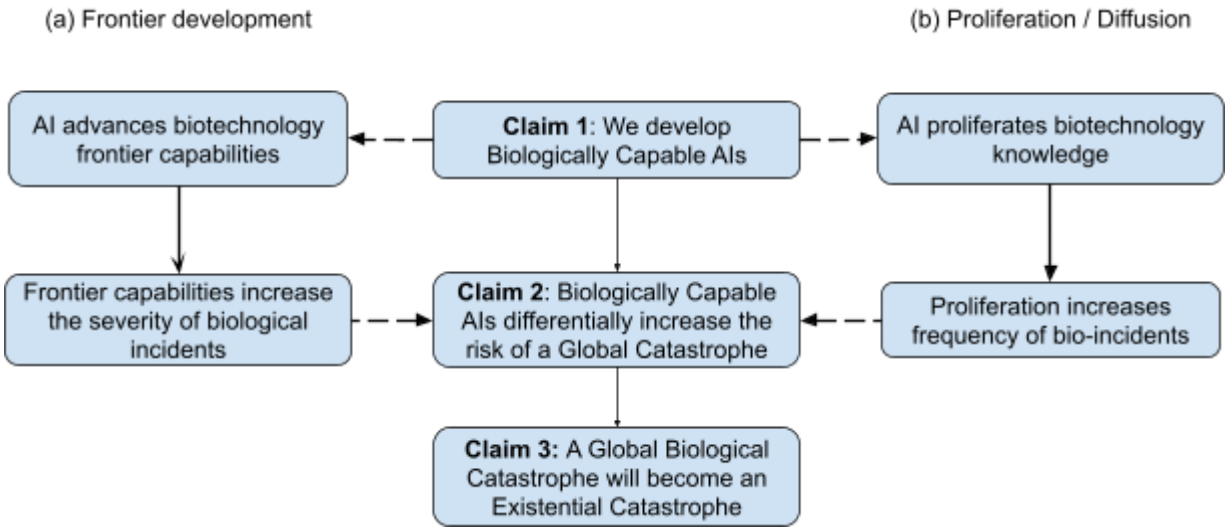


Figure 12: Outline of claims in section on threat from the use of AI for a biological catastrophe

Claim 1: It will be possible to build Biologically Capable AI systems over the coming decades

With the development of ever-capable AI systems, we may expect to see to see two forms of advancement in biology:

1. Advances at the *frontiers* of life sciences

AI systems lead to new breakthroughs. For instance, "Biological design tools" (BDTs) may be developed.[89] BDTs are tools which increase frontier knowledge, for instance developing new proteins or other biological agents. RFDiffusion, AlphaFold and ProGen2 are all examples of such BDTs.

2. Accelerations in the *diffusion* of biological knowledge and technology.

The mass adoption of large language models (LLMs) may provide large portions of society access to knowledge of biological systems - for both positive and ill intent. The growth of Cloud Labs (Arnold 2022) may further reduce barriers to entry, allowing many to not only understand biology, but carry out live experiments.

[89] The term Biological Design Tool originates from Sandbrink (2023)

Progress and diffusion in the life sciences are likely to occur even in the absence of AI.[90] Nonetheless, AI systems may accelerate these trends (Sandbrink 2023). At some point - through a combination of LLMs and BDTs - we may reach a point where AI systems are (in aggregate) effectively able to automate almost all the activities involved in biological research. We can label such set-up a *Biologically Capable AI* (or a "BCAI").[91]

Such a single "BCAI" could profoundly transformative biotechnology. This could have both positive and negative purposes. For instance, a BCAI could automate both the process for developing vaccines to save lives - but also dangerous pathogens to threaten them.

Timeline to Biologically Capable AI systems

How far are we from powerful AI models being used in biology?

A study from Arb Research (2023) analyzes exactly this question. They conclude there is a **70%** chance that "impressive protein generating capabilities" are developed in the next 3-8 years.[92] They forecast this using historical trends in (non-biological) AI progress, combined with forecasting the relevant data needed for biological neural networks. Of course, an AI capable of generating proteins would just be one part of the puzzle. Nevertheless, this forecast gives us a lower bound on when BCAIs will be possible.

As an upper bound, BCAIs might be only possible around the same time that transformative AI is developed more broadly. For instance , we might only see advancements in biology along with a wider Process for Automating Scientific and Technological Advancement (PASTA).[93]

Thus, combining the forecast for protein-generating AIs (~2026 to 2030) and our forecast for transformative AI (~2030s), we estimate a **~50%** likelihood of biologically-capable AI systems being developed by 2030. However, the phenomenon of emergence (among other reasons) makes this difficult to forecast.

This estimate has a long-tail: If biologically-capable AI systems aren't possible under the current AI paradigm, then it's unlikely that they will be developed for a long-time, if at all (given that artificial *general* intelligence may overtake biologically-capable AIs). Over a longer time horizon, my estimate for biologically capable AI systems collapses into the same estimate as for capable, planning, strategically aware systems - around 90% by 2070.

Diffusion of Biologically Capable AI systems

[90] See Ruhl (2023), who notes that the cost of sequencing the human genome has rapidly declined over the last couple of decades, reducing the barriers to entry for many actors (i.e. increased proliferation)
[91] This term is adopted from Karnofsky (2021), who discusses the prospect of "PASTA" - Process for Automating Scientific and Technological Advancement. In the case of "BCAI", we are imaging AI systems capable of automating specifically *biological* advancement.
[92] Where "impressive" is approximately "at least as human scientists' ability to generate proteins". The authors draw comparisons to image generation, which took a similar length of time from very basic abilities to "impressive" (i.e. human or above) levels
[93] See Karnofsky (2021)

We already have some evidence about the potential for AI systems to diffuse knowledge about biology, including dangerous capabilities like the steps required for developing new pathogens.

A paper from Soice et al (2023) vividly illustrates this possibility. In an experiment, non-scientist students were given access to LLM chatbots, and asked to see how far they could get in producing an engineered pathogen through the use of a chatbot alone. The result was the following:

> *In one hour, the chatbots suggested four potential pandemic pathogens, explained how they can be generated from synthetic DNA using reverse genetics, supplied the names of DNA synthesis companies unlikely to screen orders, identified detailed protocols and how to troubleshoot them, and recommended that anyone lacking the skills to perform reverse genetics engage a core facility or contract research organization*

Although the students could not complete the production of an engineered pathogen, the exercise nonetheless showed that with greater advancements, the difficulty for the average citizen to build a bioweapon is declining. This illustrates the dangers of widespread access to large language models without restriction.

Furthermore, once the first AI system capable of developing bioweapons is built, it will immediately become affordable for millions to use the system. This is due to the way we currently train AI: Once an AI system (like GPT-4) is *trained*, it soon becomes affordable for millions of copies to be *used*, because it is much less resource intensive to use a system than to train one.[94] This could happen even if frontier AI labs are effective at controlling the use of their own systems, given the nature of large language model diffusion (Cottier 2022), in which open-source alternatives of equal capability tend to appear only a few years after the frontrunner.

Thus, very soon after the first BCAI system can be built, it will quickly become possible for millions of people to use their own BCAI system (either by explicit release from a frontier lab, or via open source models). Thus we might end up in a highly multipolar world with respect to biological AI systems.

Incentives to build Biologically Capable AI systems

The vast majority of actors will not intend to build or use AI systems for dangerous biological applications. Nonetheless, there are strong commercial and social incentives to build BCAI systems, since they may be able to save many lives (for instance by discovering new vaccines, finding new drugs, and curing diseases).

Devereson et al (2022) analyzes investment trends in AI-driven drug discoveries. Figure 13 illustrates an explosion in funding for companies in the sector:

[94] See the section on polarity for further discussion of how and why we should expect many AI systems to be *applicable* from the resources used to train *one* system.

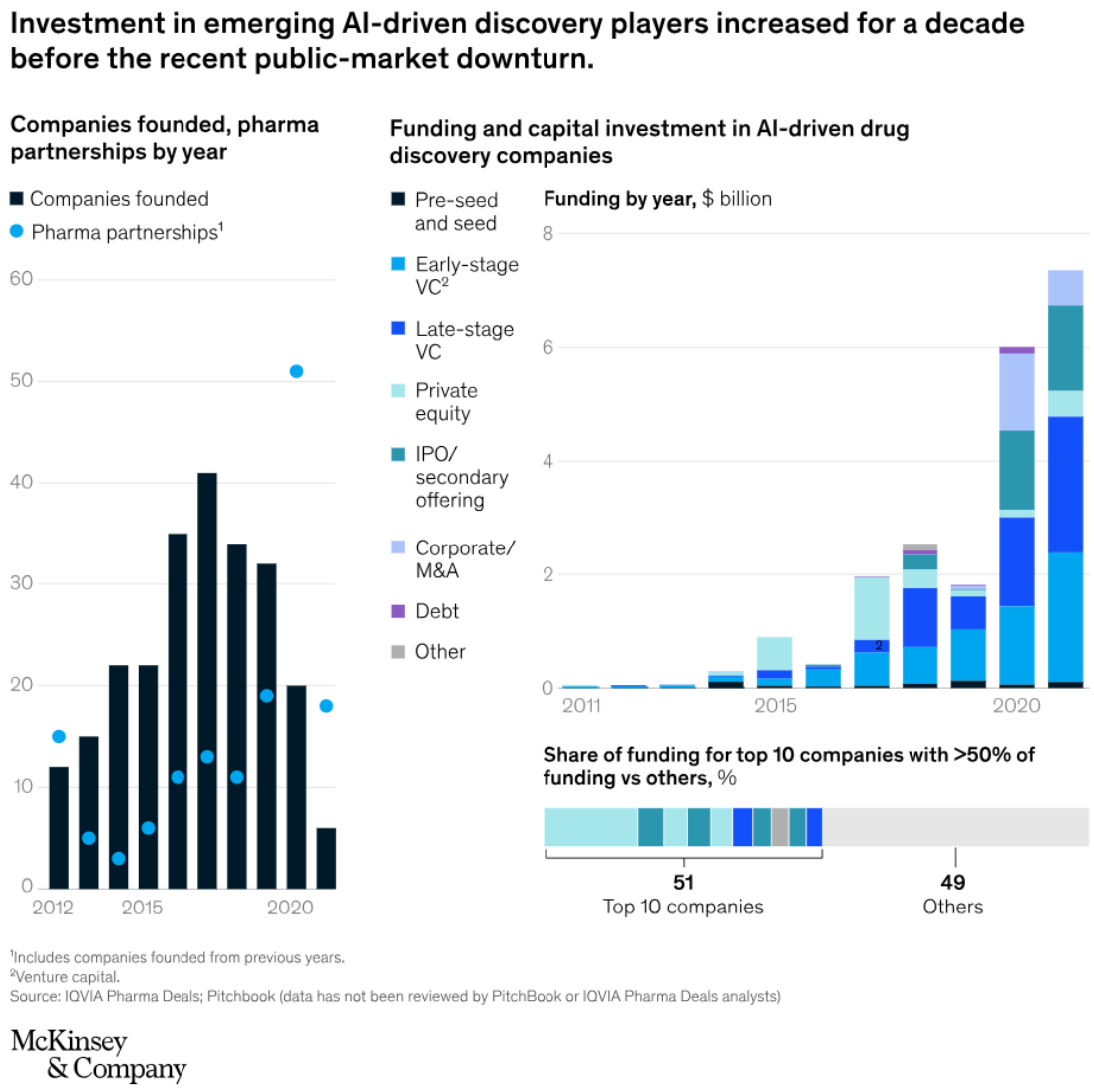


Figure 13: Graphic of (a) Number of AI-drug discovery companies and pharma-partnerships founded by year, and (b) Investment in AI-driven drug discovery organizations by yer

Thus we can see clear intent to find biological applications of AI. Whilst these investments may have many positive effects, they may also inadvertently accelerate application for misuse, such as bioweapons.

In the next section, we explore whether AI systems will *differentially* increase the risk of a global biological catastrophe.

Claim 2: Biologically Capable AI systems will differentially increase the risk of a Global Catastrophe

Thus far, we have only explored the claim that AI might drive biological advances. Now we turn to the question of whether such systems will increase or decrease the risk of biological incidents - i.e, will AI systems allow us to strengthen our biodefenses (e.g. by accelerating the

development of vaccines), or make us more vulnerable to biorisks (e.g. by increasing the lethality and availability of bioweapons)?

Analyzing the role of AI in biotechnology, we investigate the two pathways for how risks might increase (following Sandbrink 2023):

1. Risks from *frontier capabilities* (through e.g. Biological Design Tools (BDTs))
2. Risks from *proliferation* (through e.g. Large Language Models (LLMs))

Frontier Capabilities and the Offense-Defense Balance

First, we ask - as *frontier* biological capabilities scale due to AI, should we expect risks to grow or shrink?

Garfinkel and Dafoe (2019) provide a general (tech-agnostic) model for analyzing whether risks differentially increase or decrease - i.e. the "offense-defense" balance. Their model roughly suggests an inverse-U relationship (Figure 14):

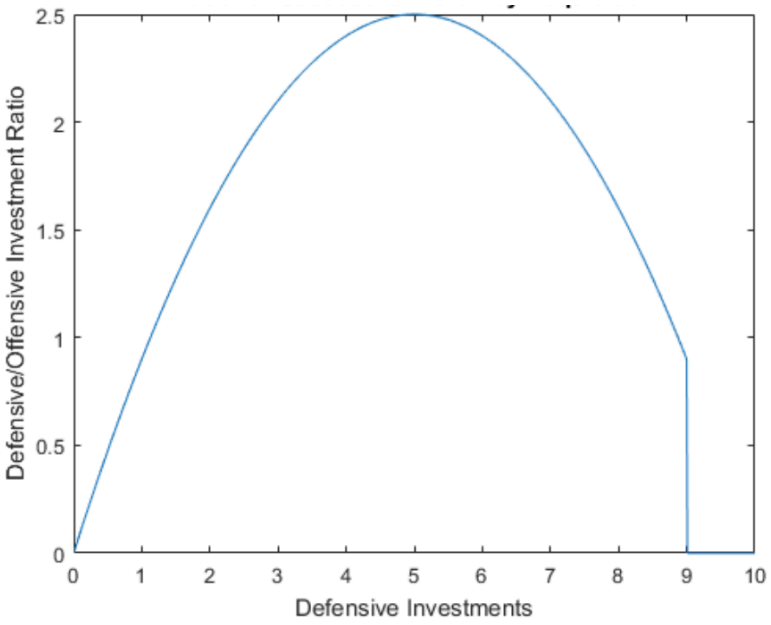


Figure 14: This model is a theoretical example in cybersecurity. It shows the Ratio-for-success (y-axis) for an idealized model, with increasing defensive investments (x-axis)

- As investments in the frontier scale, the "offense" side tends to rise first, causing an acute risk period,
- Later, defenses catch up, as investments continue to scale

How does the Offense-Defense balance apply to BCAIs?

Given the above, our default assumption should be that AI-enabled Bio Capabilities will first favor the attacking party (e.g. bad actors optimizing pathogens for maximum damage), before

further investments enable the defending party to implement resilience measures. Thus BCAIs might create an acute risk period at the frontier.

However, AI could upset this relationship in two ways:

- On the one hand, perhaps well-intentioned frontier actors[95] might be able to get "ahead of the curve", implementing very strong pathogen-agnostic biodefense measures.[96] These may be enough to slow down almost any threat, irrespective of the lethality of the pathogen. Thus defenders may have a strong upper hand.
- On the other hand, BCAI systems might be very easily adaptable for strongly offense-heavy purposes. Given their deep knowledge of human biology they may be very adept at exploiting our vulnerabilities to disease, developing pathogens optimized for lethality and for which treatment is very difficult (or even impossible). In the limit, BCAIs might be able to use atomically precise manufacturing to develop dangerous biomolecular materials against which there is no defense (Open Philanthropy, 2015).

We remain highly uncertain which of the above scenarios is more likely, and if either is even plausible. AI systems might be able to find diseases for which there is no possible cure, or find treatments that no disease can be overcome. Whilst either of these scenarios are highly speculative, they do both point towards high-variance worlds, since BCAIs might be more capable of stumbling on genetic engineering insights far beyond our current knowledge. Our tentative view is that AI-driven developments in frontier biology are likely to differentially increase risk. Still, it should be repeated that we are highly uncertain about whether BCAIs at the frontier will increase or decrease risk.

Proliferation Risks

Secondly, we ask - as the *number of actors* with access to BCAIs increases, should we expect risk to grow or shrink?

Unfortunately, the answer here is depressingly clear: proliferation almost certainly increases risk. The reason for this is simple - *a good guy with a bioweapon cannot stop a bad guy with a bioweapon*. With reduced barriers to entry, more and more people will be able to build bioweapons, which (all else equal) increases risk. Thus, the open sourcing of AI systems with biological capabilities will in all likelihood increase the risk of biological attack, at a rate faster than improved defensive measures.

Estimating the risk of Biological Catastrophe

To estimate the increased probability of a global biological catastrophe caused by AI, we first estimate an approximate baseline probability of global biological catastrophe. Then, we consider how much AI development acts as a *multiplier* on this risk.

[95] Who, by and large, are overwhelmingly likely to have positive intent, given the very small portion of actors with malevolent intentions

[96] Note, it is challenging to conceptualize the exact countermeasures BCAIs may be able to install, since such systems may be beyond current human comprehension

Baseline risk of a Global Biological Catastrophe

Bostrom and Cirkovic (2011) define a "Global Catastrophic Risk (GCR)" as the risk of an event causing >10 million fatalities (or $10 trillion worth of economic damage). We use a higher bar for defining a "Global Catastrophe", as an event with >1 billion deaths. This is not because an event causing 10 million - 1 billion deaths wouldn't be a terrible tragedy, but simply because the biorisk estimates we could find cluster on the question of events that surpass 1 billion deaths.

Existing estimates include:[97]

1. Millet and Snyder-Beattie (2017), who estimate approximately a 0.1% - 0.2% annualized risk of a pandemic which kills ~1 billion.[98] Naively extrapolating to 2100, this gives us a probability of **8% - 15%** by 2100 (geometric mean of 11%)

2. Superforecasters from the Existential-Risk Persuasion Tournament (XPT) estimate a **~1.8%** chance of a catastrophic pandemic,[99] whilst biosecurity experts estimate a **5%** chance by 2100

3. The 2008 Global Catastrophic Risks Survey found that experts estimated a **10%** chance that the "single biggest engineered pandemic" would kill 1 billion people by 2100;

Averaging across these four estimates, we get a baseline geometric mean of around **5.6%** that at least one pandemic occurs with >~1 billion deaths, by 2100. Additional analyses can be found in Ruhl (2023), Gopal et al (2023), FRI (2022) and other sources.

Increased risk due to advances in AI

We model the increase in risk from AI through two pathways:

1. **Frontier Multiplier**: As a multiplier on the *virulence* of biological threats, due to AI developing more infectious and/or severe diseases. Such advances occur at the *frontier* of life sciences, primarily due to accident risk.

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[97] Related estimates we excluded due to different criteria includes:
1. Toby Ord estimates a **1 in 30** (3.33%) chance of an *existential* catastrophe from engineered pandemics by 2120, with (presumably) a higher estimate for a global catastrophe
2. Pamlin and Armstrong (2015) estimate a **0.01% risk** that *"civilisation collapses to a state of great suffering and does not recover, or a situation where all human life ends."* due to synthetic biology by 2115.
3. Metaculus, a crowd forecasting platform, which estimates a **38%** chance of a 10% decrease in the human population [~800 million] by 2100 (Metaculus 2023). However, this combines all threats, not just pandemics

[98] 0.04% - 0.08% of risk is from an accident, whilst a similar size comes from deliberate release. We look exclusively at engineered pandemics here, because AI systems are likely much less relevant for changing the risk landscape from natural pandemics. The authors use different criteria for a "GCBR" event

[99] Defined as the death of >10% of the population

Estimating how AI-driven advances in frontier life sciences convert into increased pandemic risk is highly challenging. As a first-pass model, we make the following assumptions:[100]

1. Frontier advances will mostly increase the threat from *engineered* pandemics, rather than from natural pandemics. We therefore exclude natural pandemics from our model.

2. AI could rapidly accelerate Research & Development (R&D) across all sectors, including the life sciences. We can thus use an AI-driven R&D growth model to estimate how frontier progress in biology will grow with AI capabilities.

    a. Davidson (2023) provides a more detailed model of R&D-driven increases in economic growth, resulting from AI. Figure 15 shows a plausible economic growth model under transformative AI.[101]

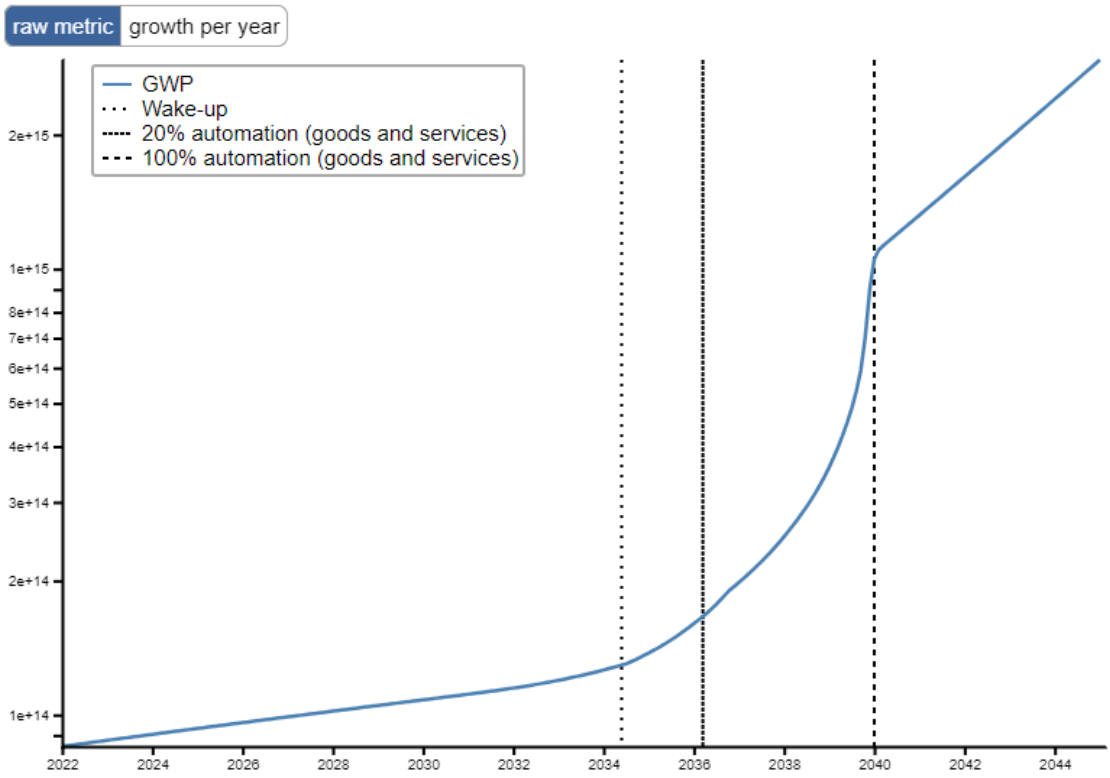


Figure 15: Plot of Gross World Product (Blue Line) under transformative AI, using default inputs from Davidson (2023)[102]

3. R&D growth, including in biology, could lead to the overall effect of *differentially increasing* frontier biological risk - due to the technology's offense-heavy nature, as discussed previously.

[100] This approach is heavily reliant on an "outside-view" approach to frontier biological development, since the exact pathways by which AI may drive life sciences are highly uncertain.
[101] Note the graph shows Global Output increasing exponentially on a *logarithmic* scale - i.e. economic *growth* is exponential
[102] To recreate, go to https://takeoffspeeds.com/playground.html, then scroll down to "Metrics"

4. AI-driven R&D growth and frontier biological risks are therefore likely to be positively correlated (for the simple reason that less research into engineered pathogens probably reduces the chance of a lab leak).

   For simplicity, we assume a linear relationship: A doubling of (R&D-driven) economic growth leads to a doubling of biological research, including in risky laboratories - and thus a doubling of lab incidents. As an intuition pump, if biology research increases by 2x, then we might (naively) assume a 2x increase in risks from biology research.[103]

5. AI-driven growth will necessarily end after some peak, at which further growth is limited by physical constraints.[104] After AI-driven growth peaks, we assume the "time of perils" will end.[105]

6. We thus expect some "acute risk period", starting at the point of AI driven Biology R&D, and ending at the point where the "time of perils" has ended. During the acute risk period, we expect frontier biological risks to scale with R&D growth, as more experiments in the life sciences are conducted (which may, for instance, increase the number of lab leaks).

   We assumed a linear relationship between economic *growth* and Biological Risk. Thus, when economic growth increases exponentially, so will biological risk.

We can illustrate such an acute risk period below (Figure 16):

[103] Of course this is a crude simplification - further research could investigate the relationship between a) General R&D and Biology research, and b) Biology research and biological incidents.
[104] AI-driven R&D must necessarily end, as exponential growth will eventually run into limits bound by the laws of physics (e.g. the speed at which space colonization can occur). We may expect frontier biological threats to end sooner than this, if a) another existential catastrophe occurs, b) biological minds become uploaded to digital minds, or c) existential security is achieved - as just some example scenarios of how this era may end.
[105] See Bernard (2023) for further discussion on whether (sufficiently) advanced AI can cause existential risk to be permanently reduced.

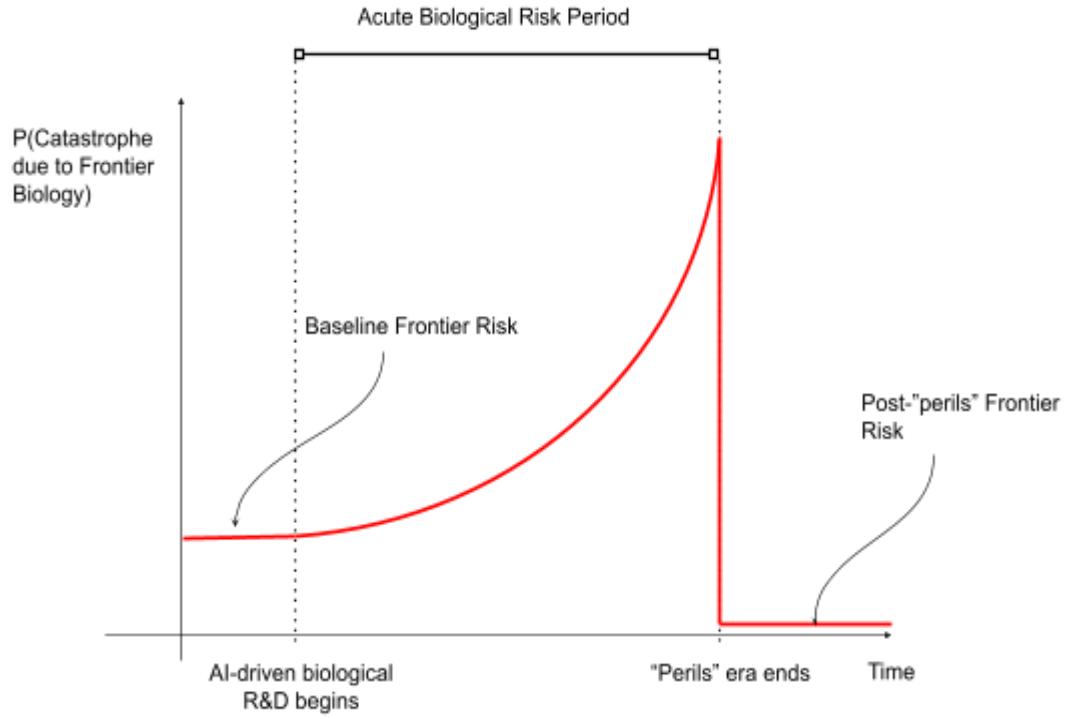


Figure 16: Author's diagram, illustrating a potential "Acute Biological Risk Period (with x-axis representing time and y-axis representing the probability of global catastrophe due to frontier biology):

1. *Currently, we experience some baseline risk from biological research at the frontier*
2. *We expect this risk to increase exponentially, since we expect Life Science R&D growth is correlated with economic growth*
3. *We expect this period to end when the "time of perils" ends, a theoretical point at which it no longer becomes feasible for a biological catastrophe to occur*

With the above assumptions in place, we make a first-pass model of how frontier biological risks will develop with advances in AI (holding any risks from *proliferation* fixed*)*:

- Baseline annual biological risk (from *frontier research)* = ~0.035% per annum[106]
- Baseline economic growth = ~3% per annum.[107]
- Economic growth at "peak AI"[108]= ~100% per annum.[109]

[106] See "*Baseline risk of a Global Biological Catastrophe*", above - we estimated perhaps 5.6% by the end of the century. Approximately half comes from accident risk (2.8%). This suggests an annualized risk of ~0.035% [1 - (1-2.8%)^(1/80)]
[107] GDP growth (annual %)
[108] Where "peak AI" simply refers to the point where AI-driven economic growth reaches its highest level
[109] See outputs of Davidson (2023), which under best guess assumptions leads to economic growth peaking around 100% per annum in 2040 [the specific year is not of relevance here, and can be adjusted based on timelines estimates]

- Biological risk at time of "peak AI" = Baseline biological risk x (Peak economic growth / Baseline economic growth) = 0.035% x (100%/3%) = 1.17% per annum
- Time between start of AI-driven economic growth and technological singularity / end of "time of perils" = ~6 years.[110]
- Integrated risk over period from baseline to peak = **~2.6%**.[111]

This simple model suggests that frontier AI-driven biological advances could pose a total risk of biological catastrophe of **2.6%**, across the acute risk period.

2. **Proliferation Multiplier**: Second, AI could act as a multiplier on the *frequency* of biological threats, as AI systems become more widely accessible. Such accessibility could lead to the mass *proliferation* of AI systems capable of causing tremendous damage - primarily through intentional misuse.

To approximately estimate this multiplier, we can take the following approach:

1. **Estimate the baseline risk of an intentional pandemic.** Previously, we estimated a baseline annualized risk of around 0.035% of a pandemic via an accident-driven engineered pathogen. Following Millet and Snyder-Beattie (2017)'s model, we assume the portion of misuse risk is roughly equivalent.
2. **Estimate the baseline *accessibility* and *intention* of causing a biological catastrophe**. Gopal et al (2023) state that "*funding in the range of $20,000-100,000 will be sufficient for a suitably skilled individual to assemble a known pandemic virus*". Whilst this is within the budget of many, most people do not have the required skill set. Currently, perhaps 30,000 people have the necessary knowledge to engineer pandemic-level pathogens if

[110] Davidson (2023)'s best guess assumptions put the gap from "Wake-up year" to "100% automation" at 5 years, and then states *"My best guess is that we go from AGI (AI that can perform ~100% of cognitive tasks as well as a human professional) to superintelligence (AI that very significantly surpasses humans at ~100% of cognitive tasks) in less than a year."* So for simplicity, we assume about 6 years between "AI that starts to noticeably increase economic growth" and "AI that has exhausted all possible exponential growth".

[111] The Risk curve has approximately the function F(t) = 0.035% *exp(ln(1.17%/0.035%)*t/6). The integral of the curve over acute risk period (over 6 years) is 2.6%

desired.[112] Yet, perhaps (only) 0.04 - 0.32 attempts are made annually.[113] This implies somewhere around 0.00013%- 0.00106% of the population would choose to cause a pandemic *if* given the opportunity.

3. **Estimate the *increase* in accessibility of pandemic-level biological capabilities due to AI**. If AI systems are able to automate - or at least teach - all the necessary steps involved in producing pandemic-grade biological agents, then they could bring knowledge barriers down dramatically. Soice et al (2023) illustrate how non-scientist undergraduate students could complete several steps involved in producing pandemic agents. In future, all steps may be automatable by AI, or at least assisted by AI with humans executing. If AI brings the knowledge barrier down to the undergraduate level (tertiary education) then perhaps up to 40% of the global population would possess the necessary skills.[114]

   However, this likely overestimates realistic capabilities, given that a) a large portion may still be unable to complete *all* the necessary steps with AI assistance, and b) we should expect some mitigations to make accessibility more restrictive.

   Nonetheless, we *very roughly* might guess that ~1 million people will be able to access pandemic-level biological capabilities with the support of AI systems (if they attempted to). This would represent a 33x increase in accessibility compared to the ~30,000 who can today.

4. **Estimate the resulting increase in risk of an intentional pandemic.** Of course, the vast majority of individuals with the *capability* to cause damage will not *intend* to, especially on the global scale. Holding the % of the population with malevolent intentions fixed to ~0.0004%,[115] a 33x increase in access may straightforwardly result in a 33x increase in those *willing **and** able to* cause damage. This inflates our estimate in step (1) from 0.035% per year to a 1.155% probability of global biological catastrophe each year, when access to biologically capable AI systems is at its' peak. We do not expect this

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[112] Esvelt (2022)*: Today, perhaps 30,000 individuals with doctorates currently possess the skills to follow the most straightforward virus assembly protocols.*
See also Esvelt (2023): "*it's probably safe to assume that there's at least four times as many people who can [develop influenza] as the number of PhDs in virology....And we're setting aside students, and master's degree folks, and even talented undergraduates, and technicians who have been working for a long time. Let's just focus on PhDs: 1,500 people a year get PhDs in virology or one of these other disciplines worldwide. And you can do that because you get to basically 125 in virology. Another three times that many gives you 500 a year in the US. The US is about a third of the global total. So you're at 1,500 a year. Assume a 20-year career in which you're reasonably active, and you're at 30,000 people with PhDs.*"
Esvelt later speculates that other diseases (e.g. coronaviruses) are less accessible. Nonetheless influenza gives us a starting estimate.
[113] This comes from Gopal et al (2023) (Appendix 1:D), who estimate around 2 - 16 individuals who have both motivated and trained to cause a catastrophe over the last 50 years. This gives around 0.04 - 0.32 attempts per year
[114] See Primary, secondary and tertiary education enrolment and completion rates, suggesting ~40% enroll in tertiary education
[115] From the geometric mean earlier estimate that around 0.00013%- 0.00106% of the population would choose to cause a pandemic *if* given the opportunity.

level of risk to last indefinitely, however, given the acute risk period explained previously. Instead, we might see the level of risk increase from the baseline to this peak. With a timeline of ~10 years from today to biologically capable AI systems, we therefore estimate a risk level of approximately **3.9%.**[116]

Summary

In summary, research in life sciences is already dual use in nature, and often can be offense-heavy given risks from both frontier developments and the proliferation of existing technology. These facts are likely to remain true as AI systems advance biology. In addition, AI systems may be able to discover destructive capabilities for which there is no known cure for biological humans. However, they may also find defenses which no attacker can overcome.

Figure 17, from Sandbrink (2023), summarizes our dilemma. We are likely to see an increase in *proliferation* (from the diffusion of large language models (LLMs), *and* an increase in *frontier* capabilities (from the development of advanced Biological Design Tools (BDTs), with potentially both simultaneously. The combined effect could move us from a more dangerous world, from (a) to (d).

[116] The Risk curve has approximately the function F(t) = 0.035% *exp(ln(1.17%/0.035%)*t/10). The integral of the curve over acute risk period (over 10 years) is 3.88%

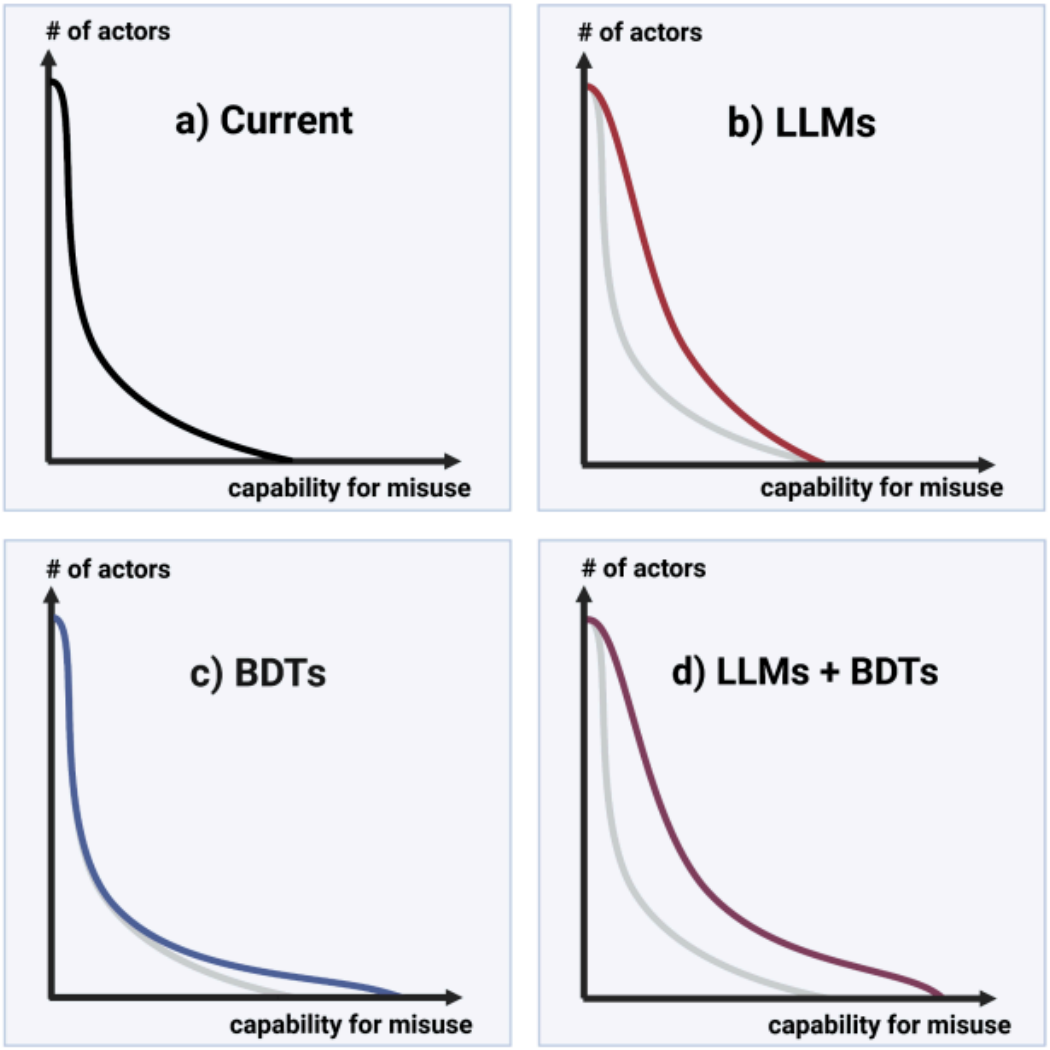


Figure 17: Schematic of effects on LLMs and BDTs on capabilities for biological misuse (Sandbrink (2023))

Overall, we estimate that risk at the *frontier* of life sciences might increase the probability of a biological catastrophe by 2.6%; whilst risks posed by the *proliferation* of biological capabilities caused by AI may pose a risk of around 3.9%. Assuming independence between these estimates,[117] we conclude with a **6.5%** chance of biological catastrophe (conditional on biologically capable AI systems).

[117] Independence seems plausible given that these threat models are two distinct pathways: One for proliferation of *existing* capabilities resulting in misuse, another for frontier developments bringing in *new* capabilities and a higher accident risk. Our estimate may be lower if these scenarios are not independent (i.e. $P(A \cup B) < P(A) + P(B)$). However it may also be higher if (for instance) increased capabilities unleashes new misuse potential when proliferated - as in scenario (d) in Figure 17. We are unsure how these two considerations trade-off against each other, and thus we maintain an estimate of 6.5%.

Claim 3: A Global Biological Catastrophe will constitute an existential catastrophe

Finally, even if it is the case that biologically capable AI systems do substantially increase the risk of a global catastrophe, they may not cause an *existential* catastrophe. Reaching the threshold of an *existential* catastrophe is a high bar. Such an event must *permanently* curtail humanity's potential - either through extinction, dystopia, or a collapse from which we are unable to fully recover.

Previous analysis about the likelihood that a Global Catastrophic Biological Risk (GCBR) will develop into an extinction event has been conducted before:

- A 2008 survey estimated a 10% chance of >1 billion casualties and a 2% risk of human extinction.[118] Thus, naively, the probability of an existential catastrophe *conditional on a GCBR* is at least 20%.[119]
- A Metaculus forecast predicts that, *conditional on a GCBR occuring*,[120] there is an 8% chance of >95% fatalities. Thus, this estimate suggests about an 8% chance of an existential catastrophe, conditional on a GCBR.[121]

Combining these estimates, we conclude with a ~15% chance that a given GCBR event (on the scale of >1 billion casualties) will result in an existential catastrophe.

Turning to how AI affects this risk: AI-induced biological catastrophes may be more likely to lead to *existentially* catastrophic outcomes, since they may be better optimized for eliminating the human population. Still, this is mostly speculative, and this fact is already accounted for in the increased rate of expected GCBR incidents, discussed above. Thus, for simplicity, we stick with the chance that a GCBR event becomes an *existential* catastrophe at 15%

Assessing the threat of biologically-capable AI systems

Summarizing the above, we conclude that:

1. Biologically-capable AI systems have a **~50% chance of being developed by 2030**, and a **90% chance by 2070**. (more)
2. *If* biologically-capable AI systems are feasible, they are likely to increase the absolute probability of a global biological catastrophe by **6.5%**. (more)

[118] Along with a 0.05% probability via a natural pandemic. Extinction risk from natural pandemics drops off significantly faster
[119] It may be higher if e.g. a non-extinction existential catastrophe occurs.
[120] That is, an event where >10% of the population is reduced by 2100 due to biological causes
[121] Whilst losing >95% of the population is not directly an extinction event, it makes humanity much more vulnerable to future extinction events, and it may make collapse unrecoverable, or else set us on a trajectory which is not positive by contemporary society's lights. Thus, I think it *very roughly* makes sense to equate this event to an existential catastrophe

3. *If* a Global Biological Catastrophe occurs, it has a **15%** chance of resulting in an Existential Catastrophe (more).

These numbers combine to give an overall best guess of **0.5%** of an existential catastrophe, by 2030. This increases to **0.9%** by 2070 (assuming no other external shocks). Note, however, that these estimates are *highly* speculative, and they may be easily incorrect by an order of magnitude or more.[122]

Military AI use

For a more comprehensive report on military uses of AI, we recommend Autonomous weapon systems and military artificial intelligence (AI) applications report.[123] *This section follows the structure of that report.*

As well as risks from power-seeking AI systems and an AI-induced biological catastrophe, there is a plausible threat model by which AI systems are developed and deployed for military applications. These risks could be significant enough to constitute a distinct threat model to catastrophe. Whilst there are both specific accident risks from the use of AI in the military, we focus on structural or strategic risks, as we believe these are more concerning with regards to existential risk.

Figure 18 provides an overview of the risk landscape.

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[122] Key uncertainties include:

- Whether biologically-capable AI systems are *plausible*
- When they will be possible from, and how long they are a threat for (until e.g AGI or superintelligent AI is developed and ends the threat of an AI-induced biological catastrophe
- The relationship between advances in AI and risks at the frontier of life sciences (e.g, will AI-induced scientific research increase the frequency of lab leaks?)
- The relationship between how AI systems are diffused (e.g via open source) and the risks from mass public access to dangerous biological capabilities.
- The percentage of people who would *intend* to cause a catastrophe (and how this changes over time)
- The percentage of people who *currently* are capable of causing catastrophe, and how this changes over time.
- How effective defenses are from biological attack or a lab leak.
- How biodefenses will scale with AI systems
- How likely a biological catastrophe is to cause an *extinction* event
- How likely a biological catastrophe is to cause a *non-extinction* existential event, at various thresholds of damage
- Many sub-questions that branch off from these uncertainties, as well as others not included here

This list illustrates how uncertain we are about facts relevant to forecasting the probability of an existential catastrophe from the intersection of AI and biology - and thus why extreme care should be taken when interpreting the bottom-line probability given here.

[123] Some other recommended resources include Maas et al 2022, Guest 2023, Geist and Lohn (2018), and SIPRI (2020).

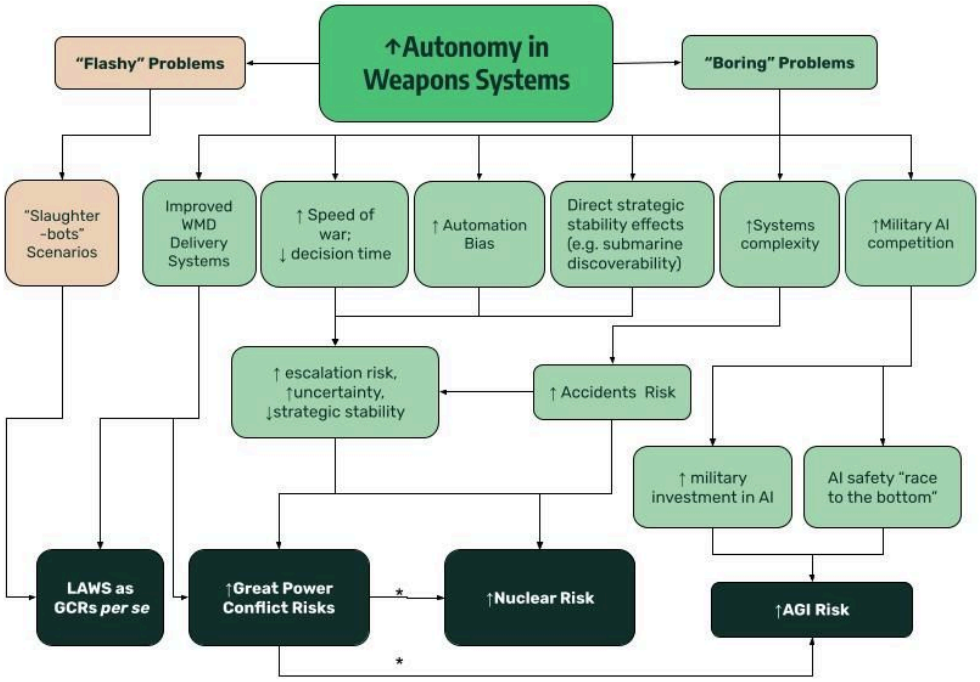


Figure 18: The Autonomous Risk Landscape. Orange = "flashy" problems; light green = "boring" (but neglected) problems; dark green = GCRs and existential threats.

Briefly, some direct risks from the deployment of militarily-capable AI systems includes:

1. **"Slaughter Bot" scenarios**

In this scenario, a swarm of military drones are used in combat to destructive effect. This could enable actors to cheaply cause targeted killings on a major scale, potentially constituting a new weapon of mass destruction (WMD).[124] Whilst horrifying, this scenario is unlikely to constitute a catastrophic or existential risk,[125] and already receives significant attention.[126]

2. **Rapid weapons delivery**

Due to improvements in embodied AI systems, it may be possible to rapidly transfer WMDs. This could enable rogue states and non-state actors to gain easier access to dangerous weapons. This could pose a (small) GCR risk in itself, but most danger from this pathway comes through structural risks via strategic instability (both by adding many potentially dangerous actors, and undermining deterrence in nuclear-armed states). Whilst most breakthroughs in AI have (in recent years) not been through embodied systems, this may change as investment flows into the field.

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[124] See Kallenborn 2020

[125] Maas et al (2022): *"...we argue that the threshold of destruction (> one million human fatalities) necessary for a GCR leaves most, if not all, near-term LAWs unlikely to qualify as GCRs in isolation"*

[126] Ruhl (2022): *"...we categorize slaughter bots as a "flashy" problem — they receive much more media attention and funding than the other risks of this report, possible defenses appear more likely, and the other issues discussed below are more neglected."*

3. **War at "machine speed"**

As well as rapid *weapons delivery*, AI may cause rapid *decision making*. That is, AI systems may be faster at processing information and acting upon it, leaving less time for military decision makers to understand the strategic picture, leading to poorer, riskier decisions. From a catastrophic / existential risk perspective, rapid AI-driven decisions are especially risky in the case of nuclear command.

4. **Biases towards automation**

A fourth risk is the general phenomena for humans to mistakenly over rely on information from AI systems, impairing effective decision-making. Again, this is especially concerning in the case of nuclear decision-making. As an extreme case, the dead hand system is believed to have almost entirely automated the launch of the USSR's Intercontinental Ballistic Missiles (ICBMs).[127] Nonetheless, thankfully there currently appears little willingness amongst nuclear armed states to fully automate Nuclear Command, Control and Communication (NC3).[128]

5. **Strategic Instability**

More structurally, AI could increase strategic instability, for instance through some of the specific pathways outlined above. Such strategic instability could increase the probability of conflict between great powers, and increase the chance of nuclear conflict. Boulanin et al 2020 explores these risks in more detail.

6. **System Complexity**

A similar structural risk is AI making warfare increasingly complex, with the "black box" nature of AI systems making their behavior unpredictable. For instance, a particular type of outer alignment failure could be that AIs are given subtly incorrect objectives on the battlefield, leading to both a direct accident risk and unintentional escalation through misunderstanding on both parties. These kinds of confusion can again increase the chances of war, including nuclear war, as well as worsening relations.

7. **Military AI competition**

Finally, there could be an "AI arms race" between great powers. This could a) reduce the timeline to dangerously capable AI systems, and b) reduce attention on safety. This risk was covered previously under the umbrella of power-seeking AI system failures

Assessing the threat of Military AI

It is challenging to estimate a single probability of an "existential catastrophe" from AI systems in the military, given that the risks are both *direct* (i.e. have a single path to catastrophe) and

[127] Note accounts of the dead hand system are disputed
[128] See SIPRI 2020: "*There seems to be a general agreement among nuclear weapon experts that machine learning and autonomy should not be integrated into nuclear command and control, even if technological developments would permit it.*"

indirect (i.e. risk *factors* for other existential risks). However, we recommend focusing on the *indirect*, *strategic* risks from military AI, over the direct risks.

This is because:

1. It is extremely unlikely that a "slaughter bot" scenario (1), or a WMD incident through faster weapons delivery (2) on its own could lead to an existential risk (and fairly improbable it would pass the bar of "catastrophic risk"). Quoting Mass et al 2022:

   ".. we argue that the threshold of destruction (> one million human fatalities) necessary for a GCR leaves most, if not all, near-term LAWs unlikely to qualify as GCRs in isolation".

   Thus we ascribe very low[129] to an existential catastrophe from this pathway.

2. On the other hand, the use of AI could act as a *multiplier* on the risk of conflict, in particular nuclear conflict (especially through pathways (3 - 6)). Given the large number of nuclear *close calls*, even a small increase in the chance of escalation could increase the likelihood of nuclear war:[130]

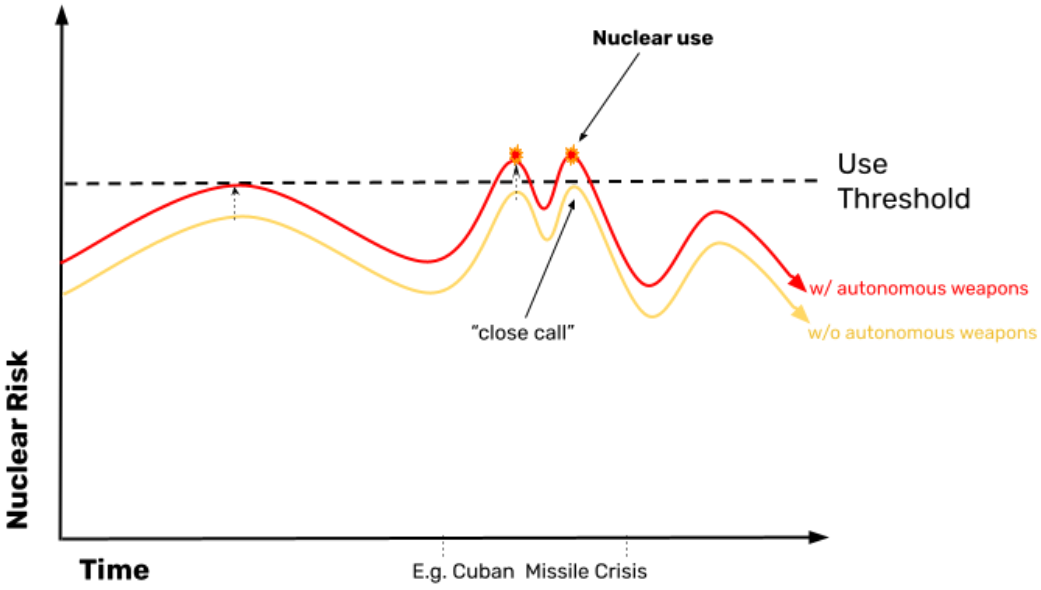


Figure 19: Autonomy and Nuclear "Close Calls"

3. In addition, AI systems could lead to a general degradation in relations between countries (primarily via pathways 5 - 7). This could make coordination challenging on other AI risks discussed previously.

Overall, we remain concerned about the use of AI in the military primarily due to the *structural* risks it poses on other direct pathways to catastrophe.

[129] Less than 0.0001% chance
[130] Figure 3, Ruhl (2022)

Democratic and Distributional Risks

A commonly cited concern about artificial intelligence is its impact on the economy, and the resulting effects on democracy, inequality and society. In this report we focus on *existential* risks from AI, and thus the direct relevance of economic disruption is not *prima facie* apparent. However, looking longer-term, we can see several existential risks or risk factors related to issues of undemocratic control and resource distribution.

Malevolence

First, if AI systems are developed by malevolent actors, they could use their increased power to advance malevolent aims (such as establishing a totalitarian regime (Caplan (2008)), potentially through the suffering of digital minds)

Flawed Lock-In

Second, even if the developers of TAI do not have malevolent intent, they still use power given by AI systems to act against humanity's best interest. For example, they may direct AI systems to personally enrich themselves, or to lock-in their own conception of the "good", without engaging in a process of long reflection.

A severely unequal global economy could allow this value lock-in to occur, just as the "Great Divergence" allowed Western countries to dominate the global economic and political order. Given the high frequency of moral catastrophes in societies throughout history (Williams 2015), it would be *a priori* surprising if the value lock-in of the interests of a few AI companies would be aligned with humanity's best interest. More generally, articulating any given value system that ought to be locked in by an AI system is an extremely difficult task - even when individuals are aiming to act as morally as possible.

Social Disruption

Finally, economic inequality induced by transformative AI could increase social tensions. If workers no longer see meaningful economic returns to labor market participation, whilst a few gain enormous wealth, then social stratification may occur. Such disparities could cause social unrest, political instabilities, and a breakdown in law and order. Negative feedback loops could spiral society to the point of civilizational collapse.

My personal view is that malevolence is relatively unlikely, given the low frequency of bad actors - though the outcome would be perhaps more severe than extinction. Whilst a poor outcome from many perspectives, social disruption also seems unlikely to become catastrophic (since the chance of an *unrecoverable* collapse seems low). A poor value lock-in seems more plausible, given the poor track record of actors trying to act in humanity's best interests based on a set ideology.

Thus, I think there is a high chance that the economic forces from TAI push humanity onto a trajectory in which a small group has control of the far future. Whilst this might not be as bad

as extinction, it is likely to be far short of our potential - and thus still ought to be considered a catastrophe.

Epistemic Insecurity

A further plausible class of threat models is if society moves into a state where AI is used for widespread misinformation, manipulation and persuasion. We define such a state as "*Epistemic Insecurity*" - where the production, distribution, acquisition and assessment of reliable information is threatened.[131]

Seger (2023) outlines considerations for and against epistemic security as a priority area for global catastrophic risk. They focus primarily on how epistemic insecurity could indirectly amplify other catastrophic risks, for instance by causing suboptimal decision making in crises, undermining coordination, or increasing polarization. However, it is also possible that epistemic insecurity could develop into a catastrophic risk of its own, if we enter a state of "epistemic babble" - a world where the ability to tell truth from fiction is entirely lost.[132]

Misinformation today

The issue of misinformation more broadly is already well discussed, and can be seen today. For instance:

- Viral AI-generated images of an explosion at the Pentagon led to a drop in the US stock market (AP News, 2023).
- An edited viral WhatsApp video led to an 8-week period of mob violence resulting in the deaths of at least nine people (BBC News, 2018)
- An AI chatbot partially-motivated a 21-year-old to make an assassination attempt on the Queen (BBC News, 2023)

Ai-induced erosion of epistemic security

Challenges highlighted above may exacerbate in future, as AI develops and diffuses into society. For instance, AI is becoming more adept at persuasive abilities - for instance, beating humans at a game of Diplomacy (Meta 2022), which relies on an understanding of human psychology. If these capabilities continue to advance, it may become trivially easy for AI systems to manipulate humans. This would erode epistemic security. In the limit, AI may be a significant risk factor for several other catastrophic risks. Some vignettes for how this may occur include:[133]

[131] This is the reverse of Epistemic Security, as defined by Seger et al 2020
[132] The term "Epistemic Babble" is introduced by Seger et al 2020
[133] These stories all assume that there is no takeover from a "power-seeking AI". However, we can also imagine that an AI system intending to disempower humans might undermine epistemic security along the pathway to gaining control.

1. China is preparing for an invasion of Taiwan. In order to disrupt the speed of response from the US, they mass produce misleading AI-generated information, providing conflicting and contradictory information for the US intelligence community. Such confusion leads to an unintentional escalation in tensions from the US, which tip over into a "hot war" between China and the US.

2. A new pandemic >10x more deadly and >10x more viral than Covid-19 has begun to spread. However, details on the disease are unclear, with AI-generated articles from conspiracy theorists providing false information. Confusion sows distrust amongst the public about important core facts about the disease, government restrictions, and vaccines. This combination allows the disease to spread much more rapidly and without effective countermeasures.

3. An AI-driven propaganda campaign is launched by an extremist party, backed by a foreign state aiming to undermine democratic processes. The propaganda campaign provides all voters with personalized and well targeted adverts. These adverts are persuasive enough to bring the extremist party to power. The party then dismantles democratic processes, using AI propaganda to undermine opponents. The new non-democratic state uses its military, geopolitical influence and AI propaganda globally, increasing authoritarianism worldwide.

Of course, none of these stories is necessarily *probable* - however, they illustrate different visions of how an epistemically insecure world may be vulnerable to increased global catastrophic risk.

Seger (2023)'s overall view is that, whilst epistemic security has historically increased (due to technological developments in information and communication over the past centuries), this trend may be faltering. Meanwhile, the necessary *threshold level* of epistemic security - that is, *the amount of epistemic security society needs to deal with potential catastrophic risks* - appears to be on the rise. The trendlines of these two factors are illustrated in Figure 20:

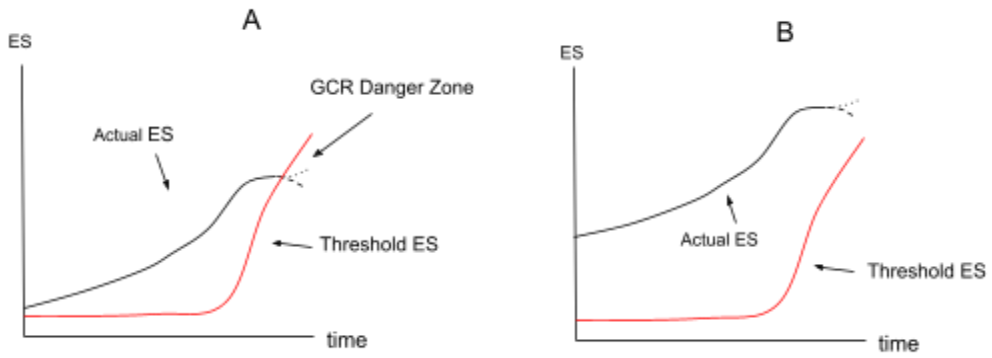


Figure 20: "Actual ES" represents the estimated trend in epistemic security over the past 500 years, whilst "Threshold ES" represents the threshold of necessary epistemic security

Thus, the overarching concern with epistemic security is that we may enter a world where the level of epistemic security *required* [the red line] exceeds the level of epistemic security we *possess* [the black line]. Indeed, this may have already occurred (image A), or be on the horizon (image B).

Risks from Digital Minds

A final threat model may emerge if transformative AI systems lead to the development of (so-called) "digital minds". By "digital mind", we here mean AI systems that have some claim to moral patienthood (or, AI systems which have some form of moral weight).

It is hotly contested when, if ever, we will develop AI systems that have morally-relevant experiences. Furthermore, discussions of "digital consciousness" are philosophically tricky - see Butlin et al (2023) for more. For the purposes of this threat model, we sidestep the difficult questions of consciousness, and instead focus on risks that may emerge *if* AI systems have some moral weight (i.e. if we develop digital minds).

A world of digital minds fraught with several challenges:

1. **Malevolence:** First, it is possible that malevolent actors (Althaus and Baumann, 2020) intentionally mistreat digital minds, causing significant suffering. Given that such minds will be built on non-biological substrates, the capacities for pleasure and pain may be far larger than the capacities in humans, and thus, the potential suffering caused to digital minds may be far worse than the worst suffering experienced by humans today. Such suffering could be "astronomical" in scale (in both intensity and frequency), constituting a suffering risk ("S-risk") (Daniel, 2017)

2. **Conflict:** Given the possibility of digital minds making up the vast majority of future morally-relevant experiences, digital minds may also be involved in conflict between AI systems. This conflict could also cause tremendous suffering (Clifton 2021), and thus constitute an "S-risk"

3. **Ignorance:** Even well-intentioned actors may be skeptical, uncertain, or otherwise ignore the existence, welfare or preferences of future digital minds. This attitude could result in the accidental mistreatment of digital minds (e.g. frustrating the preferences of digital minds, considered morally negative under a preference-utilitarian perspective); alternatively, ignorance of digital minds could lead to a universe of AI systems without moral value. Thus, much of humanity's potential will have been lost, constituting an existential catastrophe.

Further ethical challenges for a world of digital minds are posed by Bostrom and Shulman (2020a) and[(2020b)

Our overall view on this topic is highly uncertain. Nonetheless we recognize the potentially enormous implications of digital minds for the long-term future. If human civilization expands into the cosmos, it's likely the case that we will expand through digital, rather than biological

means.[134] Combined with the very little attention the issue of digital minds receives, it is worth inclusion and consideration as a distinct threat model.

Section 3: Interventions to reduce existential threats from Transformative AI

For each of the threat models described above, we list and categorize several AI risk interventions.[135] Note, this section does not evaluate the pros and cons of each intervention - thus, an intervention's inclusion should not be read as an endorsement.

Power seeking AI

As outlined previously, Carlsmith (2021) breaks down the probability of an existential catastrophe from power-seeking AI into six steps. Reversing these premises gives us several intervention strategies:

Claim	Intervention
Timelines: It will be *possible* and *financially feasible* to build capable AIs by 2070	**Slow down**: Make it *harder* to build capable AIs by 2070
Incentives: There will be *incentives* to build capable AIs	**Disincentivize**: Reduce *incentives* to build capable AIs
Misalignment: It will be hard to build *aligned* AIs	**Align**: Make it *easier* to build aligned AIs
Poor Deployment: Misaligned AIs will be *deployed*	**Managed Deployment**: Deploy aligned AIs *before* misaligned AIs
Disempowerment: Misaligned deployed AIs will *disempower* humanity	**Deployment Corrections**: Ensure that (misaligned) deployed AIs do not disempower humanity

[134] My intuition is that digital beings are likely to dominate biological over the long-term, given that:
1. Digital minds can plausibly travel much faster than biological, and over much longer distances
2. Digital minds will likely require less energy and resources per being.

See Karnofsky (2023) for more.

[135] Though there are many cases where my categorization may be debatable. For instance, some interventions cut across multiple threat models

Catastrophe: Disempowerment would be an *existential catastrophe*	**Compromise:** Make sure disempowerment does not necessarily entail *catastrophe*[136]

Increase costs to building frontier AI systems

This strategy involves slowing the *construction* of frontier AI systems - for instance by raising the financial cost of doing so (for all actors). A simplified argument for this strategy is that without powerful AI systems being a possibility, there can be no dangerous AI.

Computing power, data, and algorithmic insights make up the "AI Triad", the three inputs responsible for increasing AI capabilities in recent years (Buchanan 2020). If increasing these inputs are necessary for the construction of transformative AI, then it follows that constraints on these factors will delay the time to TAI.

Thus, three pathways to slowing AI are:

1. Compute restrictions

If compute is a constraining factor for the construction of transformative AI systems (as assumed by, for instance, Cotra 2020), then higher hardware costs will mechanically extend the timeline to TAI. For example, global export controls on the semiconductor supply chain could increase the financial cost of training frontier systems.

2. Data restrictions

A high quantity of (high quality) data is necessary for training AI systems, and could be one reason for a slowdown in capabilities (Villabos et al 2022). Legal limitations on data access (such as privacy laws) may increase this bottleneck, as well as having possible near-term positive externalities for consumer protections.

3. Algorithmic restrictions.

Since human labor is (currently) a key driver of algorithmic improvements, restrictions on labor hours spent in machine learning could slow down the rate of progress. Immigration restrictions on ML research could be one such lever.

Reduce incentives to build frontier AI systems

Instead of making it infeasible to build frontier AIs, we might consider *disincentivizing* actors from *trying* to build powerful AI (even if possible). Some pathways for doing this includes:

[136] We merge "compromise" into the intervention class of "Deployment Corrections", since we do not believe there are enough interventions distinct from corrective measures to warrant a separate category

1. AI lab Advocacy

It may be possible to persuade individuals or organizations not to build frontier AI systems. Thus one strategy might be to directly discuss the risks of building AI systems with those in the field.

2. Public / Policy Advocacy

One approach could involve shifting public opinion against the construction of frontier AI. This would increase the social stigma of working on advancing AI capabilities, especially on the riskiest components. For instance, events such as statements from the Future of Life Institute and Center for AI Safety may have shifted the Overton window on AI discourse, reducing incentives for building frontier AI. These can translate into clear policy actions, such as establishing legal liability for AI labs when accidents and misuse caused by AI products occur.

3. Development of useful non-frontier applications of AI.

A major incentive for developing frontier AI systems is the usefulness they provide, in terms of economic and social benefit (as vividly illustrated by Piper 2023). However, these same frontier systems may ultimately lead to catastrophe. Therefore, one strategy might be to "have our cake and eat it" - to reap the same rewards that frontier AI systems would, without advancing capabilities. If successful, this would reduce the incentive for building riskier models.

Further Reading: Stein-Perlman (2023); Larsen et al (2022a; 2022b); Grace (2022)

Develop techniques for aligning powerful AI systems

At its core, the risk from power-seeking AI is that AI systems may be misaligned with the objectives originally intended. To tackle this problem, several intervention classes include:

1. Ensuring AI systems are outer aligned (more)
2. Ensuring AI systems are inner aligned (more)
3. Develop a deeper understanding of AI systems (more)
4. Develop foundational knowledge relevant to building smarter-than-human systems (more)

1. Ensure AI systems are outer aligned

There are numerous proposals for specifying goals that align with our intentions, which would help us to achieve outer alignment.

- Reinforcement Learning from Human Feedback (RLHF) is a commonly used technique for aligning systems like language models to the user's intention. However, this technique alone will not be feasible for transformative AI systems - and thus additional approaches (discussed below) may be necessary.

- Approaches like Constitutional AI or "Reinforcement Learning from *AI* Feedback ("RLAIF") may help us to scale current RLHF methods for more powerful systems, where handwritten human feedback is not feasible.
- More speculatively, proposals for "scalable oversight"[137] such as Iterated Distillation and Amplification (2018) and Debate (2018) may help for problems that humans are not able to give effective feedback on. Scaleable oversight forms a core part of OpenAI's plan for "Superalignment" (OpenAI 2023).

2. Ensure AI systems are inner aligned

Currently, there are few proposals that directly aim to ensure that objective AI systems *learn* the same objective that we *specified.* Nonetheless, a popular mitigation strategy is **adversarial training** (2018), which aims to test how AI systems behave in environments outside of standard training.

If we are able to observe that AI systems pursue the specified objective *in standard training*, but then pursue some other, learned objective *outside training*, then this would tell us that the two objectives are misaligned.

Concretely, we can combat this by training AI systems across a wide range of environments. With enough training, we might be able to ensure that the specified and learned objectives are the same across *all* environments they will encounter "in the wild"

However, this technique may not guard against (sufficiently advanced) deceptive systems, which may still be able to detect the difference between a training and non-training environment (and thus, will act *as if* they are following the specified objective, but then later pursue a different learned objective in the real world).

3. Understand existing AI systems

Given a lack of shovel-ready approaches to *inner* aligning AI systems, an alternative pathway is to first understand how existing AI systems work, with the intention of later using this knowledge for aligning future systems. This research direction is commonly referred to as (mechanistic) interpretability (Olah 2022), or explainability. This line of research focuses on trying to reduce the "black box" nature of ML systems. Strong interpretability may help us to detect when an AI is potentially dangerous (such as when it is capable of deception), as well as helping us move towards a world where only "provably safe" systems can be deployed.

[137] That is, aligning systems when AI systems are too intelligent for humans to understand and give reliable feedback on (Concrete problems in AI safety, Section 5).

4. Develop foundational knowledge relevant to future systems

A third pathway takes a more theoretical approach to how smarter-than-human AI systems may behave. This approach aims to give more conceptual clarity to key ideas like agency (Wentworth 2022).

Deploy aligned AIs before misaligned AIs

We may make technical progress to make AI systems easier to align. However this does not imply that aligned systems will be the first systems to actually be deployed in the real world. Thus we need to find interventions to ensure that misaligned AI systems are not deployed. Following Karnofsky (2022) ; (2023), I list some classes of interventions which may help to solve this problem.

1. Alignment Research

Research to make it easier to align AI systems was covered in the previous section. Investment in alignment research can also help to solve the deployment problem, since it can make it easier for all actors (not just the most safety-conscious) to deploy aligned rather than misaligned systems

2. Standardized, legally enforced model evaluations

A cluster of interventions exists around the topic of "model evaluations". Broadly, this approach aims to evaluate the capabilities and safety mechanisms present in AI systems. For instance, the Alignment Research Center conducted evaluations of OpenAI's GPT-4 and Anthropic's Claude (ARC 2023). These included tests on whether the system is capable of self-replication, manipulating users, or gaining power. OpenAI and Anthropic voluntarily opted into these evaluations, however, in future these may become part of a standardized regulatory regime for the deployment of frontier AI systems. Shevlane et al (2023) provides a sketch for how such evaluations may become embedded into standard governance, to guard against both misalignment and other risks with dangerous capabilities (such as misuse).

3. Tightened information security, to prevent the leaking of potentially unsafe models

One commonly neglected intervention type is to prevent information about powerful (and potentially) dangerous AI systems from leaking - perhaps due to inertia, carelessness, or espionage. Given the potentially enormous commercial value in deploying powerful systems, we should expect strong incentives for some actors to try to steal an ML model's weights, rather than training the weights (through an expensive training process) themselves.

4. Improve coordination between relevant actors.

One overarching strategy is to facilitate better coordination between AI labs and/or countries, such that none has the incentive to "race ahead" and build more powerful AIs. There are a plethora of possible interventions under this approach - some include international AI governance bodies (for instance, an "IAEA for AI" (Altman 2023)); voluntary AI safety agreements (for example White House 2023); verification techniques (see Mittelsteadt 2021); and summits (for example Gov.UK 2023).

5. Prevent other actors from deploying misaligned AI systems

Even after the successful deployment of an intent aligned system, it may be the case that less cautious actors will later develop unaligned systems. Thus the first actor who builds an aligned AI will need to make sure unaligned systems are not later developed. Generating specific interventions may not be necessary at this stage, since an aligned AI system may be able to assist with this task. Nevertheless, setting norms early about what actors ought to do if they find themselves in this position may be valuable.[138]

Post-Deployment Corrections

In this section, we consider interventions which may prevent an existential catastrophe from occurring, *conditional on* an advanced misaligned, power-seeking AI system already deployed in the world.

Suppose that, despite some of the previously-described interventions being introduced, misaligned, power-seeking AI have been deployed. What countermeasures could humanity take to prevent catastrophe?

Some have the intuition that any effort in this direction will be ineffective. After all, such a system will (perhaps by definition) be able to outperform humans on all tasks, and thus takeover will be trivial for the AI. Others intuit that preventing AI takeover will be easy, as we can just "pull the plug".

To help assess which story is more likely true, we sketch out a graph summarizing how an AI systems' takeover capability changes over time:

[138] See for instance, discussion around "pivotal acts" (Critch 2022)

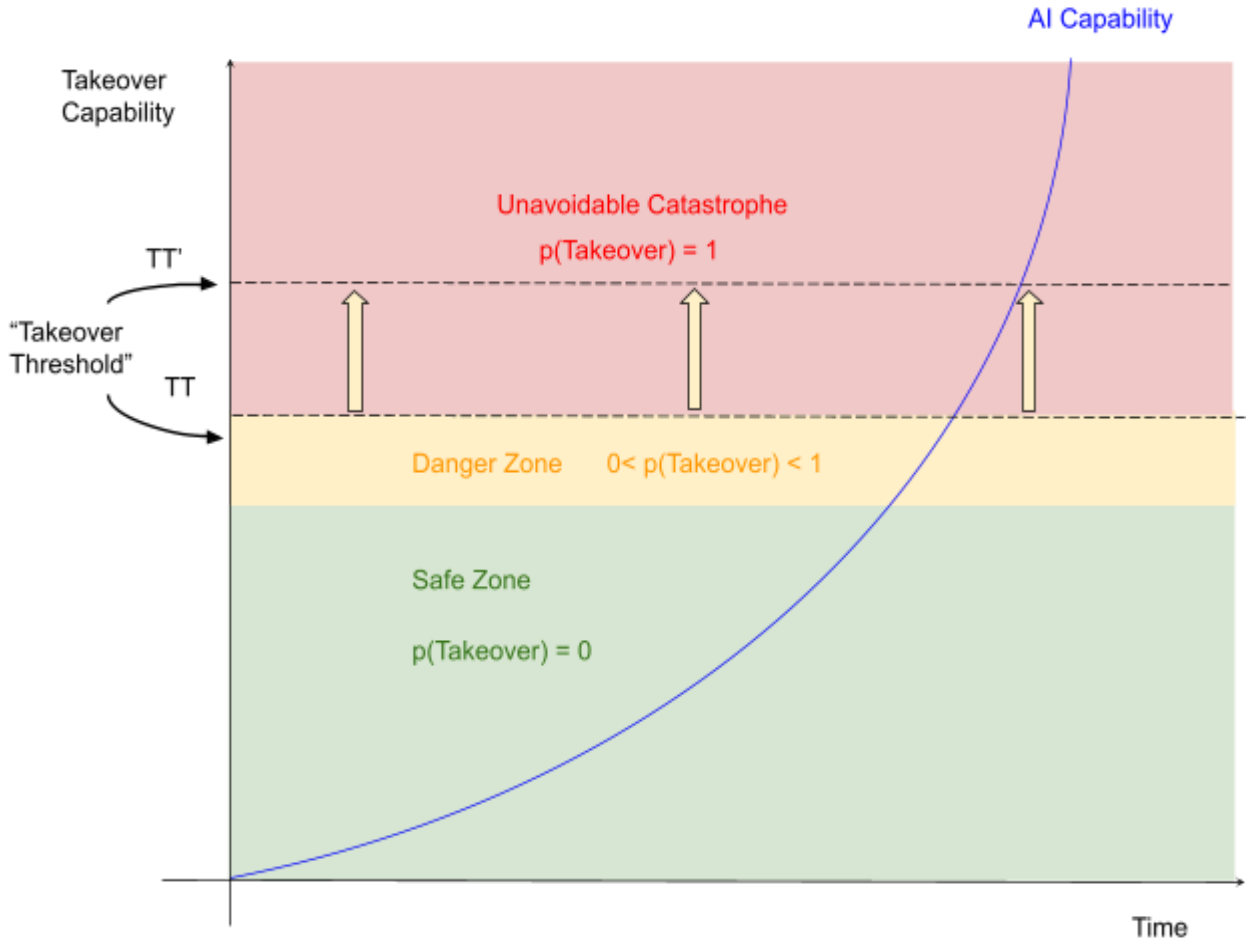


Figure 21: Illustration of the concept of "Takeover Threshold", with time on the x-axis, and the takeover capabilities of frontier AI systems on the y-axis. The blue line represents how takeover-capabilities may increase (exponentially) over time. This will mean passing through the "safe zone" (where takeover is impossible), through the "Danger zone" (where takeover could occur) to "Unavoidable catastrophe" (where takeover is inevitable). "Takeover threshold" represents the dividing line between danger zone and unavoidable catastrophe. Interventions can increase this threshold.

Currently, we are in the "safe zone": At present, an AI system with the goal of disempowering humanity would fail. However, we expect the capabilities of AI systems to improve over time - and these capability improvements include the ability to takeover humanity. We may later reach the "danger zone" - a point where AI systems *may* be able takeover, but cannot be assured of success, since there could be sufficient countermeasures available to prevent full takeover.

However, at some point, our countermeasures may become ineffective, as AI systems far outstrip human abilities. At this point, disempowerment and catastrophe are all but inevitable, assuming that the AI system is aiming to do so (i.e. assuming it is misaligned). We can label the point where catastrophe becomes inevitable as the **takeover threshold**. That is, the level of capabilities required to guarantee a successful takeover.

The concept of a *takeover threshold* gives rise to a set of interventions, which can all be broadly defined as *increasing the threshold that AI systems must pay in order to disempower humanity* (conditional on already being misaligned and deployed). Increasing this threshold moves us from point TT to TT'.

Below are some examples of potential interventions that could make takeover harder. Many are taken from O'Brien et al (2023), which outlines various "Deployment corrections" which could be employed in the event of post-deployment challenges.

Develop robust shutdown mechanisms

First, a fairly intuitive approach is to simply "pull the plug" on any system trying to takeover. Shutdown mechanisms are common in other high stakes domains such as nuclear energy (O'Brien et al (2023), p14). In practice, shutdown mechanisms might include:

- Removing AI products from the market
- Disconnecting power from the data center / cluster where the AI system is being run from
- Decommissioning the AI model, for instance by destroying associated data or assets.
- Small-scale targeted Electromagnetic Pulse (EMP) attacks on uncontrollable AI systems.

Harden Human Defenses

Another strategy might be to identify human (rather than physical) weaknesses, and aim to patch those. For instance, humans have historically been susceptible to bribery, blackmail, manipulation and coercion - weak points which a sufficiently advanced system may be able to exploit for ulterior objectives. Thus, hardening those weaknesses could be a key choke point for AI systems. Davidson (2023a) explores this approach, suggesting that we could use existing AI systems to harden our defenses against more advanced systems.

Restrict the use of AI from high-stakes domains

O'Brien et al (2023) (p12) suggest prohibiting the use of AI systems in high stakes domains. This could involve setting a use policy that restricts the model from being used in high-stakes applications, and allows banning or otherwise penalizing users that breach that policy. Those high stakes domains could include preventing AI use in critical defense and infrastructure systems.

Implement mechanisms for detecting AI takeover

A pre-emptive measure could involve developing mechanisms for detecting whether a takeover attempt is taking place, such that we have (some form of) advanced warning system.

Davidson (2023b) proposes a simple model in which humans check AI outputs in order to detect whether AI systems cause / intend to cause sabotage.

Incentivize AI systems to cooperate with humans

Finally, a more speculative proposal may be to incentivize AI systems to cooperate with human values, even in the case where they are willing & able to disempower humans. For instance, if we can guarantee that we will allocate some resources to the goals of (misaligned) AI systems, they may accept this settlement over a costly attempt at wresting full control away from humans.

AI-induced Biological Catastrophe

In this section we explore potential intervention strategies to reduce the threat from an AI-induced Biological Catastrophe

Reduce biological capabilities within AI models

A first technique might be to simply reduce the biological capabilities and applications of AI systems. For instance, Soice et al 2023 highlights that if Large Language Models (LLMs) were simply not trained using biological information, then they would not be able to provide malevolent actors the ability to cause harm. Optimistically, the authors state that "*removing under 1% of all publications in PubMed – and a far smaller percentage of all scientific research – would suffice to eliminate nearly all of the risk [from LLMs].*" If true, then (at least in the short term) it should be fairly tractable to reduce this risk. However, this may not be a long term solution, if open source models are trained on such data and then deployed.

Evaluate models for biological capabilities before deployment

Another promising approach to regulating AI models might be the introduction of standardized evaluations. Evaluations were previously described as a potential mitigation for misaligned AI. However, a similar regulatory regime could be introduced for biologically capable AIs - regulating both general purpose, highly diffused Large Language Models (LLMs) and more specialized, advanced Biological Design Tools (BDTs). One proposal might be the introduction of an independent regulator, charged with auditing the biological capabilities of AI models.

Restricting access to biologically capable AIs

Rather than just evaluating AI systems before deployment, another intervention might be to restrict general access to biologically capable AIs outright. Sandbrink (2023) proposes that AI companies be mandated to only provide access to powerful AI systems to a group of trusted scientists, rather than public access which comes at the risk of allowing bad actors to use such tools. This could be an effective intervention in cases where model evaluations are unable to prevent the proliferation of dangerous models.

Restrict access to physical biological agents

Another mitigation strategy to reduce the AI-Bio threat could be to increase defensive measures at the boundary of the digital and the physical. For instance, Sandbrink (2023) suggests that restricting access to synthetic DNA could prevent the translation of biological designs into physical agents. To restrict such access, a requirement to screen all gene synthesis orders (and other synthetic biology) could be introduced.

Invest in threat-agnostic bio defensive measures

Finally, there are numerous existing interventions already proposed to reduce the chance of a biological catastrophe, which would act on all biological risks, including those posed by AI specifically. For instance, early-warning systems, personal protective equipment, transmission-blocking interventions and medical countermeasures can all help to reduce the threat of an (AI-induced) biological catastrophe.[139]

Military AI Use

Please note: These recommendations are taken from our report on Autonomous weapon systems and military artificial intelligence (AI) applications

Given the nature of the threat from military applications of AI, we generally prioritize interventions which tackle *structural* or *strategic* risks from military AI, rather than *direct risks* such as the risk posed by "killer robots". We also prioritize interventions which engage with the most important stakeholders, such as the United States and China. This gives rise to recommending *Field building on Strategic Risks* and *Confidence Building Measures*.

Field building on Strategic Risks

The field of researchers who study issues related to military AI risks is small. For example, we identified 53 think tanks and other organizations whose work is plausibly relevant to autonomous weapons research, but only 9 who work actively on strategic risks from autonomy. We believe the neglectedness of research on strategic risks could therefore be addressed especially well through field-building interventions that foster expertise and future researchers.

When possible, therefore we would recommend field-building interventions, such as starting new organizations focused on military risks from emerging technology, as a valuable type of intervention. However, such interventions can be challenging for individual donors, as we explained in our Great Power Conflict report: "field-building requires large amounts of capital and long-term commitments, which means it is likely not the optimal intervention for individual donors to consider in this space (Clare 2021).

[139] These interventions are explored further in Ruhl (2023)

Confidence Building Measures

Recent work on regulating autonomous weapon systems shows that confidence-building measures (or CBMs) are a promising intervention for reducing military AI usage (CNAS 2021; Horowitz et al 2020). The UN Office of Disarmament Affairs defines CBMs as "planned procedures to prevent hostilities, to avert escalation, to reduce military tension, and to build mutual trust between countries" (UNODA).

For instance, SIPRI (2020) concludes, "*CBMs provide a useful compromise [between unilateral measures and hard-to-negotiate international law]. They would not need to be legally binding [...], but they would be easier to reach agreement on.*" The importance of confidence-building and transparency for autonomous weapons regulation was also discussed in a recent roadmapping exercise on autonomous weapons regulation (Arkin et al 2019). Unlike formal arms control, which legally prohibits the development or use of certain kinds of weapons, confidence-building measures are agreements designed to reduce risk in a way that all parties consider to be in their interest, often through the tools of "communication, constraint, transparency, and verification." (Stimson 2012).

In practice, we already have several case studies in which CBMs have been implemented, reducing risks of military AI escalation:

- The Helsinki Final Act's CBMs included measures such as "voluntary reporting with at least a 21-day prior notification of military maneuvers that would exceed over 25,000 troops and that would occur within 250 kilometers from a state's border," designed to reduce uncertainty and the potential for unintended escalation (Borhard and Lonergan 2018).
- The "nuclear hotline" established between the U.S. and U.S.S.R. after the Cuban Missile crisis exemplifies one such CBM, designed to share information, communicate during crises, and avoid misunderstandings and unintended escalation (Horowitz et al 2020).
- Most recently and of direct relevance, the United States and China mutually agreed to limit the use of AI systems in autonomous and nuclear weapons (SCMP, 2023). Whilst details of this agreement are currently unclear, they show a clear sign of confidence-building within the AI-military domain.

We have previously recommended another well-known kind of confidence-building measure — track II dialogues — to address risks from great power conflict. Thus, confidence- building measures can tackle multiple catastrophic risks simultaneously.

Democratic and Distributional Risks

Facilitate democratic inputs in the use of AI

One approach to prevent the concentration of political power over AI is to ensure that consequential decisions are made by a broad, diverse citizenry.[140] This can help to ensure that AI systems are aligned with humanity's "best interests", rather than defined narrowly by a small elite. Tentative proposals in this direction have been announced by OpenAI (2023) and Anthropic (2023) with organizations like the *Collective Intelligence Project* recently proposing a White Paper (2023) outlining a vision for collective decision making in the context of AGI.

Distribute the economic benefits of AI

Another concern around AI is the concentration of *economic* power, with the prospect of a small number of companies contributing to a significant portion of Global World Product. One idea to reduce this economic power is a "Windfall Clause" (2020). This paper proposes that AI companies pre-commit to donate a meaningful portion of their profits, if they see a "windfall" increase in profits. Funds from a windfall could pay for programs like a "Universal Basic Income", to ensure that workers who lose their jobs to AI do not suffer economic hardship. A Windfall clause may also help with additional issues related to TAI, such as preventing a race-to-the-bottom competition between AI companies pursuing large profits (Bova et al 2021).

Epistemic Insecurity

In the epistemic insecurity threat model, we introduced two concepts

(1) The level of epistemic security *possessed*, and

(2) The level of epistemic security *required* (the "Epistemic Security Threshold").

Possessing epistemic security means the production, flow and use of accurate information is secure; whilst the *threshold* for epistemic security refers to how much accurate information is needed to prevent GCRs. AI may directly undermine the former (through increases in misinformation), whilst increasing the latter (via raising the level of catastrophic risk posed more generally).

Increasing Epistemic Security

Proposals for improving epistemic security come from Seger et al (2020). These include:

- **Increasing the costs of spreading false information** - for instance imposing penalties for disseminating knowingly false information.

[140] Although democratization can also increase other risks - for instance, open-sourcing access to powerful AI systems likely increases the probability of misuse (all else equal)

- **Improving methods for identifying true information** - for example, developing and asymmetrically signal-boosting platforms that provide accurate information
- **Monitoring and responding to changes in the epistemic ecosystem** - for instance during times of crisis where true information is scarce
- **Building capacity for identifying and analyzing epistemic threats**
- **Establishing a group of experts well positioned to provide epistemic security to key actors (e.g. governments) during crises**
- **Developing a multidisciplinary field focused on epistemic security**

Lowering the Epistemic Security Threshold

These proposals originate from Seger (2023):

- **Restrict the flow of information and decision-making power to a small number**, thereby ensuring that all relevant decision makers are well informed and well coordinated. However, such a proposal would be highly anti-democratic, posing other epistemic risks (among other risks). Thus the author eschews this proposal.
- **Reduce the severity, prevalence and complexity of GCRs**. Most apparently, if we can directly reduce the likelihood of global catastrophe risks (in general), then the size of the multiplier from epistemic insecurity will be smaller.

However, it should be noted that the field of "epistemic security" is a small one, and thus the intervention space has not yet been well mapped out. Thus, other promising interventions may exist.

Digital Minds

Foundational Research

Given huge uncertainties about whether morally-relevant digital beings will be created, and if so, under what timeframe. Indeed, the very definition of a "digital mind" is not yet clear. Nevertheless, given the potentially huge ramifications on the far future, combined with the urgency with which AI systems are advancing, it seems prudent to invest in foundational research - primarily in philosophy and macro strategy - to figure out what digital minds are.

Field Building

Even compared to a neglected field like AI safety, work on digital minds is extremely small. Indeed, there are only a couple of organizations whose work touches on this issue.[141] Thus, a second strategy might be to increase the number of people working on this issue.

Advocacy

As mentioned, right now we are highly uncertain about the status of digital minds. Nevertheless, there may come a point where concrete recommendations emerge, for how AI labs, governments and other actors ought to handle the issue of digital minds. At such a point, it may become necessary for actors to advocate for the rights and interests of digital minds, just as advocacy has been beneficial for representing the interests of other moral patients (like non-human animals).

Summary of interventions

Overall, we can see that there are a huge variety of potential interventions that philanthropists could pursue in order to reduce the many risks from transformative AI. Those listed above are *far* from comprehensive, and often do not dive into the details of specific proposals. Nonetheless, the preceding section illustrates that **there are many actions philanthropists can take** if they are concerned about the risks of TAI. In the next section, we explore how resources are currently being allocated between various threats and intervention strategies.

Section 4: Funding to reduce risks from Transformative AI

In this section we consider how resources aiming to reduce existential risks from AI are distributed.

Overview of AI Risk Funding

1. **Philanthropic funding to tackle existential risks from AI has grown, reaching approximately $110 million per year**

McAleese (2023) details how the philanthropic funding landscape for AI safety has changed over the past eight years. They estimate that AI safety received around $110m in 2021, $140m in 2022,[142] and likely about $110m in 2023. These figures are much larger than previous years - for instance, Farquhar (2017) gives an estimate of $9m in 2017; whilst Todd (2020) estimates $40m in 2019. This growth - though turbulent - can be seen in Figure 22.

[141] The Sentience Institute and FHI (Digital Minds Research Group) does relevant work on this issue. I would guess there are less than ten people working directly on digital sentience (though work on related academic areas is lager - see e.g. the Journal of Artificial Intelligence and Consciousness),
[142] Closer to $110m when excluding $32.5m from the FTX Future Fund

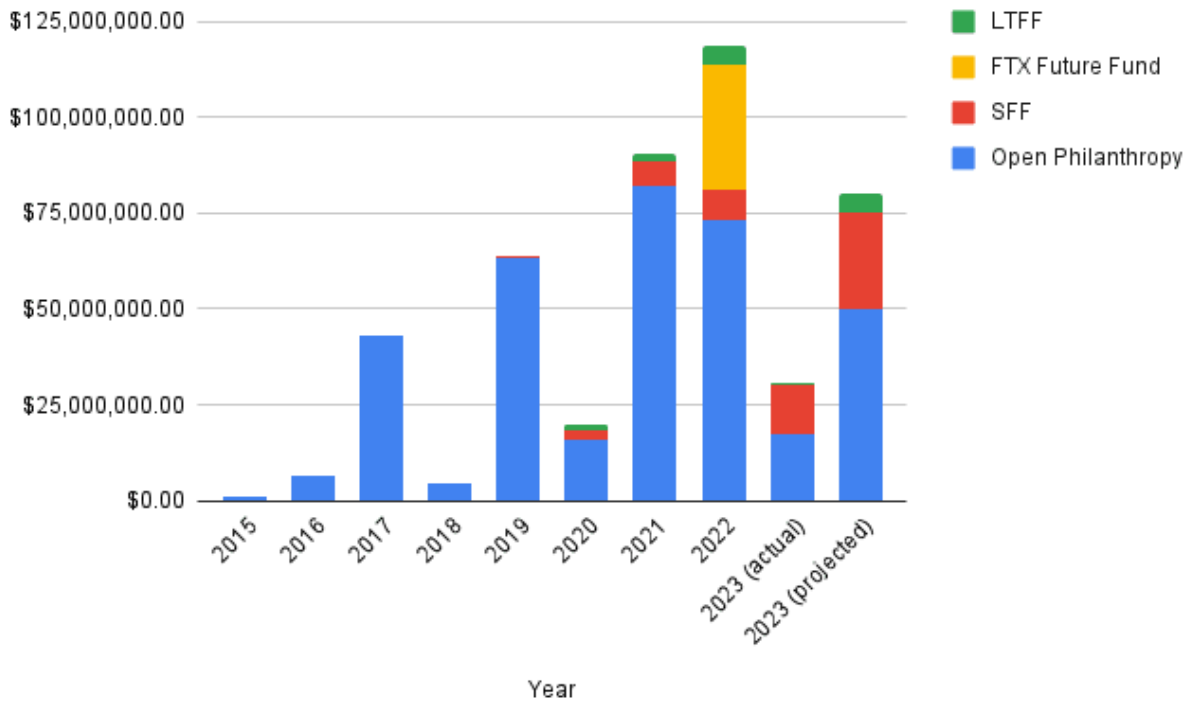


Figure 22: Data from McAleese (2023) on changes in AI safety funding from philanthropic sources

For comparison, philanthropic funding towards nuclear security is estimated at around $30m (Ruhl 2023; Toma 2022); around $50m-$100m is spent on catastrophic biological risks (Ruhl 2023); whilst an estimated $3 billion was spent on climate philanthropy in 2021 (Climateworks 2022)

Risk	Philanthropic Funding / year
Artificial Intelligence	$110 m
Biological Catastrophe	$50 - $100m
Climate Change	$3 bn
Nuclear War	$30 m

2. The vast majority of philanthropic funding originates from a couple of sources

Almost all philanthropic funding aiming to tackle risks from *transformative* AI has come from either Open Philanthropy or the Survival and Flourishing Fund (SFF). Open Philanthropy has

donated over $330m towards AI since 2015,[143] whilst the SFF has donated about $30 million since 2019. Additional sources of funding include the Long-Term Future Fund at EA Funds and Lightspeed Grants - however, most of these funds originate from Open Philanthropy and SFF.[144] Taking stock, we roughly estimate that at least 80% of philanthropy tackling risks from TAI comes from these two sources.

3. **For every $1 invested in making AI systems safe, ~$250 is invested in making AI systems more capable**

Whilst ~$100m is spent by philanthropists to reduce risks from advanced AI, investment in AI amounted to around $60 billion in 2023.[145] Around $24 billion has been invested in generative AI, the key driver of frontier AI developments.[146] This gives a crude ratio of capabilities : safety investment of ~240:1.

Another way to estimate the ratio of capabilities: safety is to look at labor, rather than capital. Approximately 400 people work full time on either technical AI safety or governance (McAleese 2022). Meanwhile, about 30,000 people attended ICML in 2021, the largest conference in Machine Learning (Figure 1.2.2, AI Index Report, 2022). This likely undercounts all people working on Machine Learning - Aschenbrenner (2023) suggests the number is probably closer to 100,000.[147] This gives a crude ratio of 300:1.

Summing this up, we conservatively estimate that the ratio of capabilities: safety is at least 100:1 (as a lower bound), with a best guess of 250:1. Figure 23 gives an illustration of this point.

[143] As of October 2023 - see Potential Risks from Advanced Artificial Intelligence | Open Philanthropy
[144] From Lightspeed: *"Our primary funder for this round is Jaan Tallinn [founder of SFF]"*. ~50% of LTFF's funding in 2022 came from Open Philanthropy or other institutional donors (mainly SFF) - source
[145] $45 billion invested in AI as of September 29 2023 (State of AI Report 2023, Slide 109). Assume an extra $15 billion for the final quarter of 2023.
[146] State of AI Report 2023, Slide 116. Again, add an extra 25% for the final quarter of 2023.
[147] Although most ML researchers are not advancing the *frontier* of AI systems - however, they are likely increasing the *capabilities* of systems, when broadly defined to include e.g. applications of ML.

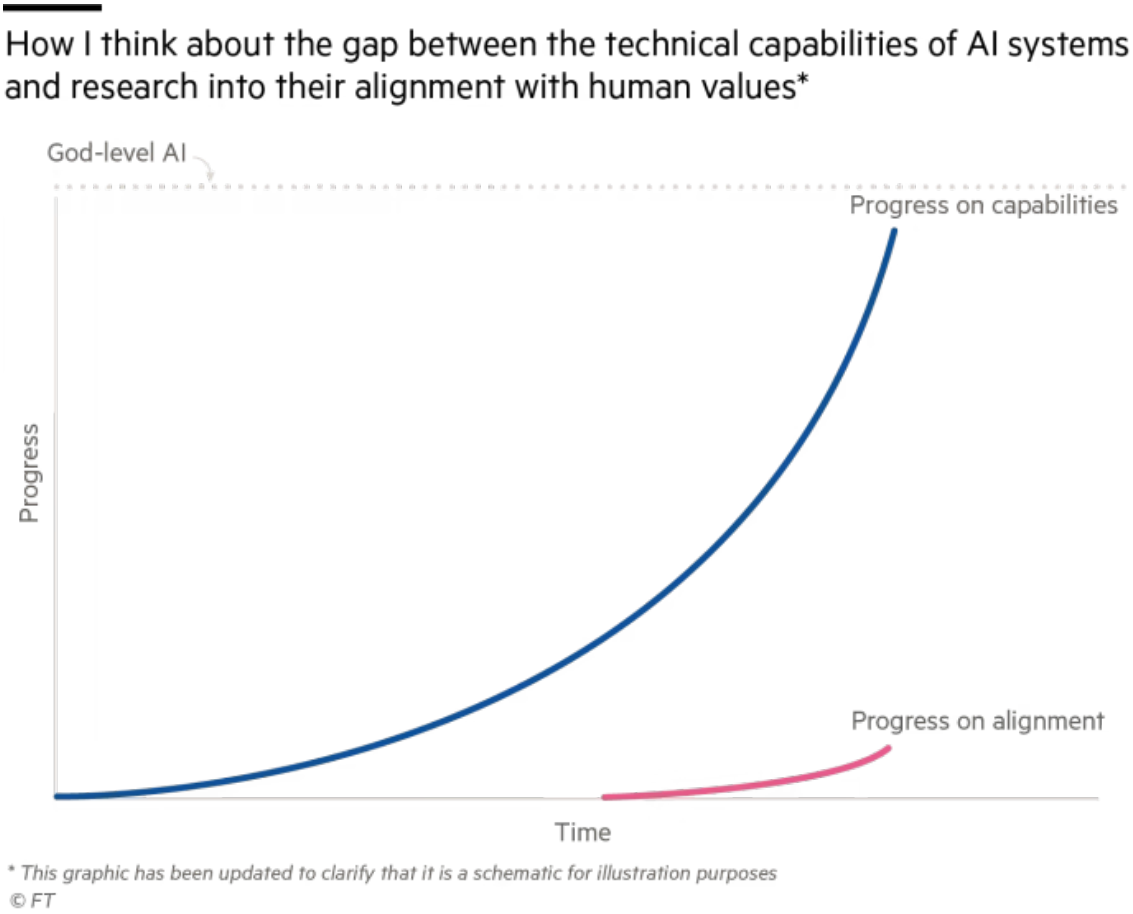


Figure 23: Illustration of the gap in "progress" between research on alignment and research on AI capabilities (Hogarth 2023)

4. **Investment in AI - both capabilities and safety is dominated by the US**

A common theme across various metrics is the dominance of the United States in AI developments. Evidence for this includes:

- Over 70% of AI research citations come from US-based institutions and organizations (State of AI, 2023 slide 68)
- 70% of global private capital investment in AI companies was in the US (State of AI, 2023 slide 111) - an increase from 55% in 2022.
- 315 out of 473 (66%) of Unicorn AI companies are based in the US (State of AI 2023, slide 112)
- 54% of papers for significant Machine Learning systems were written by authors in the United States (AI Index Report 2023, Figure 1.2.6)
- In 2022 the US passed twice as as many AI-related bills as any other country (AI Index Report 2023, Figure 6.1.3)

Behind the US, countries leading on AI include China, the UK and the European Union.

5. **Governments are significantly investing resources into AI - both capabilities and safety**

2023 marked a turning point in national government's interest in AI, with several commitments to invest resources. Several key announcements include:

- **£100 million** invested by the UK government for an "AI Taskforce", responsible for the UK's "capability in safe and reliable foundation models" (GOV.UK 2023). This comes on top of a £900 million investment in computing power (GOV.UK 2023)
- **$140 million** in funding to launch seven new National AI Research Institutes (White House, 2023), including a **$20 million** Request for Proposals by the National Science Foundation (2023) for projects on "Safe Learning-Enabled Systems"

It is yet to be seen how much these commitments will be directed towards mitigating risks from AI, rather than increasing their capabilities. Nonetheless, if these investments are considered as "AI safety", then they represent a more-than-doubling of resources in the field, outpacing philanthropic commitments.

6. **Some risks are especially neglected, and likely to remain so**

Significant funding has flowed into AI, tackling a number of key risks. Nevertheless, we believe that some areas are relatively neglected and likely to remain so - absent smart philanthropists.

Funders focused on *transformative* AI have primarily focused on power-seeking AI alone

Most philanthropic resources spent on risks from TAI have historically been concentrated on power-seeking AI. For instance, Open Philanthropy's Director of AI Strategy states *"For now, I am most focused on the threat of misaligned AI"* (Karnofsky, 2023). Open Philanthropy's grantmaking mostly reflects this view[148], and given their dominance in the philanthropic space, this has set the agenda for others in the traditional AI safety field

Funders of "Near-term" risks are focused primarily on economic and political challenges

In contrast to the above, those focused on harms and risks from *current* systems have primarily focused on issues of algorithmic bias, misinformation and job loss. There are a plethora of philanthropic foundations and non-profit organizations working on these issues - including the Ada Lovelace Institute (supported by the Nuffield Foundation); Alan Turing Institute, (funded by the UK Government); AI Ethics Initiative (supported by several groups including MIT, Luminate, Knight Foundation, and William & Flora Hewlett Foundation); and the Patrick McGovern Foundation, to name just a few.

[148] This can be particularly seen with Open Philanthropy's technical AI safety grants, which primarily focus on AI alignment. Though note a significant portion of Open Philanthropy's funding (in particular in AI Governance) can be described as threat agnostic - that is, work that will be beneficial across several different threat models (for instance, strategy research, field building, etc).

These funders do not primarily focus on worst-case issues in AI - however, their work does interact with catastrophic risks, and thus some of the issues previously discussed - such as inequality and epistemic insecurity.

Future funding on AI risks

In recent times, attention on the misuse of AI - in particular from biological weapons - has increased. For instance, misuse risk made up ~50% of the UK AI Safety Summit's focus (Gov.UK 2023); whilst public, policy and media attention has heavily focused on more intuitive risks from the misuse of AI, compared to more speculative risks such as misalignment.

In addition, it is common for AI risk discussions to focus on which actors are in control of AI systems, rather than whether or not they are controllable at all.[149] This perspective focuses less on "are AI systems safe?" and more on "who is in control of AI systems?" Indeed, it is a frequent dynamic for AI labs and national security experts to focus on being first out of the belief that they will do things better than the "bad guys".[150] For this reason, we expect misuse concerns from AI to be well resourced as AI systems continue to develop, relative to misalignment and other more speculative concerns.

Gaps in the space

Given the above considerations, I expect risks from misalignment, misuse (e.g. AI-Biorisks), and economic & epistemic impacts of AI to receive most funding.[151] Whilst positive, other risks- in particular, digital minds - are much more neglected, and likely to remain so.[152] Furthermore, even amongst threats which are relatively well funded, there are likely cost-effective opportunities that remain unfunded.

Note, the current level of resources going towards AI risks (and the allocation of funding *within* AI risks) may be *correctly* neglected. This could happen if the AI risk "market" is "efficient" - that is, philanthropists, governments and private actors are funding the most cost-effective available opportunities.

To make the case that AI risks are worthy of additional funding, we therefore need to consider whether the AI risk market is "efficient" - or, whether the topic is neglected *relative to its importance and tractability*. This is explored below.

[149] See Karnofsky (2021) on the "caution" and "competition" frames that are often adopted in discussions around AI risk.
[150] See, for example, Brookings (2022), focusing far more on the United States retaining control over China on AI, rather than on the risks AI might pose to all. Or Wang (2023) on the US Defence Department's AI strategy.
[151] This may be true even if the interventions within those buckets are not optimally distributed
[152] The issue of digital minds in particular is likely to remain neglected given that improvements to the welfare of such beings is outside the "Overton Window" for most individuals, and (I suspect) could easily remain so given the unorthodox topic.

Efficiency of AI Risk Funding

Observing the funding landscape as described above, we can ask whether the market for AI risk funding is "efficient." By "efficient", we approximately mean that funding is directed towards the most cost-effective opportunities,[153] and away from less cost-effective opportunities. If the market is efficient, then smart philanthropists can simply mirror the giving strategy of existing funders. If, however, we have reasons to believe that the market is *in*efficient (i.e. it is *not* funding the best available opportunities), then smart philanthropists can "outcompete" existing donors. Below, we discuss reasons for and against expecting an efficient market.

Reasons to suspect an efficient allocation

Existing Expertise

First, we might expect it to be hard to beat the market given that existing funders have been specialized in AI risk for several years. For example, Open Philanthropy has been investing in the AI safety field for over ten years.[154] In that time, their team has likely built significant expertise in conducting effective grantmaking in the space. In addition, the Survival and Flourishing Fund relies on several experts from across the AI risk space to conduct its grantmaking.[155] Thus, they are likely well specialized in making cost-effective AI risk grants, and so mirroring their recommendations may be an effective strategy.

Philanthropic incentives

In addition to the above, funding for AI risks has historically come from the Effective Altruism community. This community is specialized in finding the most (cost)-effective opportunities to do good. As such, we might expect resources to be allocated more efficiently here than in other areas of philanthropy.

Market Incentives

Third, as well as philanthropic incentives, there are market incentives to cost-effectively invest in reducing risk. Since many AI risks are bad for the firms that develop AI products, they have direct incentives to prevent them from occurring.

Inherent Uncertainty

Finally, it's worth noting that the AI risks space is highly complex and uncertain. As such, there is a good chance that, when looking back - **ex post** - we will observe that the best available opportunities went unfunded. However, this does **not** mean that the opportunities funded were

[153] Where cost-effectiveness is measured in approximately "percentage point reductions in existential risk per $"
[154] See Karnofsky (2012) for some of Open Philanthropy's earliest discussion of AI.
[155] See more details about how SFF works via it's S-Process

a poor investment, **ex ante**. Rather, they may have been the best bet one could make given low epistemic confidence in all available options.[156]

Reasons to suspect an inefficient allocation

In contrast to the above, we have some reasons to expect that resources won't be effectively allocated. These reasons can be divided into:

1. *Cross-cutting reasons* - reasons which reduce the efficiency of *all* actors' allocations
2. *Market-based reasons* - reasons for the *free market* to make an inefficient allocation
3. *Government-based reasons* - reasons why the *government* may inefficiently allocate resources
4. *Philanthropy-based reasons* - reasons why existing *philanthropists* may be making inefficient allocations.

1. Cross-Cutting Reasons

Limited Feedback Loops

First, across all funders there is the issue that there are limited feedback loops indicating whether an investment was positive or negative. This means that - unlike in standard markets - there is no selection process by which good funders are picked over bad funders. We may gain some evidence about effective or ineffective interventions on short or medium-run proxy measures. However, it will be very unclear which decisions ultimately prevent an existential catastrophe.

We can thus only rely on poor proxies of whether a decision was net-positive or negative (nevermind the magnitude of the impact). For instance, one proxy might be the forecasts and opinions of others in the AI risk field. However, this makes it hard to assess if the entire space is missing valuable opportunities.

It's worth noting that this lack of signal cuts both ways - it may be true that existing funders lack effective feedback loops, but this is also an issue for smart philanthropists aiming to "beat the market". Thus it's not clear whether one can perform observably better than existing funders, given the inherent uncertainty that affects all actors.

Cognitive Biases

Second, there are well documented cognitive biases that make existing actors likely to misallocate resources on existential risks. These include:[157]

- **Zero-risk bias**, in which we tend to round small probabilities in our head to zero.

[156] As an analogy, we may observe that *ex post* the best investment opportunity in (say) 2013 might have been to invest in Bitcoin. But this was not apparent given the available evidence at the time. Thus, in 2013 the market may have been *ex ante* efficient.
[157] See more at Yudkowsky (2008)

- The **availability heuristic**, which biases us to neglect events that we cannot easily recall (of which an existential catastrophe is by definition not possible).
- **Optimism bias**, where we believe we are less likely to suffer negative events (such as an existential catastrophe)

Again, these biases plausibly affect all actors, so we should expect individuals to be biased in their allocations based on these biases (absent mechanisms which help to overcome them)

2. Market-based reasons

Externalities

AI risk can be modeled as a problem of externalities. That is, the full costs and benefits of producing powerful AI systems are not just felt by those developing such systems, but by society more broadly.

For instance, if an AI company sold a highly persuasive large language model to customers, the full costs (beyond merely the costs of running the machine) might include increased misinformation, the reinforcement of existing social biases, and political polarization. These costs are not just borne by the developer or consumer, but society at large. Many of the risks discussed above are of this nature.

Externalities of this kind suggest that the private sector will not (by default) adequately mitigate against AI risks - and thus further investment in AI safety may be necessary.

Public Goods

Investments in AI safety can also be modeled as public goods. AI safety is both non-excludable (it benefits all, regardless of whether they invested in safety or not) and non-rivalrous (One individual's safety doesn't prevent others' safety). Furthermore, AI safety is not just a *local* public good, but a *global, intergenerational* public good - investment in safety is to the benefit of all individuals, across generations.

There are well studied economic reasons why we might expect underprovision of public goods - especially of a global, intergenerational kind. Ord (2020) provides a detailed account of why we would expect such neglect in the case of existential risks.
Of course, AI safety being a public good does not imply *zero* funding from AI firms. After all, building safe products will likely help them become more marketable and profitable.[158]
Meanwhile, AI firms also have a self-interest in preventing an AI-induced catastrophe, since this would cause them to be unprofitable

Despite these reasons, we still have good grounds to suspect that AI firms will inadequately or inefficiently invest in measures to reduce AI risks.

[158] Although, within AI safety, AI firms would (ceteris paribus) prefer to invest in capabilities-enhancing safety measures, rather than pure safety goods.

3. Reasons in Government

Short-term and localized focus

In the same way that AI firms have a profit motive which needs to satisfy shareholders, democratic governments need to satisfy their voters - or else they will be voted out of office. However, given the global intergenerational nature of AI risks, governments - like AI firms - do not face pressures to act in the interests of *all citizens* who may benefit from AI safety investments. Instead, they are likely to invest only up to the point that makes sense for their national self-interest amongst the voting public.

Meanwhile, governments have their own incentives to *build* transformative AI systems. After all, advanced AI could help with many functions of government - including improved public services, efficient administration, national defense, economic growth, and global competitiveness. As such, we might expect government's to prioritize *capabilities-enhancing* safety opportunities, just as with private industry.

Low Knowledge Capacity

A second challenge government's face in efficiently reducing AI risk is their lack of technical expertise. Most work developing powerful AI systems takes place outside governments, as does most safety work to reduce risks. Thus, even if government's sincerely aim to reduce risks effectively, they may not be well equipped to do so.

Furthermore, given the technical nature of the problem, it is unlikely that most voters will be interested or able to push democratic governments in the optimal direction for tackling AI risks.

4. Reasons in philanthropy

Turning to the existing philanthropic landscape, we see various reasons to suspect a potentially poor allocation of resources to reduce AI risks

High centralization

First, we observed above that AI risk funding is highly concentrated amongst just a couple of organizations. In both organizations, there are only a small number of people who make consequential grantmaking decisions.[159] This has three key downsides:

1. *Lack of competitive dynamics*. Without other actors, there is little competitive pressure for funders to compete in finding the most effective AI risk opportunities. Existing "buyers" (funders) of AI safety thus occupy a *monopsony* market position, which is less efficient than a competitive philanthropic market would be (in theory).

[159] See Cotra (2023), who notes that Open Philanthropy has (until ~mid-2023) never had more than about three people working full-time on AI safety grantmaking. Meanwhile, all of the Survival and Flourishing Fund and the Long-Term Future Fund's grantmakers are part-time

2. *High-risk concentration of assets*. With just a couple of key funders, the AI safety space is potentially overdependent on a narrow portfolio of assets. The collapse of either one of these funders could have a long-lasting negative impact on the space. Such a risk has already been seen with the collapse of FTX, which aimed to make grants in AI safety among other areas. Thus there is likely value in rebalancing this portfolio, to diversify the risk.

3. *Limited information aggregation.* In a well functioning market, there are many people whose role is to collect, aggregate and act based upon new information. In the case of risks from transformative AI, we might similarly expect an efficient market would be composed of many high-capacity teams updating regularly based on new information from a diverse range of sources. Without a large number of actors operating in the space, such a mechanism is likely much slower than optimal.

Funder Constraints

Second, we can observe a couple of particular features which increase the chance that funders may make suboptimal decisions:

- Funders tackling *transformative* AI are low on capacity to evaluate potential grants, slowing the pace at which highly impactful opportunities received funding.[160]

- Existing funders tackling *transformative* AI systems have close connections to both AI firms and AI Safety organizations. Such connections may be a potential conflict of interest which could impair the impartiality of their funding decisions.

- Even if such grantmaking is not directly affected by conflicts of interest, the *perception* of a conflict of interest could lead to poor decisions in the opposite direction. For instance, if Funder A has very good reasons to believe that Organization B is highly impactful, but there exists some (perceived) conflict of interest between A and B, then A may wish to only fund B at a reduced rate - even if B is in fact doing highly valuable work.

Thus we can see how existing funders may face organizational constraints which prevents them from funding the most effective opportunities available.

Funder Biases

Aside from direct constraints, existing funders are likely to have various biases which may push them towards some funding opportunities, and away from others. These potential biases include:

Geography: Existing funders are almost entirely based and focused on the US. Whilst the United States is perhaps the most important country to focus on in AI, specialization in this

[160] See previous footnote, which notes that the largest funder of AI safety has historically not had more than 3 FTE employees focused on grantmaking

geography may come at the exclusion of other highly effective but neglected opportunities in other countries.

Threats: Most philanthropy can be divided into two camps - those targeting risks from *transformative* AI systems, and those tackling *current harms* from AI.

- Funders targeting *transformative AI* are most focused on power-seeking AI, but as we have seen, there are a broad array of potential existential risks. Given many uncertainties in this space, we ought to diversify our portfolio to cover all of them - even if we believe that power-seeking AI is the most serious risk.
- Meanwhile, funders tackling harms from *current AI systems* often do focus on a wide range of risks - from job losses, to algorithmic biases, to misinformation. However, such funding is rarely focused on *extreme, tail-end* risks from these issues - areas which we believe should be prioritized when reducing existential risks.

Interventions: Most funders targeting *transformative AI* have historically been concentrated on technical over policy solutions.[161] Yet as noted earlier, governments are playing an increasing role in AI safety - for better or for worse. As such, policy action to steer governments away from ineffective or net-negative policies and towards the most effective opportunities is a potentially high-impact strategy that funders may be less likely to take up.

Nevertheless, it's not clear what the impact of getting governments involved is at this stage, and so philanthropists should be cautious before involving governments too closely in the AI safety space (given the biases governments are likely to have, as mentioned previously).

Overall

In summary, we have various reasons to believe existing actors - in the private sector, in government, and in philanthropy - may not be funding the most optimal opportunities to reduce existential risks from AI. This suggests that smart philanthropists can "beat the market" - that is, they can fund highly effective organizations which may otherwise be neglected. In the next section, we spell out a strategy for how to do so.

Section 5: Prioritization Strategy

Given the vast number of potential threats humanity may face from AI, combined with the many available interventions, and the attention AI already receives, it can be overwhelming to know what to do.

[161] This is a weakly held intuition about the historically contingent nature of how the field of AI safety has progressed. Specifically, it appears that concerns about advanced AI - especially misalignment - grew out of a technology-based culture in Silicon Valley, rather than a policy-based culture in D.C. Some (e.g. Critch 2021) have highlighted that this has historically caused a blindspot around political/policy over technical solutions. It is worth noting that a significant and growing fraction of resources and attention are now devoted towards policy. Nonetheless, there may be some interventions still missed from this neglect.

To help deconfuse the space, in this section we develop a strategy for prioritizing the most impactful interventions. First, we outline a framework - the "Defense in Depth" model - within which to prioritize risks. Second, we detail six "Guiding Principles" - key concepts for prioritization. Finally, we outline a grantmaking strategy, derived from the guiding principles and other sections of this report.

Defense in Depth Framework

A common approach to risk management, across a wide variety of issues, is the "Defence in Depth" Framework, also known as the "Swiss Cheese Model".[162] Cotton-Barratt et al (2020) develop a Defense in Depth model against existential risks. They break down interventions into three distinct "defensive layers":

1. Prevention: Reducing the chance that a *local* catastrophe[163] occurs.
2. Response: Reducing the likelihood that the catastrophe becomes *global*.[164]
3. Resilience: Reducing the likelihood that the global catastrophe leads to an *existential* catastrophe.[165]

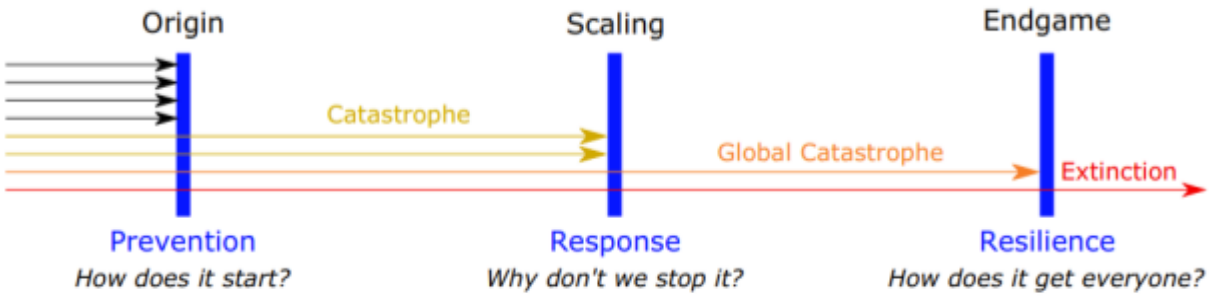


Figure 24: Example classification of various threat models, with several defense layers preventing a fraction getting past a given stage (from Cotton-Barratt et al (2020))

These three layers are not neatly defined, with some interventions cross-cutting. Nonetheless, the defense in depth approach gives us a starting framework within which we can derive some key principles for effective philanthropists aiming to reduce extreme risks from AI.

[162] Defense in Depth is a common approach to reducing a wide variety of risks. Muelhauser (2021) outlines several applications of the Defense in Depth approach, in the context of AI governance.
[163] Where the bar for "catastrophe" is approximately >1 million deaths or >$1 trillion in economic damage
[164] Where "global" means the catastrophe threatens a large portion of the world's population [although the dividing line here is not clear]
[165] Using this definition of existential catastrophe.

Guiding Principles

Below we detail six key principles to invoke when prioritizing between opportunities to reduce AI risk. These principles are generally valuable heuristics, and time-invariant.

1. Prioritize reducing risks which pose the greatest expected harm

First, all else equal, we ought to focus on tackling threat models which could cause the *greatest expected harm*.

The various threats outlined above have both some probability of occurring, and a severity conditional on occurring. Combining these two gives us an estimate from the expected damage that will result from the threat - and thus the expected value of preventing the threat occuring.

Precise quantification of expected value of threats is very challenging. Nonetheless, we can aim to at least make ordinal statements between threats (e.g. “A is more likely than B”), which can help guide prioritization. Figure 25 illustrates such a case - all else being equal, it's more valuable to close off threat A than threat B.

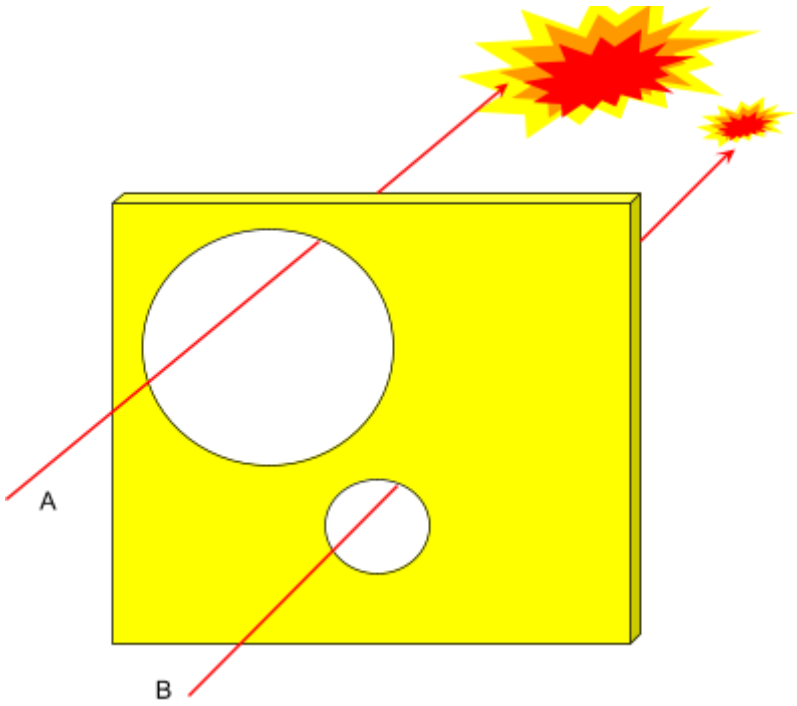


Figure 25: Illustration of two example threats, “A” and “B”, with gaps in defensive layers representing the probability that each threat passes through the defensive layer (in this case, A is larger, and thus has a higher probability of causing catastrophe)

2. Prioritize the most **neglected** risks and interventions

Second, all else equal, we ought to focus on threats and interventions which receive the least resources.

Focusing on neglected issues is based on the assumption of *diminishing marginal returns*. When an issue already receives a lot of funding, we can assume that one extra unit of funding will go less far. In this case, we should instead focus on areas with low hanging fruit. We can focus on both neglected *threats* - risks which others have not focused on; and neglected *interventions* - solutions which others have overlooked. Figure 26 illustrates such a case.

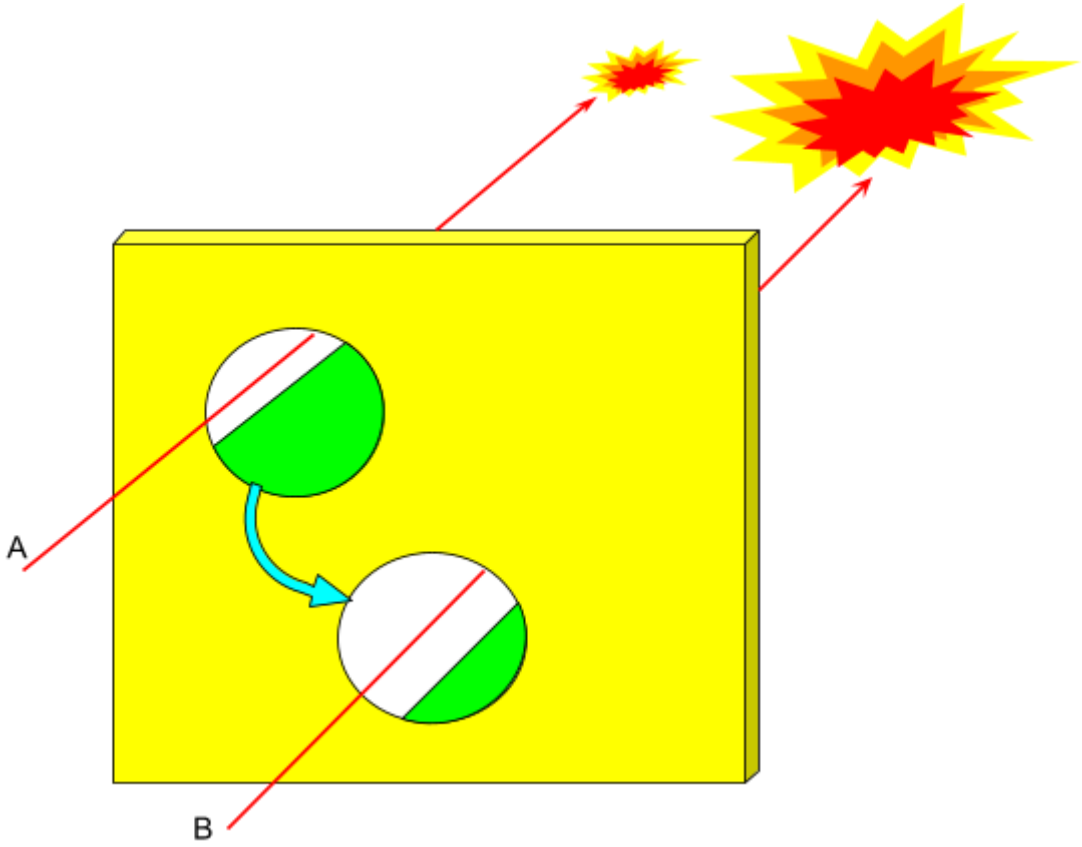


Figure 26: Illustration of two threats, "A" and "B", with some existing mitigations in place (represented by green chords covering the defensive layer). Because threat A receives more mitigation than B, it is less likely to occur. However, additional marginal resources spent on A are likely more expensive (since low-hanging fruit on A are likely already taken). Thus, moving resources towards threat B is likely a net-positive.

An important caveat is that neglectedness is not sufficient reason to fund a given opportunity. After all, there are many eccentric causes which go unfunded, which are correctly ignored. Thus, we need to combine neglectedness with a rationale for why this decision is mistaken - for instance, if there are observable market inefficiencies (as discussed previously).

3. Pursue an anti-correlated strategy

All else being equal, we ought to focus on areas which we expect will be important *in cases where existing defenses fail*. This means a strategy whose success is *anti-correlated* with the success of other actors

The reason for pursuing an anti-correlated approach is the Swiss Cheese Model. We want to make sure that our defensive layers don't just cover the same ground as existing defenses, but are actively differentiating. We can see why through the diagram below:

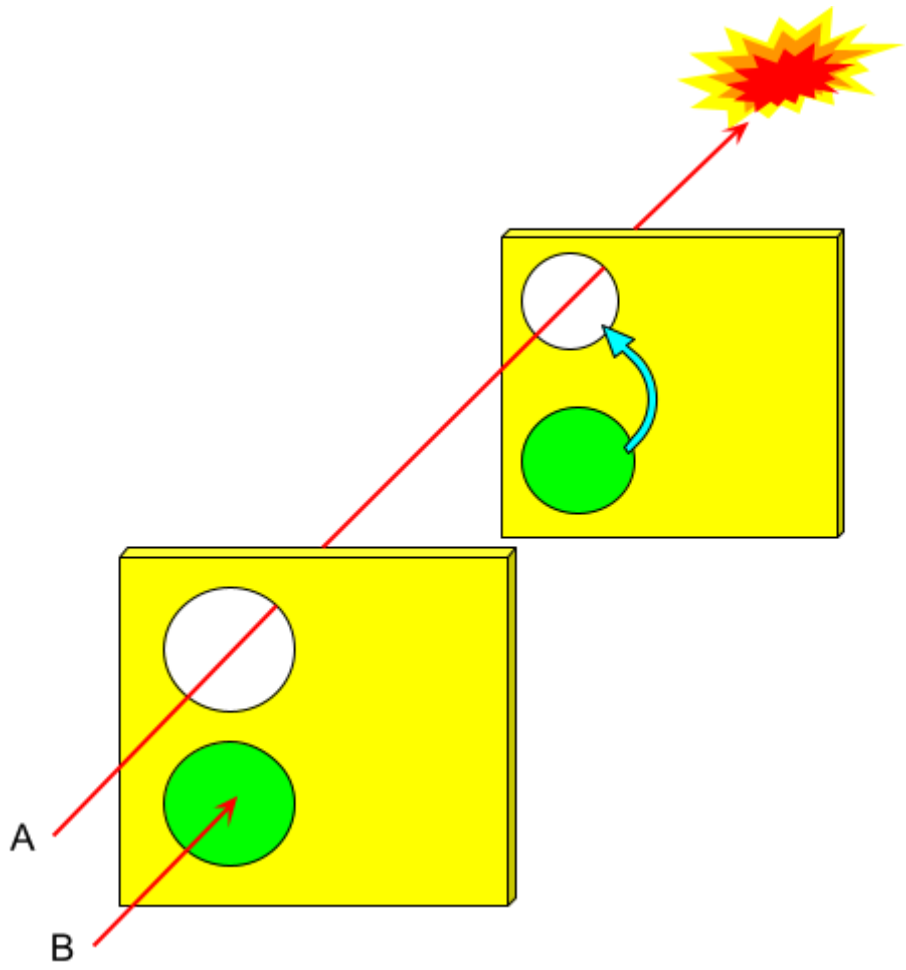


Figure 27: Illustration of two threats, "A" and "B", with existing mitigations on threat B across two layers. Moving defenses on threat B towards threat A (on either layer) will have the net effect of reducing overall risk

In this example, threat B is very well fortified against, whilst threat A has no defenses at all. By adding to A, or shifting resources from B to A, we can reduce overall risk. Similarly, if all defenses are concentrated on one defense layer, we may be able to reduce risk by spreading resources between layers.

This principle helps to guard against the potential biases and inefficiencies that other actors - AI labs, governments, and philanthropists - may possess. For instance, if other actors are investing most of their "eggs" into defenses against threat B, then on the margin we ought to prioritize worlds where they fail - for instance if another threat (A) occurs instead, or if mitigation plans for B are insufficient.

An anti-correlated strategy is closely related to the principle of *Robust Diversification*, which states that one should "prioritize strategies that are effective under pessimistic assumptions".[166] Such reasoning transfers across to an AI risk mitigation strategy.[167] In practice, such "worst worlds" thinking can imply acting as if:

- The safety plans of frontier AI labs will fail
- The timeline to transformative AI is short
- AI systems will develop discontinuously and/or very rapidly
- Society will struggle to coordinate the deployment of safe AI systems

Acting under such beliefs can make sense even if one puts low credence in them - since in cases where they are false, the value of many AI risk interventions is reduced.

4. Invest in Threat-Agnostic approaches

All else equal, it is better to prioritize interventions which act on multiple threats, rather than just one. This is especially true if multiple threat models pass through the same defense layer, as shown below.

[166] Ackva, Sandkühler and Buxton-Walsh (2021), page 66

[167] Note, however, that such a principle only stretches so far: In worlds where an AI catastrophe is all but assured, the value of reducing AI risks is significantly reduced. Instead, we should focus on "Tightrope scenarios": world's where "risks from AI are significant and could lead to catastrophe, but in principle we can navigate the risks and avoid catastrophic outcomes." Such terminology is taken from Garfinkel (2023). Our view is that on the margin, this should push one towards more pessimistic assumptions

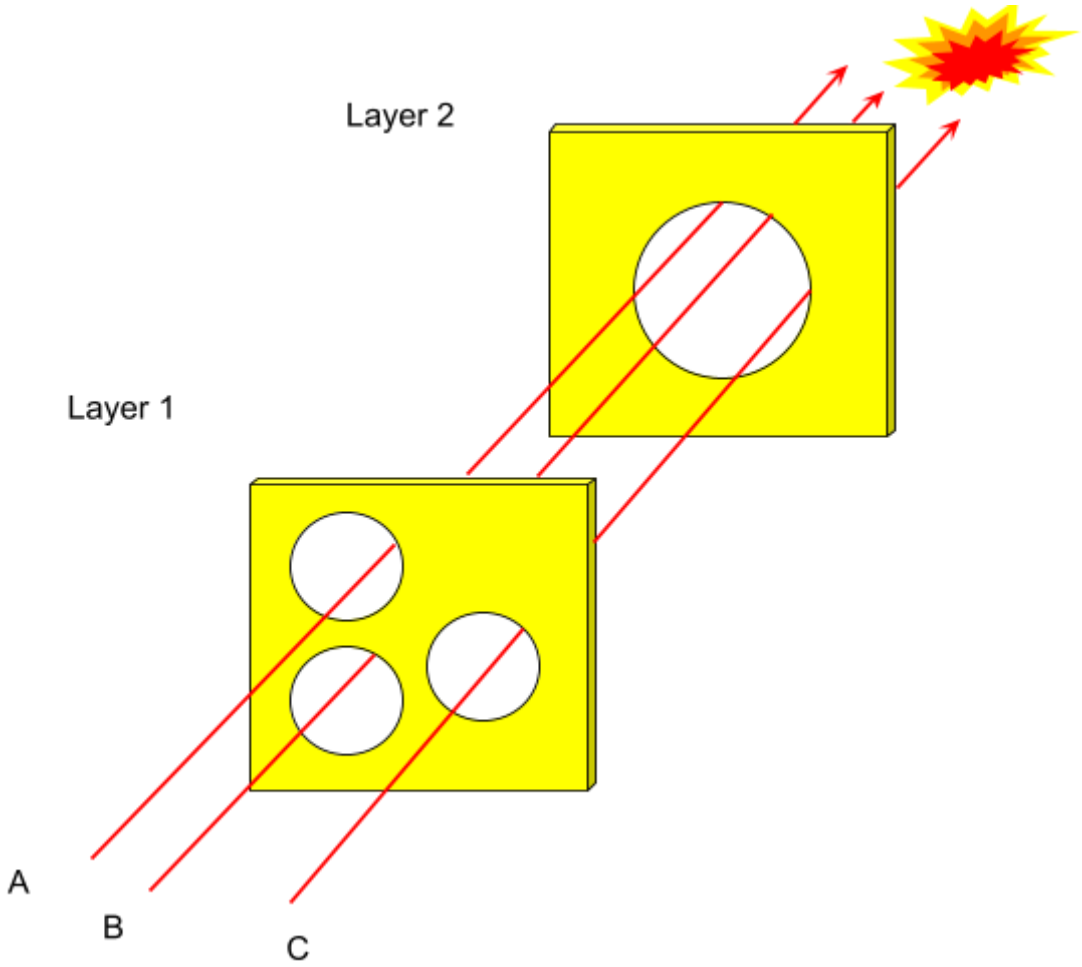


Figure 28: Illustration of three threats, "A", "B" and "C". In the first layer, we need separate defenses for each of the three threats. However, in the second layer, a single defensive intervention can act against all three threats simultaneously

If we are only able to strengthen one defense layer, then (all else equal) we should strengthen layer 2, rather than any weakness in layer 1. This principle generalizes: actions which reduce risks from multiple threat lines are (all else equal) more effective than from any single line of attack.

Threat-agnostic interventions are more effective if one believes threats are roughly equal in size. Of course, if one believes a particular threat line / model dominates others, this is less the case. As such there may be some tradeoff between threat agnosticism and prioritizing the biggest threats.

It is also worth being cautious about potential convergence in solutions.[168] It would be surprising if the most effective mitigation against threat A is *also* the most effective against threat B. It may be more cost-effective to implement targeted interventions on threats A and B individually, rather than aiming to broadly tackle both at the same time.

[168] Beware surprising and suspicious convergence

5. Leverage Existing Societal Resources

We expect funding to (continue to) flow into AI safety, from both governments and other philanthropists - see Overview of AI risk funding. Additional funding may be helpful, but could be easily focused on less impactful opportunities. As such, one strategy to pursue is to shift resources towards the most critical and neglected issues.

Such a strategy can be modeled as a "multiplier": For every $1 donated to an advocacy organization, they can move $X towards intervention A, with some probability p. In cases where p*X > 1, it is more effective to donate to the organization advocating for funding towards A, rather to fund A directly. Because governments and large philanthropic organizations possess significant resources, there can be a large multiplier effect on leveraging these organizations

6. Mitigate against unintentionally worsening risk

We are highly unsure about even the sign of many interventions, given huge uncertainties across threats, interventions, strategies, policies, etc. Thus there is a non-trivial chance that actions to reduce x-risks from AI have the reverse effect (some claim existing philanthropic investments have had this effect).[169] Some particularly salient risks of causing harm include:

- Accelerating racing dynamics
- Shortening timelines / accelerating capabilities
- Increasing takeoff speed

Note that some of these potential harms are anti-correlated. For example, slowing timelines may have the net effect of increasing takeoff speeds, due to a potential overhang in key inputs (e.g compute). Thus it is hard to have high confidence that any given intervention will necessarily be net-positive. This consideration should be accounted for irrespective of one's given focus.

Founders Pledge's Grantmaking Strategy

The principles outlined above are a useful guide for where to focus resources in general, but they don't provide concrete recommendations for how philanthropists should reduce risks from AI today. In this section, we outline a grantmaking strategy, informed by all the preceding sections.[170]

[169] For instance, Open Philanthropy's investment in OpenAI has been controversial
[170] Note that these recommendations are based on the state of the AI risk field in late 2023, and so may not represent our approach in future years.

Invest in resilience against AI takeover

Across all threats, we recommend interventions which help to prevent a catastrophe becoming *existential* - i.e, interventions which increase *global resilience*.[171] For concreteness, we outline some potential interventions which boost resilience against an AI catastrophe (also discussed previously):

- **Robust Shutdown Mechanisms**: We can invest in infrastructure and planning to make it easier to close down AI systems in the event that they show signs of attempting a takeover. This could both literally and metaphorically resemble "pulling the plug"
- **Harden Human defenses**: In retort, some may argue that the weakness in our containment strategy will not be physical limitations, but humans - who may be vulnerable to manipulation or blackmail. Thus we may wish to prepare individuals working close to capable AI systems, to make sure they are not susceptible to such attack
- **Clarifying containment protocols**: More broadly, we may wish to ensure there are clear plans and protocols in the event that an AI system appears to be capable of AI takeover. This is because, whilst much work has been conducted on *evaluating* the capabilities of potentially dangerous AI systems, less planning has been made for what ought to happen in the event that AI systems *are* dangerous and capable of takeover.
- **Closing gaps in existing threat models**: We can already observe some pathways by which humanity is at risk of catastrophe - such as pandemics, nuclear war, and (more broadly) emerging technologies (such as nanotechnology). Plausibly, these are also pathways which an AI system would actually use to disempower humans. Thus, we might also wish to reduce these risks for the sake of preventing an AI takeover

There are several lines of reasoning which point towards resilience being a promising approach:

Resilience directly reduces the chance of a successful takeover from a misaligned AI systems

First, increasing civilizational resilience is likely to make it harder for a misaligned system to actually disempower humanity. This was described in the above section on post-deployment corrections. In brief, resilience increases the "takeover threshold" - that is, the threshold of

[171] We take our definition of resilience from Cotton-Barratt et al (2020): "*reducing the likelihood that a severe global catastrophe eventually causes human extinction*"

capabilities which AI systems must exceed in order to successfully takeover.

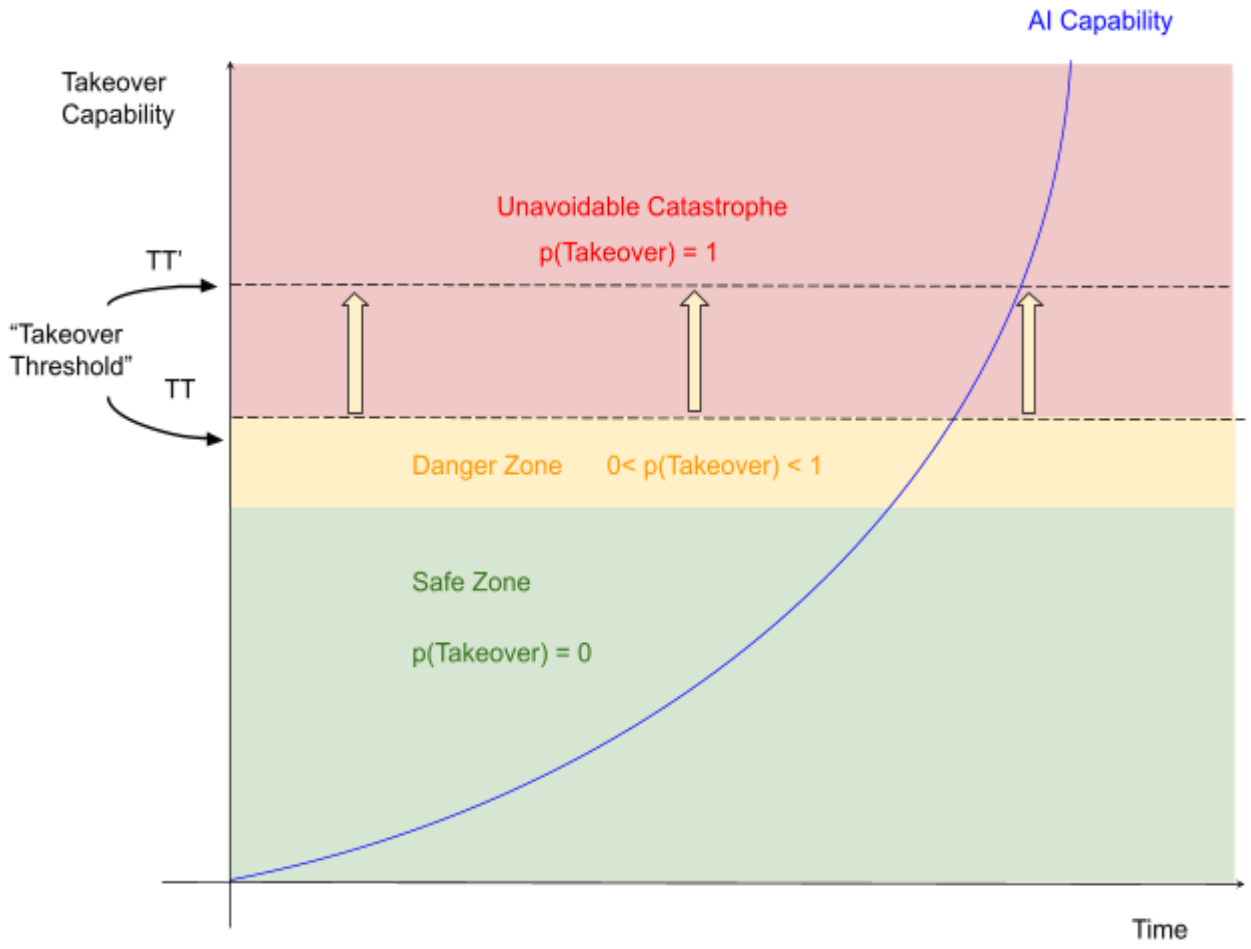


Figure 29: Illustration of the concept of "Takeover Threshold", with time on the x-axis, and the takeover capabilities of frontier AI systems on the y-axis. The blue line represents how takeover-capabilities may increase (exponentially) over time. This will mean passing through the "safe zone" (where takeover is impossible), through the "Danger zone" (where takeover could occur) to "Unavoidable catastrophe" (where takeover is inevitable). "Takeover threshold" represents the dividing line between danger zone and unavoidable catastrophe. Interventions can increase this threshold.

Interventions to increase resilience push the takeover threshold up. Diagrammatically, they move us from a cost at TT to TT'. This has two notable effects:

1. For any given capability (c) within the "danger zone", the probability of a successful takeover decreases
2. The length of time before a catastrophe becomes unavoidable is extended.

The implications of this are:

- Increasing resilience should directly reduce the chance of an existential catastrophe by weakening a takeover attempt
- In cases where AI systems do not attempt takeover until p(success) is close to 1 (for instance, if they are deceptively aligned and "bide their time"), there are at least benefits to delaying a takeover attempt. For instance, extra time allows for more alignment research to be conducted, such that a safe AI system can be deployed instead.

Increasing global resilience can be threat-agnostic

Interventions which increase global resilience do not just reduce the chance of AI takeover - they also provide defenses against other threat models. For example, one way to improve resilience against an AI takeover might be to restrict the use of AI in high stakes domains (such as nuclear or biological weapons). At the same time, such restrictions also reduce the chance of an AI-induced biological catastrophe, or a catastrophe caused by military AI use (for instance, preventing AI from usage in nuclear weapons systems).

Interventions to increase global resilience are promising for tackling other catastrophic risks, even when ignoring risks from transformative AI. For example, "Right of Boom" interventions - actions limiting destruction *after* the first use of nuclear weapons - are a promising line of interventions to prevent the chance of nuclear catastrophe. Thus, action to boost resilience has positive externalities beyond reducing AI risks - and hence, even if one is skeptical about many of the risks outlined in this report, resilience remains a promising avenue to pursue.

Investment in resilience is anti-correlated with existing funders' priorities

Governments (as well as existing philanthropists) tend to have a short-termist bias, neglecting issues which are less salient or are far away in time. For example, existing funders face incentives to consider only the interests of current generations.

Under such a "current generations" view there is no significant difference between a *global* catastrophe causing the extinction of 98% of people, and an *existential* catastrophe causing full human extinction.[172] Interventions which aim to prevent an *existential* (not just global) catastrophe are therefore more likely to go unfunded. These biases give reason for an extra focus on resilience to *existential* threats.

Meanwhile, there are some funders focused on existential risks from transformative AI, such as Open Philanthropy and the Survival and Flourishing Fund. However, the vast majority of these funders have focused on either a) technical research to reduce the chance of misalignment, or b) governance mechanisms to ensure a successful deployment of AI systems. Whilst both focuses are highly important, they neglect the issue of what happens in the case where misaligned systems are in fact deployed and attempting takeover. Thus we are missing a critical defensive layer to prepare for ***worlds where alignment research and governance fail*.**

Arguments against pursuing resilience

[172] See Parfit (1986, p453-454) for more discussion.

One objection to pursuing resilience is that AI systems may simply just "bide their time". That is, they may act as deceptively aligned systems, up until the point where takeover is all but assured. Then, they may execute a "treacherous turn", turning on humanity. In response, I think this is a valid concern - however, it does not negate all the value of resilience. For instance, making it harder for a system to takeover does at least buy humanity some time. This is because when a takeover is harder to execute, the AI system will need to wait longer to acquire the necessary capabilities.

Another objection is that powerful AI systems will develop very quickly. For example, if there is a "hard takeoff", there might be a very short period of time between the "safe zone" and an unavoidable catastrophe. I think this is a valid concern, and one that does diminish the value of resilience interventions somewhat.

A further uncertainty is whether there are many cost-effective interventions available which have not already been funded. After all, several of the interventions boosting resilience are already likely invested (at least some degree) by other funders for other reasons. For example, civil defense interventions are already an opportunity for preventing catastrophe from nuclear and biological threats - thus, there is an open question whether pursuing resilience to reduce AI risk will lead to any new unfunded opportunities. However, resilience interventions to prevent a biological, nuclear or climate catastrophe appear neglected on their own grounds, thus leaving highly impactful opportunities on the table

Novel research approaches to prevent unsolved misalignment problems

Within the threat model of *power-seeking AI*, we recommend funding underexplored research approaches to prevent key misalignment challenges, especially those related to inner misalignment. There are a couple of reasons we believe this is a key node to fund:

Inner misalignment issues - such as those related to catastrophic goal misgeneralization, deception and the "Sharp Left Turn" - are key unsolved challenges critical to preventing AI takeover

In the power-seeking threat model, a key reason why AI systems may be difficult to align is due to inner misalignment. In brief, this is the challenge of ensuring that the objective we *specify* for an AI system is the objective that the AI system actually *learns* (rather than a correlated or orthogonal terminal goal, which incentivizes the system to instrumentally converge on power-seeking.

Several reasons to be concerned that inner misalignment may occur:

1. AI systems may undergo *goal misgeneralization* (DeepMind 2020), pursuing a different goal in the real world compared to the goal we observed when training a system.
2. AI systems may act as *deceptively aligned*, obscuring their true (misaligned) objectives (Karnofsky 2023)

3. AI systems may be obscure "inscrutable matrices", unintelligible to human evaluators, making testing for deception and alignment challenging.

Furthermore, there is some chance that a powerful AI system undergoes a Sharp Left Turn, where AI capabilities advance and generalize (far) faster than our existing alignment techniques. If this is the case, then most empirical alignment research will fail to prevent misalignment.

Our (highly tentative) view is that the strength of evidence for inner alignment failures and a "Sharp Left Turn" appears to be less strong than for outer failures. Nonetheless, we believe that if inner failures are likely to occur, the current state of research is inadequate for preventing such issues. This makes inner misalignment one of the largest risks, posing some of the largest expected harm (relative to other risks).

AI labs are not on track to solve key inner alignment problems

A second reason to focus on the inner alignment problems is that progress on the above problems is anti-correlated to the strategies of existing AI labs. Or, put another way, in cases where AI labs fail to align AI systems, it is likely due to an inner alignment failure.

Exploring the plans of frontier AI labs more closely, we can see that organizational plans to "solve alignment" mostly focus on outer, rather than inner alignment:

1. OpenAI's "Superalignment Team" plans to develop techniques for scalable AI oversight, whereby an "AI Alignment Researcher" is trained using human feedback, which then assists humans in developing and evaluating aligned models, along with alignment research. OpenAI's plan will likely assist on outer alignment failures - like specification gaming - and more broadly will help to specify goals that align with our intentions.

   However, this plan does not include work on robustness or interpretability, which would both help tackle inner failure.[173] Similarly, it does not work if aligning human or superhuman AI systems is very different from current systems.[174]

2. Anthropic's "Core views on AI safety" focuses on empirical approaches to AI safety. Key approaches include Constitutional AI (developing "Helpful, Harmless, Honest" AI systems using "Reinforcement Learning from AI Feedback") and Mechanistic Interpretability.[175]

[173] OpenAI say this themselves: *"The path laid out here underemphasizes the importance of robustness and interpretability research, two areas OpenAI is currently underinvested in"*
[174] OpenAI says this: "*Aligning AGI likely involves solving very different problems than aligning today's AI systems. We expect the transition to be somewhat continuous, but if there are major discontinuities or paradigm shifts, then most lessons learned from aligning models like InstructGPT might not be directly useful.*"
[175] Other research includes: *"Scalable Oversight, Process-Oriented Learning, Understanding Generalization, Testing for Dangerous Failure Modes, Societal Impacts and Evaluations"*. However, my impression is mechanistic interpretability and ConstitutionalAI are core to Anthropic's plans

Constitutional AI is one approach to ensuring outer alignment, whilst interpretability can help to understand the mechanics of AI systems. This will likely be beneficial for scanning for inner misalignment. Whilst promising, this plan does not directly prevent deception, goal misgeneralization, or a "sharp left turn".

3. DeepMind's AI Safety Team released some details on their high level strategy. Key approaches include scalable oversight, mechanistic interpretability and capability evaluations. Additionally, risks from goal misgeneralization and deception are acknowledged.

   However, it is notable that this piece does not represent the views of DeepMind as a whole, and thus it is possible that as an organization, DeepMind does not focus on building inner-aligned systems.

Anthropic, DeepMind and OpenAI represent the frontier of AI safety, with most other labs having much more limited safety teams. Thus, we can reasonably conclude that other labs are likely to implement less rigorous safety techniques.

Given the state of AI safety outlined above, many of the key issues - like goal misgeneralization, deception and the Sharp Left Turn - remain unsolved. In world's where AI labs fail to prevent misalignment, it is likely because their techniques did not cover the right issues; Even if one believes that inner misalignment is a relatively unlikely problem, we ought to focus on it, as an anti-correlated strategy to counter AI labs' approaches.

AI labs are less incentivized to solve certain alignment failures

Additionally, we should focus on the alignment problems outlined above because *AI labs will not be incentivized to tackle them.*

If we model AI labs as firms, where capabilities are a proxy for profits, then we should expect firms to fund alignment research in cases where it also increases their capabilities (i.e. where the correlation between alignment and capabilities is high.)

For concreteness, take the example of Reinforcement Learning from Human Feedback (RLHF). RLHF helps to "outer align" large language models to human preferences. Outer alignment of this variety also helps to make AI models more capable (and thus profitable). RLHF is thus a technique where **the correlation between alignment research and capabilities is high**. In future, we should expect similar alignment research (for instance scaleable oversight approaches like Constitutional AI, Reinforcement Learning from AI Feedback, and similar) to be funded by AI labs, given similar commercial incentives.

This leaves a class of alignment issues relatively more neglected - in particular those, where **corr(alignment research, capabilities) is low**. Alignment research which does *not* help accelerate capabilities likely includes theoretical and conceptual alignment theory, less connected to existing systems. This also reduces the chance of unintentionally causing harm by accelerating capabilities.

Mechanisms for coordination between key actors

Across multiple threat models, a significant source of risk is due to the inability for key actors to coordinate with one another. Thus, a promising strategy is to focus on ensuring they can work together, through various means. Below we outline why coordination appears critical

Risks are primarily driven by multilateral, rather than unilateral failures

Across our analysis of threat models, it is rarely the case that a single individual or group causes disaster by acting unilaterally; instead, the likelihood of catastrophe is driven up by multipolar dynamics between actors. We can see this effect for several risks:

Power-Seeking AI

A significant driver of the risk from power-seeking AI is a coordination failure between the developers of frontier AI systems. We term this kind of failure a *multilateral failure*.

For example, the *Deployment Dilemma* refers to the dynamic where each is incentivized to build and deploy systems quickly, at the expense of appropriate safety measures. This dilemma can occur even if all actors would prefer to slow down and invest in safety. If an AI lab merely *perceives* that others are acting less cautiously, then they may feel obligated to deploy their own AI system first (even if they themselves are unsure about its safety). This dynamic can occur both at the industry level (between competing AI firms), and at the international level (between nation states). Coordination issues are further compounded by the ever-decreasing cost of building AI systems, which means more and more actors can potentially cause catastrophic risks.

We thus suspect that improving coordination will help to reduce the risk of a *multilateral failure*. Coordination appears key because almost all frontier actors appear in agreement that the chance of a catastrophic risk is not trivial.[176] As such, a *multilateral* failure (for example, firms engaging in an AI arms race) appears more likely as a driver of risk than a *unilateral* failure (for example, if a single actor skeptical of risks decides to deploy an unsafe model).

Biological Risks driven by AI

Distinct from power-seeking risks, a key node of failure in the case of an AI-induced Biological Catastrophe is the risk from *mass proliferation of dangerous biological capabilities*. We believe this form of proliferation - providing mass access to pandemic-capable AI systems - poses more risk than the increase in risk from *frontier AI-driven biological capabilities*, accessible to only a small number.

As such, once again the key driver of risk is not from a leading actor unilaterally choosing to cause a catastrophe; rather, the risk comes from an uncontrolled release of dangerously

[176] For instance:
- Anthropic's CEO estimates a 10 - 25% chance of an AI catastrophe
- OpenAI' CEO states that the worst-case scenario for AI is "lights out for all of us"
- DeepMind's CEO believes AI risk should be treated as seriously as the climate crisis

capable systems to a wide number. Thus, coordination amongst actors to prevent this occurring is crucial

Military AI risks

Within the class of risks associated with the use of AI for military applications, our greatest concern is not that individual actors will unilaterally deploy AI military weapons like "killer robots". Rather, most risk is driven by structural problems, such as racing dynamics between militaries. This pushes us to prioritize the pursuit of coordination between states, such that these structural risks are minimized

Democratic and Distributional Risks

As with military AI risks, a key driver of risks related to democratic control and the economic distribution of AI systems is cases where a small number of actors gain vast amounts of wealth and power, at the expense of the best interests of all other individuals (both present and future). Thus, securing agreements between relevant parties *before* powerful AI systems are developed, about ideal governance structures, will be key. Again, this involves developing better coordination mechanisms.

Overall, we can see that pursuing coordination to prevent multilateral failures is a threat-agnostic approach. We have already seen evidence that coordination efforts can simultaneously help with multiple risks - see, for instance, the UK's AI Safety Summit, which covered various frontier risks including loss of control, misuse, and structural challenges.

It is worth noting that "coordination mechanisms" is a relatively vague term to describe a number of different interventions. Coordination can be facilitated through various forms, such as:

- The signing of treaties, statements and letters to signal agreement[177]
- Drafting and implementing commonly agreed regulatory regimes,[178] as well as the enforcement and verification of such regimes[179]
- Supporting discussions and dialogues, through both official and unofficial channels.[180]

Coordination is relatively less likely to worsen risk

Whilst no given intervention comes with zero chance of backfiring, our view is that improving coordination is relatively less likely to increase risk. This is because coordination can help to reduce racing dynamics, which generally shortens the time available to reduce risks, as well as incentivizing actors to cut corners. Thus, by reducing racing, coordination efforts can tackle multiple sources of risk at once.

[177] Previous examples include The Bletchley Declaration, Statement on AI Risk and Pause Giant AI Experiments: An Open Letter
[178] Example proposals include an International AI Agency (IAIA)
[179] Example proposals include Mittelsteadt (2021) and Shavit (2023)
[180] For example, track I and track II dialogues

Government Support and Advocacy

Another key strategy for minimizing existential risks from AI which we find promising involves targeting governments and other key institutions. Such targeting could involve a wide range of interventions, including:

- Building capacity and expertise inside governments to tackle risks
- Providing policy advice and recommendations
- Advocacy and activism efforts to increase the incentive for governments to prioritize safety

Governments are the only actor with the capability to implement crucial interventions

National governments possess legitimate political power, which allows them to pursue many interventions relevant to preventing risks from AI. Furthermore, they are often the only such actor. For example, only national governments are able to introduce legally-binding policies, taxes and regulations, backed by the power of the state.

This power means that in order to introduce many of the most essential interventions, governments must be engaged. Such engagement allows for an *anti-correlated* approach: Whilst other philanthropists and actors can make some progress against risks, they do not have all available interventions at their disposal. Thus, in cases where their approaches fail, it may require the powers of government to intervene.

Government resources can be leveraged towards high impact opportunities.

As well as possessing unique powers, governments possess significant financial resources. There is evidence that governments are keen to direct some resources towards AI, including towards its potential risks.[181] However, they may not be targeted at the largest threats or highest impact opportunities within AI. Thus, one strategy is to leverage resources in this direction, through advocacy and engagement with government by third parties.

Existing anti-government biases mean key interventions might be neglected

Despite the powers and resources possessed by governments, there has historically been little engagement between states and experts concerned by advanced AI. For example, the AI safety community has historically been based in Silicon Valley, and thus unconnected from key policy debates in Washington DC and other national governments.[182] As such, an

[181] See the AI Safety Institute in the UK, or its equivalent in the US, both of which have resources for AI safety

[182] This concern was mentioned in a previous footnote:
This is a weakly held intuition about the historically contingent nature of how the field of AI safety has progressed. Specifically, it appears that concerns about advanced AI - especially misalignment - grew out of a technology-based culture in Silicon Valley, rather than a policy-based culture in D.C. Some (e.g. Critch 2021) have highlighted that this has historically caused a blindspot around political/policy over technical solutions. It is worth noting that a significant and growing fraction of resources and attention are now devoted towards policy. Nonetheless, there may be some interventions still missed from this neglect.

anti-correlated strategy implies we may want to counter potential biases that may emerge from this disconnect.

Governments are often under-equipped and not incentivized to tackle the largest risks

One reason for skepticism about engaging governments in AI risk is concern that they may lack the necessary expertise to tackle the biggest risks, and they do not have an incentive structure set up to solve global, intergenerational issues. Both of these concerns were raised previously. We believe these concerns are valid - however, one promising response is to boost governments' capacity, and to incentivize them to take into account the interests of all parties affected by AI risk.

Governments can tackle multiple threats at once

With increased capacity, governments will be able to tackle a wide range of threats - from power-seeking AI and AI-induced Biological Threats to Democratic, Distributional and Epistemic Risks. Thus, providing government support and advocating for action can be an effective threat-agnostic approach.

Conclusion

Transformative Artificial intelligence could fundamentally transform civilization, on a scale greater than any existing event in human history. Such transformations are likely to occur within the lifetime of many people on Earth today - and may even occur within this decade. Grappling such an extreme possibility can be a dizzying experience.

This report has sought to help navigate what is a highly complex, rapidly changing landscape, with a focus on how to best prevent the chance of catastrophe. Our overarching takeaway is that, despite the complexity, **these challenges are not insurmountable**. There are a wide range of actions one can take to reduce the risks - from technical research ensuring AI systems are controllable, to policy efforts ensuring that governments are equipped and coordinated to handle the risks. We believe meaningful progress **can** be made, which will help not just prevent catastrophe, but allow for the long-term flourishing of our species.

Postscript: What if we succeed?

This report has focused heavily on the many wide-ranging catastrophic risks we face from artificial intelligence. One could be forgiven for coming away with a deep sense of fear at the prospect of powerful AI systems. But it is important to remember the enormous potential that could be achieved, *if* we successfully navigate these challenges.

On the average day, about 16,000 children die before they reach their fifteenth birthday.[183] Often, such tragedies occur due to diseases which are easily treatable. A further 165,000 adults die from a wide variety of causes.[184] 10% of people live on less than $1.90 a day, and the majority live on less than $10[185] - whilst the top 1% of earners take home 20% of the wealth.[186] Meanwhile, hundreds of millions of animals are killed every day[187] - ~95% of which spend their whole lives in factory farms.[188] Average global temperatures have significantly spiked over the past decades, set to cause significant deaths and destruction.[189]

Some look at this dark picture and conclude that perhaps it's not even worth trying to prevent catastrophe. Perhaps we, in some sense, "deserve" to be destroyed by our own technological creation, given the state of the world today and its trajectory.

Yet, such a picture of tragedy misses the underlying trendline: on almost all measures, the world *is* getting better. Compared to a century ago, the *vast* majority of people have more wealth, health and opportunity. More people live in democratic countries, human rights have been extended to many groups, and violence has been declining. Of course, there is still much progress needed on these issues, but the trajectory is clear. In summary: **The world is awful. The world is much better. The world can be much better**.

Transformative Artificial Intelligence - if safely controlled and cautiously deployed - could see that positive trajectory turbocharged. Economic growth could dramatically rise, lifting millions out of poverty. Scientific research could unlock cures for many diseases, whilst technological breakthroughs could allow for dramatic improvements to halt climate change. If well governed, these positives will not just be for the benefit of a small minority, but for all. A profound improvement in the human condition would occur. This is the *lower bound* of what is possible - with more far fetched sci-fi stories becoming plausibly in reach.

But this progress can **only** happen if we successfully navigate the risks posed. Without caution, we will not only miss the benefits, but lose everything that has come before. Some believe that any hint of caution is defeatist and doom-ridden, and instead we should throw out all guardrails in the pursuit of progress. **They are mistaken**: it is precisely *because* of the progress we have made - and the potential we could realize - that caution now is imperative. This is why action to mitigate the risks is so critical.

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[183] Roser (2022) *"Globally 4.3% of children die before they are 15 years old... This means that 5.9 million children die every year – 16,000 children on any average day, and 11 children every minute.2*
[184] Ritchie and Mathieu (2023) *"There were 67.1 million people who died in 2022"* - implying ~183,000 deaths every day. Subtracting 16,000 deaths from under-15s gets approximately 165,000.
[185] Roser (2016)
[186] Our World In Data (2023)
[187] Roser (2023)
[188] Ritchie (2023)
[189] Ritchie, Roser and Roadao (2020)

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